

20th Century Antarctic Seasonal Sea Ice Concentration Reconstructions

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This thesis titled
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Abstract

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Ongoing changes in the Earth's cryosphere create positive feedback loops through changing albedo, creating a spark of interest in researchers on what is happening in regions of sea ice. Despite this conviction for research, Antarctic sea ice observations have been sparsely recorded throughout human history and have only been reliably observed since the introduction of satellite measurements in approximately 1979. Recent research has shown the ability to reconstruct Antarctic pressure and sea ice extent values throughout two different studies using principal component regression techniques based primarily on temperature and pressure observations along with large-scale climate variability indices that influences Antarctic sea ice.

This short historical span of sea ice concentration (SIC), a pixel-scale observation that outlines how much of the ocean area is covered in sea ice, have shown increases around much of Antarctica since 1979. This short historical record, comparatively, does not capture the full range of climate variability in one of the most isolated places on Earth. To improve the understanding of historical variations in Antarctic SIC, reconstructions of the five leading empirical orthogonal functions (EOF) have been conducted in this study to extend SIC data back to 1905. This new seasonally resolved dataset allows researchers to place current trends in a longer historical context. This study examines the skill of these seasonal Antarctic SIC reconstructions, the first of their kind,

along with discussing regional changes in Antarctic sea ice since the beginning of the 20th century, and the implications these historical variations may have for further Antarctic sea ice and future research.

The resultant reconstructions of SIC presented through this study outline the ability to gather skillful and meaningful results on seasonal scales through 1905 that provide overall much better skill than the climatological mean. These reconstructions allowed for analysis on sea ice concentration trends throughout the pre-satellite era spatially and temporally to understand the overall trends of Antarctic sea ice in a longer historical context than ever before, supporting previous regime changes in sea ice patterns between the pre-satellite era and observed satellite era (after 1979) suggested in previous work. In particular, the role of Pacific Ocean variability in Antarctic SIC throughout the 20th century is made even more prominent through investigating the importance of predictor networks and correlations of modes of climate variability with SIC in this study.

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Chapter 1: Introduction

The ongoing changes in Earth's climate create positive feedback loops within the cryosphere by surface albedo changes that are intricately linked to the atmosphere through observations over extended periods of time (Rohli 2018; Donohoe et al. 2020). This makes the effects of climate change on the cryosphere, especially over large bodies of ice within the Arctic and Antarctic circles, especially sought after by researchers. The presence of the Southern Ocean yields impacts on a global scale due to the influence that it has with Antarctica through thermal expansion of the ocean along with a uniform global sea level rise (Hobbs et al. 2016). This justifies the need of more research in the climate-cryosphere connections, given the global impact and ever-changing nature of the relationship between these two entities.

Antarctica, and the sea ice around the continent, play a significant role in the state and variability of the Southern Ocean with changes that occur locally on the continent. The sea ice around Antarctica provides insulation, freshwater budget, and ice albedo feedback to the Southern Ocean (Liu et al. 2004). The atmosphere around Antarctica, along with the sea ice that surrounds the continent, exerts feedbacks on the surface air temperature, where an increase in surface air temperature leads to a decrease in ice-covered area, which will lower albedo and therefore raise temperatures in this region. This positive feedback loop leads to both the large interannual variability and potential for rapid change of surface air temperatures over higher latitude locations compared to the tropical and midlatitude regions of Earth (Wallace and Hobbs 2006; Donohoe et al 2020).

Antarctic sea ice typically reaches the maximum extent around mid- to late September, with the annual minimum occurring in late February (NOAA 2021).

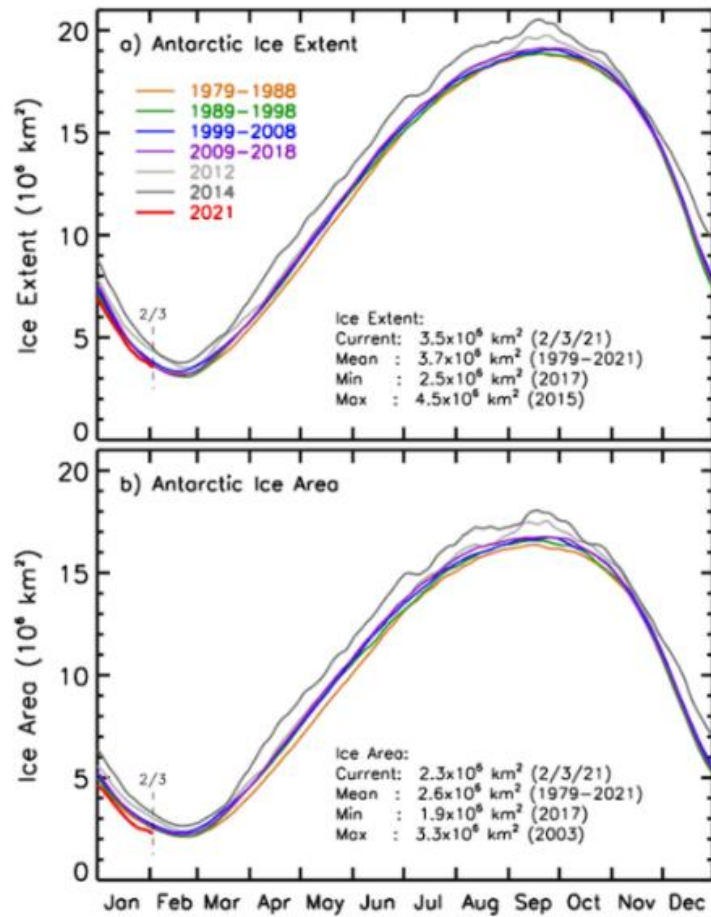
Antarctic sea ice coverage has been steadily increasing since 1978, although at a slow rate (Comiso et al. 2021). This slow rate of Antarctic sea ice growth since 1978 is in stark contrast to the Arctic sea ice coverage decreasing at a very rapid pace deduced from what observations are present in these areas. The geography, including the atmospheric and oceanic circulation, play a key role in sea ice allowing for some explanation on the differences between the Antarctic and Arctic sea ice patterns.

The continent of Antarctica is divided up into different sectors to analyze sea ice patterns based on close geographical proximity defined differently by different studies. Nicolas and Bromwich (2014) begin outlining sectors, yielding the broader sectors of the Antarctic Peninsula, West Antarctica, and East Antarctica. Raphael and Hobbs (2014) further defined these sectors pertain to sea ice extent variance and decorrelation length scale to further split Eastern Antarctica into the King Hakon Sea region along with a separate East Antarctica sector, while the Ross and Weddell seas become their own sectors in portions of West Antarctica and the Antarctic Peninsula. The Bellingshausen and Amundsen seas were combined into one sector outlining much of the West Antarctica sector (Fig. 1.2).

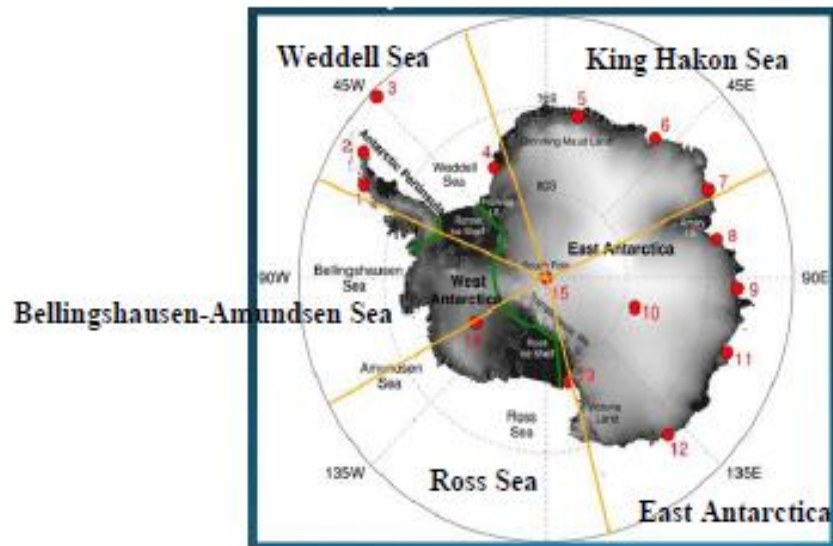
Sea ice observations, however, are sparse and do not span a long historical period, as these areas lay in harsh conditions that are typically not habitable for people. The easiest method of spatially and temporally complete observations of sea ice in these regions stems from satellite measurements, where since 1978, Antarctic sea ice

Figure 1.1

Decadal Averages for Sea Ice Extent and Sea Ice Area



Note. A: Decadal Average of sea ice extent in the Southern Hemisphere with key values;
 B: Decadal average of Antarctic sea ice area with key values (Cosimo et al. 2021).

Figure 1.2*Map of Antarctic Sectors*

Note. A map of Antarctic sectors extracted from Nicolas and Bromwich (2014) overlaid with orange lines to represent the sectors expanded on by Raphael and Hobbs (2014).

concentrations, a pixel-scale observation defined as the fraction of ocean area that is covered by sea ice, have increased in most regions (Hobbs et al. 2016; Holland 2014; Purich et al. 2016). However, the short record, especially compared to other global measures of climate, means that the observations capture a limited range of natural climate variability. To account for this, observation-based climate reconstructions have been a method that have allowed researchers to be able to account for and model conditions on the Antarctic continent to gather a better picture of historical seasonal climate variability prior to direct measurements (Fogt et al. 2016), but these primarily have only been attempted for the Antarctic atmosphere, and only recently for Antarctic sea ice extent (Fogt et al. 2022). This thesis research aims to create statistical

observation-based reconstructions of sea ice concentrations by using datasets providing information throughout the 20th century with as much accuracy as possible. Namely, the research questions are as follows:

1. How does the skill of seasonal Antarctic sea ice concentration reconstructions vary spatially throughout the 20th century?
2. How do large-scale patterns of climate variability influence the sea ice concentration reconstruction skill over the Southern Ocean?
3. From using the sea ice concentration reconstructions, how are the Antarctic sea ice changes since 1979 understood in a longer historical context spanning the entire 20th century, and to what magnitude have large-scale patterns of climate variability influenced Antarctic sea ice concentrations throughout the 20th century?

The chapters in this thesis will be structured as follows: Chapter 2 will explore literature relevant to Antarctic sea ice concentrations and phenomenon that influence these measurements. Chapter 3 outlines data collection, processing, and analysis techniques to statistically create and evaluate the skill of reconstructions. The reconstructions, their skill, and a preliminary analysis on what they unlock for understanding historical pre-satellite Antarctic sea ice variations will be discussed in Chapter 4. Chapter 5 will close by answering the research questions in detail supported from results in this work.

Chapter 2: Literature Review

2.1 Fundamentals of Sea Ice Concentrations

Sea ice concentration describes the amount of area covered by ice compared to a reference area and is typically reported in values as a percentage, where 0 percent indicates no ice coverage while 100 percent indicates the location is completely covered by ice (NSIDC 2021). These measurements are often derived from satellite data at a spatial resolution of 25 x 25km at certain time steps, where the percentage is derived from the ratio of the amount of area within the grid covered by sea ice to the total amount of area in the grid. This differs from sea ice extent, which just describes the most equatorward extent of 15% or more sea ice concentration. Sea ice concentration can also be delineated from sea ice area, broadly defined as the amount of area covered by sea ice.

Sea ice concentration can be affected by the presence of open water where ice would be expected to appear, called polynyas (NSIDC 2020). Antarctica is prone to latent-heat polynyas, where ocean water would be at the freezing point with winds pushing ice away from the coast. This allows for latent heat to be released, along with some sensible heat exchange (NSIDC 2020). Salt is released into the ocean as sea ice in the polynyas is formed, allowing for the ocean to have higher salinity at the surface, causing the surface water to be denser than other water and allowing it to sink, ultimately sinking to the bottom of the ocean and forming Antarctic Bottom Water. This sinking of water to the ocean's bottom underneath the sea ice also allows for deeper, warmer, waters to surface farther north. Polynyas will provide areas within sea ice that have lower concentration values.

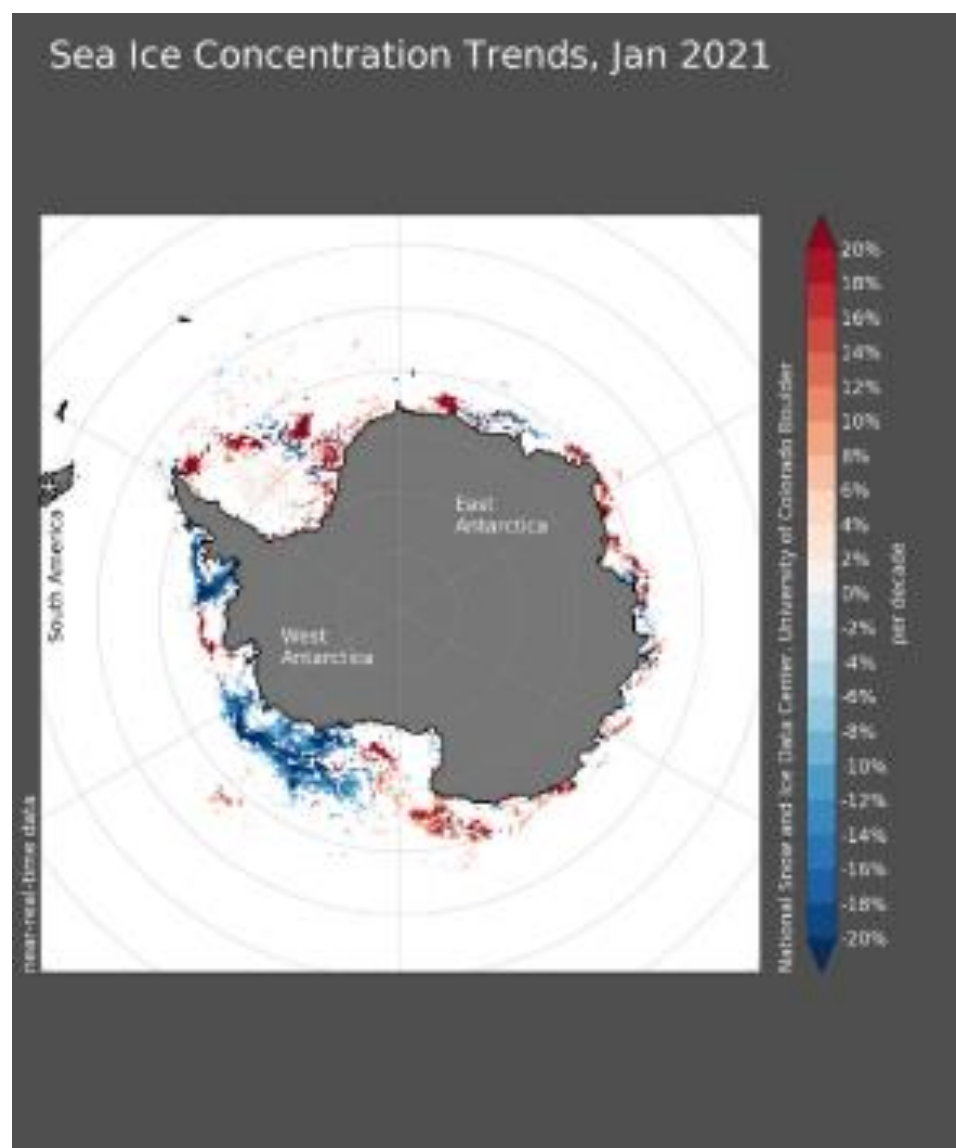
Antarctic sea ice has been incorporated into various products, but sea ice data prior to 1970 in this area are very limited (Hobbs et al 2016). In these products, it has been established that sea ice concentration has been decreasing in the previous decade off the coast of West Antarctica while increasing near the coast of East Antarctica, as demonstrated in Fig. 2.1 obtained from the National Snow and Ice Data Center [NSIDC]. Sea ice extent has been decreasing in recent years (since 2016) but as of August 2020 was on the rebound and trending more towards the average values between 1981 and 2010 (Fig. 2.2). This rebound was short-lived, as rapid declines beginning in August 2021, trending toward all-time low values in early 2022. This brings into question what is occurring on seasonal scales to gather a larger picture of sea ice concentrations and how their yearly values are affected by seasonal variability. The details of the sea ice concentration changes over the satellite record are described in more detail in the following section.

2.2 Seasonal Variability in Sea Ice

Sea ice concentration values have been positively correlated to sea ice duration—the length of time sea ice exists before melting away. With this relationship, it is possible to examine sea ice concentration in the context of retreat (after sea ice has reached its maximum extent) to advance (when it begins expanding equatorward after reaching its annual minimum). The maximum extent of sea ice concentration tends to arrive around the end of September to early October, with the minimum extent normally occurring in late February to early March, based on climatology (both the mean and interannual variability) calculated from 1981-2010 (Fig. 2.3).

Figure 2.1

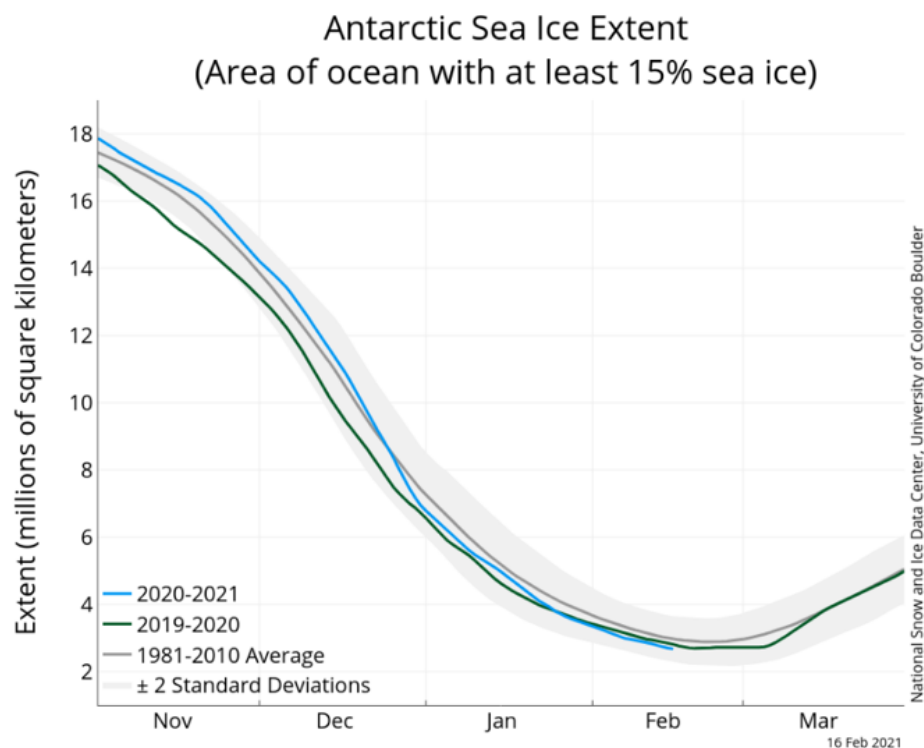
January Trends in Antarctic Sea Ice Concentration



Note. Extracted from the National Snow and Ice Data Center (NSIDC).

Figure 2.2

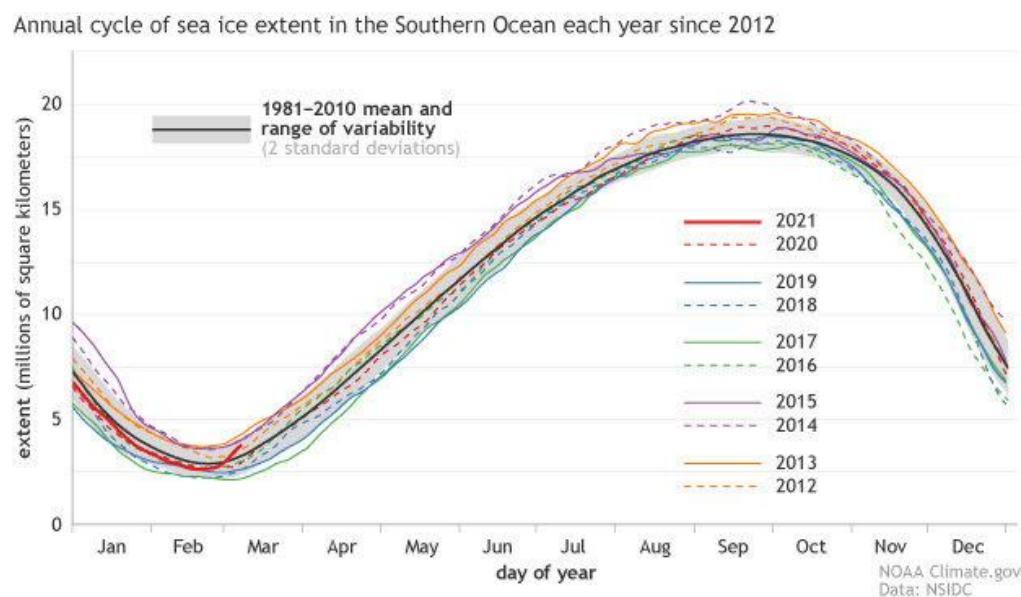
Antarctic Sea Ice Extent Trends, November through March



Note. The blue line indicates for 2020-2021, the green line indicates trends between 2019-2020, with the gray line indicating average trends of 1981-2010 with $\pm$ two standard deviations shaded in gray, extracted from NSIDC.

Figure 2.3

Interannual Variability in Sea Ice Extent between 1981-2020



Note. The black line and gray shading indicates mean and range of variability between 1981 through 2010, extracted from NOAA 2021.

The timing of ice retreat and the timing of ice advancing becomes an important concept towards the understanding of sea ice concentration, particularly factors that can drive spatial variations in sea ice concentrations from one year to the next. With Antarctica experiencing sea surface warming, which inhibits the progress of ice retreat in months after the warmest months due to water holding the heat in and warming at slower rates, sea ice concentrations will drop in areas where warm water is being funneled in by the atmospheric and oceanic circulation (Clem and Fogt 2015). The presence of oscillation patterns in both the atmosphere and ocean can make these water temperatures

even warmer compared to their usual state, along with the impacts of the greenhouse effect and the Earth warming (Li et al. 2014).

Many atmospheric oscillation patterns have been linked with dipolar patterns of sea ice concentration anomalies from the Weddell Sea and Antarctic Peninsula westward toward the Ross and Amundsen Seas in a seasonal context, with cold season months, when ice is farther equatorward and sea ice concentrations are the highest around the continent, having enhanced connectivity to atmospheric circulation (Simpkins et al. 2012). The magnitude of these effects from the Southern Annular Mode (SAM) and the El Niño-Southern Oscillation (ENSO) are comparable during seasons; however, the atmospheric changes imposed on sea ice concentration anomalies are substantially stronger during June-October due to the warm season months not possessing the enhanced connectivity that colder months possess. However, the role of both SAM and ENSO provide negligible contribution to linear trends in sea ice concentrations anomalies over a period of 1980-2008 and therefore cannot be linearly attributed to long-term changes associated with sea ice concentration trends (Simpkins et al. 2012). The connections between Antarctic sea ice variability with ENSO, SAM, and other modes of climate variability, will be discussed later in this literature review.

A dipole-like pattern characterized by out-of-phase relationships between the sea ice concentrations in the Atlantic and Pacific oceans sectors surrounding Antarctica is called the Antarctic Dipole (ADP; Yuan and Martinson 2001). This dipole pattern exists strongly in the surface air temperature and less strongly, but still noticeably, in sea level pressure anomalies (Yuan and Martinson 2001). The mechanisms for this phenomenon

are still yet to be completely understood, as they rely on satellite data that is relatively short compared to the high level of interannual variability. The ADP however, had been observed to have some of the strongest telecommunications between tropical and polar regions in sea ice along with temperature fields in the northwestern Ross (Pacific sector) and the central Weddell Gyre (Atlantic sector) and have been linked towards ENSO signals outside of tropical ocean regions (Yuan and Li 2008).

Remotely sensed data from satellite observations have been able to deduct that there is a strong seasonal and spatial dependence for sea ice concentrations trends, in which they exhibit consistent spatial patterns in past decades (Hobbs et al. 2016). There are limited satellite observations available pre-1979; however, the few that are present can provide more knowledge for complete estimates of sea ice patterns on the Antarctic continent. Even with earlier estimates, sea ice concentration values have only been known with confidence since the early 1970s, where microwave observations from satellites became available, which found that sea ice decreased between 1973-1977 before a gradual increase into 2002 with some estimates finding increases as far and possibly exceeding 2014 before alarming decreases in sea ice extent in 2016 was observed (Turner et al. 2007; Parkinson 2019). Seasonal variability in this region is impacted by several differing climatic factors that will influence sea ice concentration variability, thus influencing the sea ice concentration values for any year, as alluded to earlier, and described in more detail later.

Throughout austral winter and autumn Antarctic sea ice area (closely tied to sea ice extent) increased up to six times compared to other seasons (Roach et al. 2020; Shu et

al. 2020). The seasonality of sea ice extent translates into sea ice concentration as well; regressions of observed September sea ice extent has shown patterns of sea ice concentration change primarily close to the sea ice edge (Lefebvre and Goosse 2008). This seasonality is another layer of complexity given the recent dramatic changes from observed record highs in Antarctic sea ice extent in 2014 into a record low in 2017 (Parkinson 2019; Eayrs et al. 2021). This drastic change in sea ice over this short period of time collectively equated to an approximately observed thirty-year loss in Arctic sea ice, an unprecedented decline in ice compared to the overall trend (Eayrs et al. 2021). This was evident in the 2016 Antarctic sea ice event, where the sea ice extent decreased markedly beginning in September due to record low-pressure systems over the Southern Ocean, enhancing equatorward sea ice drift (and melt) and ocean warming (Wang 2019 et al.). Given that patterns of atmospheric variability in the tropical and midlatitudes of the Southern Hemisphere can influence regional patterns of sea ice concentration, including those that impacted the rapid changes from 2014 to 2017, there is strong potential to understand historical sea ice conditions around Antarctica since the history of these patterns is well-known throughout the 20th century. Nearly all these patterns influence the regional atmospheric circulation, especially a feature near the Antarctic Peninsula called the Amundsen Sea Low.

2.3 The Amundsen Sea Low (ASL)

The Amundsen Sea Low is a climatological low-pressure center located in the Amundsen-Bellingshausen Seas that is characterized by large monthly to interannual geopotential height and pressure variability (Emanuelsson et al. 2018). The longitudinal

position of the ASL will alter the location and intensity of the meridional component of the wind, which will in turn also impact Antarctic heat and moisture transport and sea ice distribution in the region (Hosking et al. 2013). The ASL tends to be located somewhere between the Antarctica Peninsula in austral summer and the Ross Sea (in the Southern Pacific Ocean) in winter (Yiu and Maycock 2019).

Since 1979, there has been warming prevalent over western West Antarctica in the months of September, October, and November, tied to the deepening of the ASL. The deepening of the ASL results in an increase of northerly flow over the Amundsen Sea and the South Pacific, which allows for warming temperatures, including the sea surface temperatures off the coast of West Antarctica that in turn affect sea ice concentration values through dynamic ice transport (Clem and Fogt 2015). The influence of the ASL on this portion of Antarctica reduces sea ice concentration values due to warm air advection from the maritime northerly winds that will become present over these locations (Hobbs et al. 2016).

The ASL is also important in driving Ross and Bellingshausen Sea sea ice concentration variability. Decadally-varying patterns of tropical sea surface temperatures, precipitation, and seasonality are important features that modulate regional Antarctic sea ice concentration. These decadal tropical-polar relationships have been confirmed in modelling simulations, observations from satellites, and ice core proxies (Purich et al. 2016; Thomas and Abram 2016). Since 2014, when the Interdecadal Pacific Oscillation has changed phase, the overall trend of sea ice concentrations in the Bellingshausen Sea has overall become negative on decadal time scales (Meehl et al. 2016), with increases in

the Ross sea. This links the trends in the Pacific Ocean circulation patterns with the sea ice trends in the Pacific-Indian Ocean sector of Antarctica through long-range atmospheric teleconnections in the form of Rossby wave patterns, which include changes in the ASL. Similarly, the equatorial Pacific waters influence the Pacific Ocean sectors closest to Antarctica due to the clockwise rotation of winds surrounding the ASL, changing sea ice patterns in this region.

In addition to the thermal advection and its connection of the ASL to regional sea ice concentration, the ASL provides steering mechanisms for Antarctic sea ice in the form of winds which have become stronger with the deepening of the low itself (Holland and Kwok 2012). The presence of these strong winds impacts sea ice by forcing sea ice to drift slowly into areas where it will be influenced even further by mechanisms such as ENSO, SAM, and the IPO. This allows for another dimension of sea ice drift that will impact the presence of sea ice, especially its regional concentration around the Antarctic Peninsula that can be further affected by local oscillation patterns along with interannual stress variability produced by the low itself on the ice shelf (Holland et al. 2019).

Wind variability in previous months can influence the ice anomalies several months later due to the lagging nature (i.e., slower response) of ice and ocean processes and their impacts. This has even been accounted upwards to 30% of winds in autumn effecting sea ice anomalies for the months of March through May in the western Ross Sea (Holland et al. 2017). The lagged atmosphere to ocean response pattern has been found to be smaller in model runs with different parameters, but still detecting some influence on seasonal winds leading sea ice concentration effects months after they occur.

This lagged effect is most pronounced through zonal winds that drive ice transport, allowing sea ice retreat variations throughout the differing months (Holland et al. 2017). These wind transports modify the ice around the Antarctic continent and can dominate variability in ice shelf melting along with distributing heat in the ocean around the ice shelf of Antarctica (Holland et al. 2019). As such, understanding the wind patterns, even months in advance, can help to understand changes in the subsequent Antarctic sea ice. These patterns of climate variability known to influence the winds around Antarctica, and therefore can potentially alter sea ice concentration and the ability to reconstruct its history, are described in detail in the following sections.

2.4 Large Scale Patterns of Climate Variability that Influence Antarctic Sea Ice Concentration

2.4.1 ENSO (*El Niño - Southern Oscillation*)

The impact of the El Niño-Southern Oscillation on sea ice extent and concentration varies across ocean basins. From 1979-2006, ENSO has been associated with decreases in the western Antarctic Peninsula and the southern Bellingshausen Sea with local increases in the western Ross Sea region (Stammerjohn et al. 2008). ENSO maintained a strong correlation with the Pacific-South American pattern (PSA) between 1948-2014, when ENSO primarily was operating in a central Pacific type (an unconventional ENSO event where temperature anomalies are centered in the central Pacific rather than the Eastern Pacific). After 2014, ENSO teleconnections have shifted toward an eastern Pacific type (Yu and Kim 2010; Yu et al 2015), with downstream teleconnections toward Antarctica associated with the PSA moved to be in correlation

with the Southern Annular Mode. The findings from Yu et al. (2015) implied that the forcing from ENSO patterns has become increasingly important in the past two decades and will eventually become even more vital to understand the dominance between the central Pacific type of ENSO events rather than the ENSO-PSA relationships. The changing relationships have rendered ENSO more effective in influencing Southern Hemisphere climate after the early 1990s (Yu et al. 2015).

ENSO is a driver for the ASL, in which pressure will be (in general) higher than average in the ASL region during El Niño events, while the mean pressure will be lower than average during La Niña events (Yiu and Maycock 2019). ENSO events also will affect the location of the ASL, as the low tends to drift further southeast during El Niño events, while maintaining a more northwestern location during La Niña events. Seasonal variation from Rossby wave sources comes from the teleconnection of ENSO events, with a significant correlation being found in near-surface temperatures in the Amundsen Sea region with ENSO events in austral winter and autumn (Yiu and Maycock 2019).

Further research on ENSO has found that positive phases of ENSO modulate mean meridional heat flux through changes in the regional mean meridional circulation, leading to less ice in the eastern Ross/Amundsen sector in El Niño events (Liu et al. 2004). The presence of positive phases of ENSO influencing the sea ice concentration assists with the ability to further investigate these sea ice concentration values in the Ross/Amundsen sector of Antarctica, and the effects of El Niño on Antarctic sea ice concentration in surrounding sectors. The impact on sea ice, and specifically sea ice concentrations, in one region of Antarctica can influence the regions around it (Raphael

and Hobbs 2014). Given the strong connections this climate mode has with Antarctic sea ice concentration, it serves as a powerful predictor in reconstructing its past variations. This study will employ multiple ocean and atmospheric measurements of historical ENSO variability to understand these important connections throughout the 20th century.

2.4.2 SAM (*Southern Annular Mode*)

The SAM, a measure of the strength and location of the mid-to-high latitude pressure gradient across the Southern Hemisphere, can deviate from north to south, shifting further to the south in the positive phase and towards the north in the negative phase. A positive SAM event, characterizing a strengthened pressure gradient, is typically marked by lower-than-average polar temperatures and stronger circumpolar winds near the Antarctic Circle, in contrast to the warmer than average conditions over Antarctica and weaker zonal winds for a negative SAM phase that reflect a reduced meridional pressure gradient (Turner et al. 2007).

The mean sea level pressure decrease around Antarctica has caused SAM index values to shift to more positive phases prior to 2017 with seasonal increases mostly during austral summer (Turner et al. 2016). The changes in ice concentrations associated with the SAM stem mostly from the wind circulation patterns along with thermodynamic changes of atmosphere-ocean interaction, even coupled with wind shifts (Holland and Kwok 2012). This allows for the SAM trend to play an important role in winter and allows for ice coverage to increase significantly with near-surface wind changes (Hobbs et al. 2016). These sea ice concentration changes can be determined by the presence of zonal winds, favoring further ice expansion (partly through enhanced equatorward

oceanic Ekman drift), or meridional winds associated with the circulation of the ASL, where expansion will occur when southerly flow is present or compaction/reduction with northerly flow (Fogt and Marshall 2020). The SAM also can be an important predictor but estimates of its variability in the 20th century are limited to seasonal reconstructions since it requires measurements of pressure or wind near Antarctica (which do not exist continuously prior to 1957).

2.4.3 IPO (Interdecadal Pacific Oscillation) and PDO (Pacific Decadal Oscillation)

The IPO, located over the Pacific, monitors the magnitude of precipitation and convective heating anomalies occurring in the tropics and middle-latitudes and stretches to both the Northern and Southern Hemispheres, changing in sign roughly every fifteen to thirty years. The IPO is differentiated from the PDO (Pacific Decadal Oscillation) in that the IPO spans both hemispheres, while the PDO occurs in the middle-latitudes in the Northern Hemisphere (i.e., the PDO has no signature in the Southern Hemisphere). Positive phases of IPO allow for a warmer than normal tropical Pacific, which will alter the location of the ASL, effecting sea ice concentration values as described above. The negative phase can affect Antarctica, including the deepening of the ASL and allowing for warm air advection that can also aid in lower sea ice concentrations in these locations (Meehl et al. 2016). The deepening of the ASL is strongly connected with the IPO, as the tropical convective heating and precipitation anomalies during IPO phases drive a Rossby wave towards Antarctica that influence the ASL (Clem and Fogt 2015). In winter, the convection in a negative IPO phase generates anticyclonic circulation in the upper troposphere, altering vorticity and divergence in tropical areas which locally affects the

Rossby wave source regions and the downstream Rossby wave teleconnection patterns that extend to Antarctica. Both the IPO and PDO will be explored as predictors of 20th century Antarctic sea ice concentration variability, especially for their known influence on decadal and longer timescales.

2.5 Synthesis

The increasing Antarctic sea ice extent, opposite of what one might expect in a warming world and in contrast to sea ice losses in the Arctic, needs further scientific understanding. This is especially true given sudden changes in Antarctic sea ice extent in 2016, and again in 2021. Increased understanding in processes is generally best understood by more observations to assess patterns of variability and change. However, Antarctic sea ice concentrations have only been monitored (albeit very sparsely) since the 1930s when Antarctic whalers would document where whales, who congregate near ice edges to feed, were caught all the way until the start of Antarctic climate data collection in 1957-1958 (Scott and Hansen 2016; Fogt et al. 2016). Satellite measurements provide the full spatial coverage beginning continuously around 1979. The lack of these spatially complete and continuous observations surrounding the Antarctic continent, along with the expansive region, seasonal, and temporal (i.e., 2016, 2021) differences in the sea ice variations, makes it difficult to measure and understand Antarctic sea ice concentrations historically and gather a picture of what is occurring with sea ice in this region in a longer historical context. Nonetheless, given the numerous relationships between sea ice concentration and patterns of climate variability described in this chapter, there are other measurable climatic factors that can be provided as predictor networks to derive spatially

and temporally complete sea ice concentration reconstructions for this area prior to 1979. Indeed, very recent work (Fogt et al. 2022) has proven the success of reconstructing Antarctic sea ice extent throughout the 20th century. Over 40 years of spatially complete satellite data spanning back into the late 1970s allow for reliable sea ice concentrations observations from this timeframe until the present timeframe allow for a robust establishment of statistical relationships of Antarctic sea ice concentrations in the early 20th century, which motivates the purpose behind this project (Turner et al. 2007): what can be said about the spatial patterns of Antarctic sea ice variability, depicted in changes of sea ice concentration reconstructions in a longer time span? Given that Antarctic sea ice extent reconstructions have been deemed meaningful in an independent part of this project (Fogt et al. 2022), it worthwhile to investigate if similar skill can be obtained for Antarctic sea ice concentration. A dataset that provides a longer time frame on Antarctic sea ice concentrations will assist further research, improve Antarctic climate models, and place the dynamic process of Antarctic sea ice into a greater perspective; the creation and evaluation of such a dataset serves as the impetus for this research study.

Chapter 3: Data and Methods

3.1 Area of Study

For sea ice concentration reconstruction, the area of study will be anywhere there is sea ice present in the Antarctic region. This area will go further north than 66.5°S, extending beyond the Antarctic circle, so the area cannot be bounded by a specific latitude. This will allow for a complete reconstruction of sea ice concentrations to be obtained for the Antarctic region to give a full spatial picture of sea ice in this region.

3.2 Data

3.2.1 *Sea Ice Concentration Data*

Satellite observed data from 1978 through 2020 on sea ice concentration data were collected from the National Snow and Ice Data Center (NSIDC). These data serve as a calibration dataset to assist the development of various statistical methods for modeling the relationship between sea ice concentration and atmospheric data in the satellite era. This study uses Antarctic sea ice concentration from satellite-observed sea ice measurements from Nimbus-7 Scanning Multichannel Microwave Radiometer (SMMR) and Defense Meteorological Satellite Program (DMSP) Special Sensor Microwave Imager - Special Sensor Microwave Imager/Sounder (SSM/I-SSMIS). From the NSIDC datasets, sea ice concentration data have been calculated in the Climate Data Record (CDR) of Passive Microwave Sea Ice Concentration, Version 4 (<https://nsidc.org/data/g02202>) (Meier et al. 2021). The CDR algorithm output combines ice concentration estimates from the National Aeronautics and Space Administration (NASA) Team algorithm (Cavalieri et al. 1984) and the NASA Bootstrap algorithm

(Comiso 1986). The data span the period 25 October 1978 to 31 December 2020 and are daily except prior to July 1987 when they are given every other day. Data are gridded on the SSM/I polar stereographic grid ($25 \text{ km} \times 25 \text{ km}$). Sea ice concentrations represent the percentage of this $25 \text{ km} \times 25 \text{ km}$ grid cell that is covered in ice using the two different algorithms to distinguish between open water, clouds, and sea ice. Monthly sea ice concentration data, which the seasonal means are later calculated from, was calculated as the average of the sea ice concentration for each grid cell for all days in that month. In addition to the alternate day observations from 1978 to 1987, there are several days and segments of days with no observations in the sea ice concentration data. In particular, there is no data between 3 December 1987 and 12 January 1988. These data are infilled using the method described in Fogt et al. (2022) for sea ice extent.

To reconstruct the spatially varying sea ice concentration data, the seasonal mean concentration data are decomposed into empirical orthogonal functions (EOFs) that represent different uncorrelated patterns of variability within the full spatial sea ice concentration field. Although generated mathematically, these EOF patterns representing distinct spatial configurations of sea ice anomalies are often strongly related to different modes of climate variability. Each pattern is represented by a timeseries, called the principal component timeseries (PCs), which represents the amplitude of each pattern through time. Using the expansion coefficients, the EOFs and PCs can also be merged back together to get the full spatial field through time. The EOFs will be put through principal component analysis, where five leading EOFs / PCs (which explain the most variance in the full spatial sea ice concentration field, and therefore are the most

meaningful) will be used for reconstruction, not the full spatial pattern of Antarctic sea ice concentration.

3.2.2 Reconstruction Predictor Data

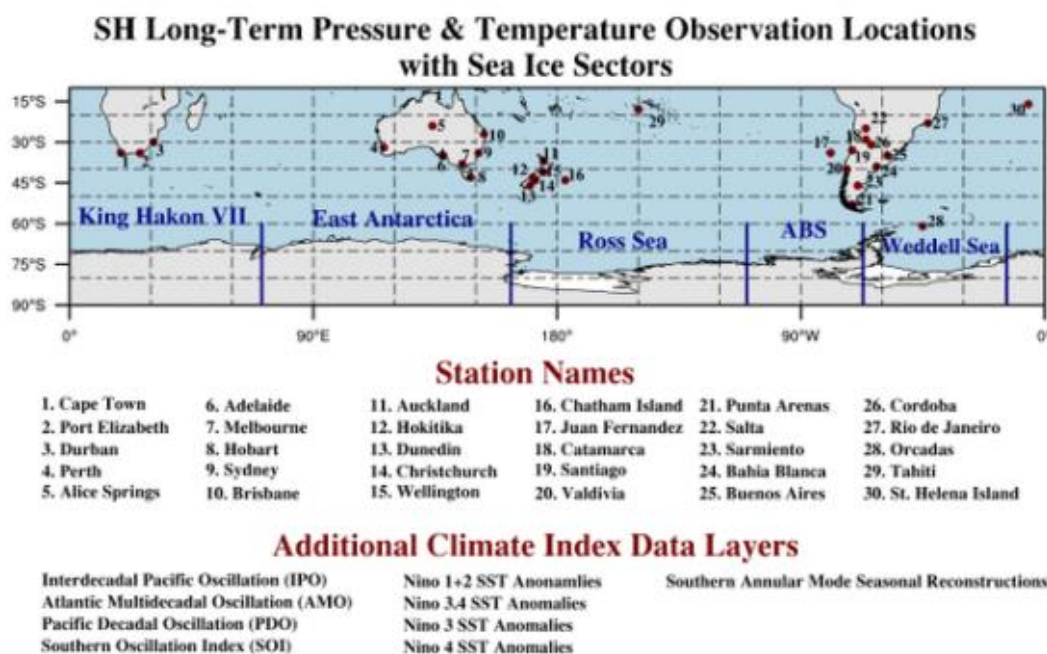
Climatological data for the Southern Hemisphere were used for sea ice concentration reconstruction at seasonal timescales, serving as the primary base of predictor data for the reconstruction. These data include monthly mean pressure and temperature across the Southern Hemisphere from 1905-2020, obtained from previous research along with from the University Corporation for Atmospheric Research's (UCAR) data archive. To fill in missing data, some of the station observations in this archive had to be patched through nearby stations using mean monthly offsets (differences); stations in South Africa were patched through creating monthly means from daily data from the National Centers for Environmental Information Global Surface Summary of the Day.

Station measurements of temperature and pressure that serve as the primarily source of reconstruction predictor data include 30 midlatitude stations throughout New Zealand, Australia, southern Africa, and southern South America (Fig. 3.1). These observations were obtained from the Global Historical Climatology Network (GHCN), the University Corporation for Atmospheric Research, quality-controlled observations from the Climate Research Unit, and public data from the New Zealand National Institute for Water and Atmospheric Research. Most of the atmospheric pressure and surface temperature data above were archived by Dr. Fogt through previous research projects, with updates to extend the data from 2018 through 2020 as part of this research. SLP and

surface temperature data from Antarctic stations were also considered through the Scientific Committee on Antarctic Research (SCAR) Reference Antarctic Data for Environmental Research (READER) dataset that creates long term climate data for Antarctica (Turner et al. 2004).

Figure 3.1

Map of Stations for Pressure and Temperature Data



Note. A map outlining the various locations of stations that pressure and temperature data measurements were obtained for reconstructions along with the sectors of sea ice in Antarctica.

Beyond the point locations of station temperature and pressure data in Fig. 3.1, the next predictor datasets are indices that represent patterns of large-scale climate variability known to influence Antarctic sea ice, such as the El Niño - Southern

Oscillation (ENSO), Interdecadal Pacific Oscillation (IPO), Pacific Decadal Oscillation (PDO), Southern Annular Mode (SAM), Southern Oscillation Index (SOI), and the Atlantic Multi-decadal Oscillation (AMO). The SOI represents the atmospheric component of ENSO, calculated as the difference in pressure from Tahiti, French Polynesia, and Darwin, Australia. This research employs the SOI data from the Australian Bureau of Meteorology due to the lack of data gaps throughout the 20th century (<http://www.bom.gov.au/climate/enso/soi/>). The AMO data are obtained from NOAA (<https://psl.noaa.gov/data/timeseries/AMO/>) following Enfield et al. (2001). The IPO is calculated by finding the difference between sea surface temperatures (SST) anomalies in the central equatorial Pacific, northwestern Pacific, and the southwestern Pacific from the NOAA Extended Reconstruction Sea Surface Temperature dataset, version 5 (ERSSTv5). The reconstruction also uses the PDO index, which coincides closely with the IPO; however, the PDO only considers SST anomalies in the northern and tropical regions of the Pacific, similarly calculated from ERSSTv5. The ERSSTv5 dataset also allows for calculation of the Niño sea surface temperature indices (to monitor various representations of the oceanic component of ENSO), such as the Niño 1+2 SSTs (averaged from 0°-10°S and 270°-280°E), the Niño 3 SSTs (averaged from 5°N-5°S and 210°-270°E), the Niño 3.4 SSTs (averaged from 5°N-5°S and 190°-240°E), and the Niño 4 SSTs (averaged from 5°N-5°S and 160°-210°E); see Fig. 3.2 for a map of these regions.

The final climate mode index in the predictor datasets is the SAM, where seasonal SAM index reconstructions extending back to at least 1905 from Fogt et al. (2009) (http://polarmet.osu.edu/ACD/sam/sam_recon.html) will be merged with the

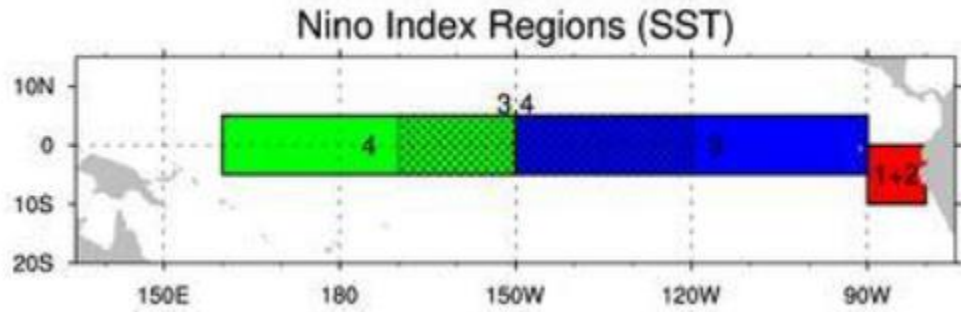
observationally based SAM index values (Marshall 2003) after 1957 (<http://www.nerc-bas.ac.uk/icd/gjma/sam.html>). The SAM index is only available at seasonal timescales while each of the other predictor datasets are available at monthly timescales. Seven of the nine climate indices in this research are based on ocean variability. The inclusion of these indices will assist in depicting patterns of dominant ocean climate variability that influences sea ice concentration in Antarctica that might not be replicated with the presence of only temperature and pressure measurements.

These predictor datasets were used to give an estimation in relation to historical sea ice concentration measurements for the reconstruction of sea ice concentrations prior to measurements to be conducted in this project. The datasets were converted into netCDF (network Common Data Format) files to be read correctly by the reconstruction code (programmed in the National Center for Atmospheric Research Command Language, NCL). Two separate programs transferred the various predictor data to netCDF, one to convert text files organized by month and the other program to convert text files organized by the season. The Niño SST dataset files did have to be extended further back to 1854 using ESSRTv5 using NCL. Specifically, for the ENSO SST data, a separate NCL program converted global mean SST data to Niño SST regional data by taking the area weighted mean of mean SST of ESSRTv5 at specified Niño coordinates based on the regions (Fig 3.2) with each region receiving its own netCDF file for analysis from 1905 onward.

Seasonal means of both the predictor datasets and sea ice concentration dataset were calculated to improve the reconstruction, as the literature review showed strong

Figure 3.2

The Four Regions for Niño SST Data



Note. This figure is from (Shea 2018).

Seasonal means of both the predictor datasets and sea ice concentration dataset were calculated to improve the reconstruction, as the literature review showed strong relationships across the Southern Hemisphere with Antarctic Sea ice concentration on seasonal and longer timescales. These seasonal scales were derived using three months with March, April, and May representing MAM; June, July, and August representing JJA; September, October, and November representing SON; and December, January, and February representing DJF.

3.3 Reconstruction Methodology

3.3.1 Reductions of Predictor Datasets using Principal Component Regression (PCR)

The same following procedure was used to reconstruct each individual seasonal mean PC from the full spatial Antarctic sea ice concentration field. The slower responses of sea ice from atmospheric and oceanic effects needed to be accounted for, so a lagging technique of data for up to one season was completed, where the predictor data lead the

Antarctic sea ice concentration PCs. An example of this would be for reconstructing a sea ice concentration PC for DJF: predictor data were then averaged in December-January (DJF, no lag), November-January (NDJ, 1-month lead), October-December (OND, 2-month lead), and September-November (SON, 3-month lead).

Using seasonal mean data, a reduction of the overall predictor datasets was performed through correlations between each predictor observation or index with the seasonal mean sea ice concentration PC at all lags. The predictor variables that showed the least correlation (determined by stations not meeting various statistical significance thresholds of the correlation coefficient at $p < 0.10$, $p < 0.05$, $p < 0.025$, and $p < 0.01$) were removed of the dataset to reduce unnecessary variables for the reconstruction procedure. One reconstruction is also performed using all available temperature and pressure observations, regardless of the magnitude of its correlation with the observed sea ice. These various significance thresholds form what are henceforth called the predictor networks, labeled as the following in decreasing dataset size: All (all observations regardless of correlation magnitude), 10% ($p < 0.10$ network), 5% ($p < 0.05$); 2.5% ($p < 0.025$); 1% ($p < 0.01$). This allowed for the most meaningful predictor data for sea ice concentrations to be put through another series of reductions through principal component regression (PCR).

The four different predictor networks will have ensemble reconstructions based on the retained temperature and pressure observations that were significant at the given threshold of the network. Additional ensemble reconstructions were created as needed by adding climate indices to the retained observations that are significantly correlated at the

threshold as well creating a layering of climate indicators. The indices layered were as follows (Fig. 3.1 for a list in correct order): pressure and temperature observations, IPO, AMO, PDO, SOI, Niño 1+2 SSTs, Niño 3.4 SSTs, Niño 3 SSTs, Niño 4 SSTs, and the SAM index. The combination of these predictor networks and the data layers (one at time) will potentially allow for a larger ensemble for each reconstruction.

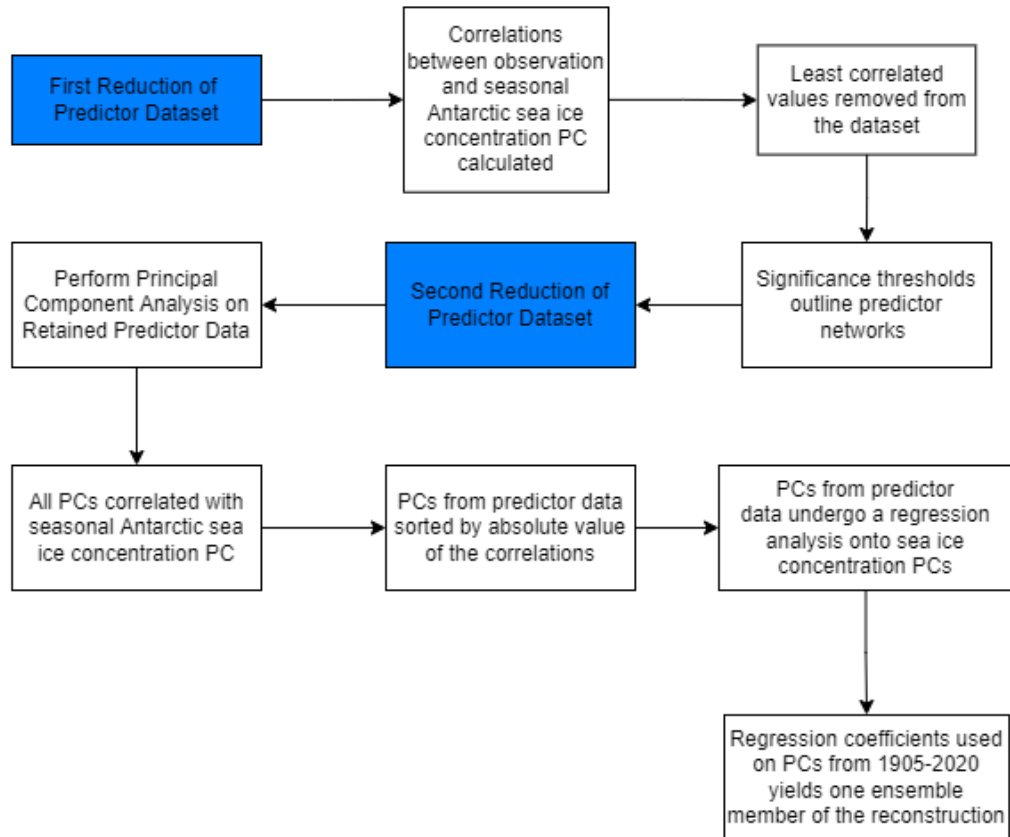
The retained data from the correlation analysis underwent a second reduction through EOF analysis to reduce noise in the predictor variables and reconfigure the predictor data to represent patterns of large-scale climate variability more precisely across the Southern Hemisphere (the EOFs) rather than individual climate variables at point locations. Principal component analysis allowed for data reduction from an extensive number of variables into a smaller subset of variables that will still cover a majority of variance within a dataset. To expedite calculations, only a maximum of 30 PCs are calculated from the retained predictor data.

All the PCs (up to 30 total) from the predictor data are correlated with Antarctic seasonal sea ice concentration PC that is being reconstructed. The PCs from the predictor data are then sorted by absolute value of their correlations with the observed sea ice concentration PC, and a subset of the predictor data PCs is then regressed (using multiple linear regression) onto the sea ice concentration PC (details described in following section). Using these regression coefficients along with the PCs (which can be extended back in time with the station data and PC expansion coefficients) yields one ensemble member of the reconstruction. The reconstruction can also be thought of as a weighted sum of each retained predictor variable, with the weights determined from the

relationships the predictor data have with the PCs, and the regression coefficients of the PCs with the seasonal sea ice concentration PC.

Figure 3.3

Flowchart of Steps of Reduction of Predictor Data



Note. The steps of reduction in predictor datasets.

3.3.2 Leave-One-Out Cross Validation and Reconstruction Ensemble Approach

The PC with the highest absolute correlation is first used to create a reconstruction, and after completing the following steps, the PC with the next highest absolute correlation is added into the regression model. The process of an independent validation method was used to determine the ideal number of PCs to retain for the model, using the proven leave-one-out cross validation procedure (Fogt et al. 2009; Fogt et al. 2016; Fogt et al. 2022). In this validation procedure, the regression procedure is repeated iteratively for each year. Specifically, the regression of PCs will be repeated one time for each year in the observation period (1979-2020, a total of forty-two times) while leaving one individual year out of the regression along with the two neighboring years. Each time, new regression coefficients are generated using the remaining data, and these regression coefficients are used to predict the central value. The central year is therefore assumed to be independent of the study in this case, and each of these independently predicted values are concatenated to construct a fully predicted and independent time series, known as the validation reconstruction.

The original (calibrated) and validation reconstructions then undergo several statistical tests to evaluate the skill of the reconstruction, and which number of PCs is the most optimal number for each ensemble member's reconstruction. The first of which is the correlation of the original, calibrated reconstruction (from the PCR model) versus observed data, known as 'cal r.' The validation reconstruction is separately correlated with the observed data to yield the validation correlation (val r). As with other

correlations (described later), the cal r and val r statistics range from -1 to 1, with +1 indicating a perfect reconstruction.

Once these statistical values are derived, the skill of this individual reconstruction is further assessed using two additional statistical metrics. These two skill metrics, the reduction of error (RE) and coefficient of efficiency (CE), analyze the reconstruction performance against the climatological average to determine if the reconstruction adds value beyond the simple use of the climatological mean to understand historical sea ice variability. Both the RE and CE yield a value between negative infinity and positive one, where the goal is to have a positive number, which signifies better reconstruction skill than the climatological mean used to estimate sea ice conditions. The RE is calculated from the following equation:

$$RE = 1.0 - \left[\frac{\sum_{i=1}^n (reconstruction - actual)^2}{\sum_{i=1}^n (sea\ ice\ data - \overline{sea\ ice\ mean})^2} \right]$$

Equation 3.1

where the calibrated reconstruction from 1979-2020 is in the numerator and the observed data (sea ice concentration PC) from the same period is in the denominator. If the calibrated reconstruction is consistently equal to the climatological mean, the ratio will be one, which will yield a RE value of zero. If the RE is equal to one, the reconstruction minus the actual observed will be zero, yielding a perfect reconstruction.

The CE is derived from the following equation:

$$CE = 1.0 - \left[\frac{\sum_{i=1}^n (validation - actual)^2}{\sum_{i=1}^n (sea\ ice\ data - \overline{sea\ ice\ mean})^2} \right]$$

Equation 3.2

where comparisons are made between the validation reconstruction versus the actual observations during the same period. The CE equation is much more difficult to pass compared to the RE equation due to the validation reconstruction being done independent of the reconstruction, and values will be found where RE will be greater than CE values in most cases.

Once the CE values are determined, the validation procedure will be repeated once again, with an added PC at a time to the regression model (the next least correlated PC, until all possible PCs have been added). If the resulting reconstruction yields a higher CE value, it will be retained as the potential ensemble member reconstruction, and the previous reconstruction discarded. This is repeated for all the ranked PCs (by correlation magnitude). The overall highest CE being selected as the reconstruction ensemble member for the data network and data layer. This retained reconstruction is one ensemble member, the entire procedure is then repeated for each data network and data layer. For each seasonal PC, at a minimum, there will be 5 ensemble members coming from only the temperature and pressure data networks. At a maximum there will be 41 ensemble members for each seasonal PC should all data layers be included in all data networks. In total across the five 5 PCs and 4 seasons, there is a possibility of 100 – 820 different reconstructions produced.

3.4 Analysis Methodology

3.4.1 Determining the ‘Best Fit’ reconstruction using an F-Test

Once all reconstructions are produced for each data network and data layer, an individual ensemble reconstruction is selected as the ‘best fit’ reconstruction. For a first

estimate, the highest CE of all ensemble reconstructions for a given seasonal sea ice concentration PC is selected as the ‘best fit’ reconstruction. However, the high CE may be artificial and represent model overfitting (cases where too much information is added to the reconstruction model, and it performs poorly outside the calibration window). To determine potential model overfitting, the variance of reconstruction was compared to the observed data using an F-test. If the variance of the reconstruction was significantly different ($p < 0.01$ of equal variance) than the observed reconstruction (both during and outside the calibration period, 1905-1978 and 1979-2020), there is a possibility of overfitting in the reconstruction occurring. The f-test statistic is calculated through the following equation:

$$f = \frac{\sigma_1^2}{\sigma_2^2}$$

Equation 3.3

where σ indicated the variance of two separate datasets. In some cases, the variances were significantly different, suggesting the best-fit reconstruction was subject to overfitting and the next highest CE value was selected as the best-fit reconstruction, so long as the variances were not statistically different from the observed variance during 1979-2020. In the absence of an alternative reconstruction ensemble member with equal variances, the highest CE would be subject to being named the best-fit reconstruction. This process was to be repeated for every season and EOF to find the best-fit reconstructions for each EOF and season for an array of twenty best-fit reconstructions.

3.4.2 Box Plots

Box plots were made to visually see the skill of the reconstructions for each season and were calculated separately over the full range of ensemble reconstructions by season as well separately for the best fit reconstructions. These box plots show the minimum, median, and maximum values, along with the first and third quartiles for each of the four reconstruction skill metrics: cal r, val r, RE, and CE values. The boxplots are helpful observing ranges of these skill metrics by season and the overall reconstruction performance, as the values closer to one (for all skill metrics) indicate perfect reconstructions, and values above zero indicate improved skill relative to the climatological average.

3.4.3 Predictor Counts, Time Series Representation, and Low-Frequency T-Series

Predictor count plots were used to monitor changes in the predictor data that are being put into the reconstructions visually by season and by PC. This assisted in explaining the relative ratios of data used in the ‘best fit’ reconstructions to determine if certain climate indices were often used as predictor data for a certain reconstruction.

To analyze the skill of the reconstructions through time, two types of time series plots were utilized, the first being a spaghetti model that displays all the ensemble members from the reconstructions along with the best fit reconstruction and ensemble mean. In these plots, the deviations between the various ensemble members provides a first estimate of the reconstruction uncertainty. The second type of time series plot contains only the best fit reconstruction and the ensemble mean, along with a 95% confidence interval. The 95% confidence intervals in the second time series plots

approximate the reconstruction uncertainty. The confidence interval is taken as the larger of two standard deviations of the reconstruction ensemble estimates at each year (varies through time), or 1.96 times the standard deviation of the residuals between the reconstruction and observed values during 1979-2020 (a constant value). The use of both time series programs allowed for additional confidence on the skill of the reconstructions and various ensemble members, before proceeding to making an eleven-year running mean smoothed low frequency time series plots of the reconstructions. The low frequency plots allowed for analysis of long-term changes of sea ice concentration patterns. Subsequent analyses further examined the low-frequency and other time variations in these reconstructions through correlations, trend analyses, and spatial analyses of the sea ice concentration field.

3.4.4 Correlations

Throughout the analysis of best-fit reconstructions, correlations such as the cal r and val r skill metrics were calculated using the Pearson correlation equation:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Equation 3.4

where r will fall between a range of negative 1.0 to positive 1.0 and represents the correlation coefficient, x_i and y_i represent the variables at each year within each dataset with $\bar{x}$ and $\bar{y}$ representing means of the separate datasets. If an r value reaches a value of positive 1.0, it will indicate a perfect positive linear relationship, a value of zero indicates

no linear relationship, and a value of negative 1.0 represents a perfect negative relationship between the two datasets. The most ideal in terms of this project is a correlation coefficient of close to positive one, where observations and reconstructions are the same, with values close to positive one being acceptable as well. The statistical significance of correlation will then be found using hypothesis testing, using a null hypothesis of no linear relationship ($r = 0$) and an alternative hypothesis that the correlation has a linear relationship, either positive or negative ($r \neq 0$). The probability of both tails in a t -distribution must be considered given no *a priori* knowledge on the sign of the correlation coefficient. The test statistic will be found using the formula:

$$t_{n-2} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

Equation 3.5

where n is the sample size (years), and r will be the correlation value found in Equation 3.4. This t -value will be used to calculate the probability that the null hypothesis is true (p-value) using $n-2$ degrees of freedom. Throughout this study, there are various probability levels used to reject the null hypothesis, ranging from $p < 0.10$ to $p < 0.01$.

3.4.5 Trend Analysis

To further understand the reconstruction and sea ice concentration data, root-mean-squared error RMSE and linear least-squares regression are used (referred to as static trends). These values are calculated over the pre-satellite period, the satellite period, and the overall period on a seasonal basis. The RMSE is calculated using:

$$RMSE = \sqrt{\frac{(recon_i - observed_i)^2}{n}}$$

Equation 3.6

where higher values of an RMSE indicate the presence of large deviations from observations, whereas low values of RMSE indicate the reconstruction and observations are much closer together. An RMSE of zero indicates a perfect reconstruction (no error). Due to the squaring in the numerator, the RMSE is much more sensitive to larger deviations from the mean than smaller fluctuations around the mean.

These regression trends use the equation for a best fit line:

$$y = mx + b$$

Equation 3.7

where y is the dependent variable, m is equal to the slope of the line, x is known as the independent variable, and b is the intercept on the y-axis. Throughout this research, the independent variable will be time in years with the dependent variable being sea ice concentration (reconstructed and observed). The slope (m) can be delineated from the following equation:

$$m = \frac{n \sum_{i=1}^n x_i y_i - (\sum_{i=1}^n x_i)(\sum_{i=1}^n y_i)}{n \sum_{i=1}^n (x_i)^2 - (\sum_{i=1}^n x_i)^2}$$

Equation 3.8

which can be further simplified into the following equation:

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Equation 3.9

where x_i will be the independent variable, y_i will be the dependent variable, while $\bar{x}$ and $\bar{y}$ will represent means of the datasets. To test the statistical significance of the slope, the standard error of the slope (s_m) is needed, and can be calculated through the equation:

$$s_m = \frac{\sqrt{\left(\frac{1}{n - (n - 2)}\right) \left[n \sum_{i=1}^n (y_i)^2 - (\sum_{i=1}^n y_i)^2 - \frac{(n \sum_{i=1}^n x_i y_i - [\sum_{i=1}^n x_i] [\sum_{i=1}^n y_i])^2}{n \sum_{i=1}^n (x_i)^2 - (\sum_{i=1}^n x_i)^2} \right]}}{\sqrt{n \sum_{i=1}^n (x_i)^2 - (\sum_{i=1}^n x_i)^2}}$$

Equation 3.10

A null hypothesis of no slope ($m_0 = 0$) can be used as a test statistic to determine probabilities of the null distribution using a t-distribution:

$$t_{n-2} \approx \frac{m - m_0}{s_b}$$

Equation 3.11

where s_m comes from Eqn. 3.10, t_{n-2} represents the t-distribution with $n-2$ degrees of freedom, m represents the actual slope, m_0 represents the hypothesized slope (zero), and n represents the sample size. The null hypothesis will agree with previous tests where there is no slope ($m_0 = 0$) and critical thresholds of $p < 0.10$ to $p < 0.01$ will reject the null hypothesis (the alternative hypothesis is two-tailed, with $m_0 \neq 0$).

The final calculation performed throughout trend analysis will be the computation of the 95% confidence interval (CI) to observe the range of values where slope has a high probability of occurring given the sample data, and is calculated through the following equation:

$$CI = m \pm t_{n-2} \times S_b$$

Equation 3.12

where m represents the slope, t_{n-2} represents the t-value corresponding to $p = 0.025$ with $n-2$ degrees of freedom, and the standard error of slope (s_m) coming from Eqn. 3.10.

3.4.6 Spatial Reconstruction Trends

While the reconstructions are seasonal PCs of sea ice concentration, the five PCs can be merged back to get a full spatially and time varying reconstruction through the usage of the ‘eof2data’ function within NCL. This spatial reconstruction however is not the full spatial field as this function only used the five leading PCs, making it impossible to reconstruct the entire field.

Trends (in various time periods) and correlations (with climate indices) can be computed for the full spatial reconstruction using the equations above, calculated at each grid point of the spatial sea ice concentration reconstruction. The correlations and regressions performed on the spatial reconstruction will allow for an estimate of spatial variations within the full (i.e., leading 5 EOFs) field that are not as easily determined when working with the PCs by themselves.

Chapter 4: Analysis

4.1 Creation of EOFs/PCs

4.1.1 *Spatial Plots of EOFs*

Antarctic sea ice is a spatially complex environment, as described in the literature review, and in particular, the variability in adjacent portions of the Antarctic sea ice is strongly related. This makes the reconstruction of the full Antarctic spatial field complex, especially given that it varies widely seasonally, with concentrations bound within a finite range (0-100%). To simplify the reconstruction of this complex spatial field, the seasonal mean sea ice concentration was decomposed into its first five leading empirical orthogonal functions (EOFs). The presence of five EOFs within each season allows for twenty plots to explain a certain percentage of variance seen in the full spatial field from 1979-2020. The five EOF patterns are shown in Figs. 4.1-4.5, plotted as regression map in units of percentage sea ice cover (sea ice concentration anomaly) per standard deviation of the PC. Smaller absolute values in these figures indicate data values closer to the mean (less variability) while higher values (in an absolute sense) indicate values spread out from mean values (stronger anomalies) – it is these regions where each EOF pattern is isolating a portion of the full spatial sea ice variability. By design, each EOF is distinct (uncorrelated, orthogonal from each other), so patterns in each EOF can be thought of as unique aspects of Antarctic sea ice concentration variability. The overall signs of the regions are arbitrary within a given EOF map, however the relative signs (regions of positive and negative values) matter and show opposing regions of sea ice temporal variability (if sign is opposite, sea ice anomalies behave together, but in

opposite directions, and if sign of two regions is both the same, they vary together temporally).

EOF1 represents the patterns in the full sea ice concentration that explain the greatest percentage of variance (spatially, and in time). The percentage of explained variance ranges from >25% in MAM to only 16.7% in DJF. Spatially, the loadings (grid points of maximum sea ice concentration amplitude, darkest color in Fig. 4.1) from EOF1 vary from region to region based on different seasons, with EOF1 explaining much of the sea ice variations over the Weddell Sea in DJF and MAM, but less variability in the same region over JJA and SON. EOF1 also experiences elevated levels of associated variability across the Ross Sea region throughout JJA and SON, which extends further into the Bellingshausen-Amundsen Sea region during SON especially. The offsetting sea ice concentration anomaly pattern between the Weddell and Bellingshausen / Ross Seas is the common sea ice anomaly pattern known as the Antarctic Dipole, which often emerges as the leading EOF in Antarctic sea ice (Yuan and Martinson 2001). In EOF1, A high pocket of variability also emerges in the King Hakon region during JJA that lessens into SON and is lower across other seasons, showing local variability in sea ice concentrations around the continent (Fig. 4.1). Thus, the notion that local variability is an important consideration when evaluating the skill of reconstructions is confirmed.

EOF2 explains ~10-15% of the total spatial Antarctic sea ice concentration variability in all four seasons. Spatially, EOF2 in all but SON (Fig. 4.2) shows low amplitude, except for areas around the Antarctic Peninsula, slightly shifted from the key regions in EOF1. In DJF, pockets of positive sea ice concentration anomalies occur near

coastal areas of the Antarctic Peninsula in the Bellingshausen-Amundsen Sea region with smaller pockets extending into the Ross Sea and an area of negative sea ice concentration anomalies in the Weddell Sea sector. This pattern is prevalent in SON; however, with large areas of negative sea ice concentration anomalies in the Antarctic Peninsula regions and positive sea ice concentration anomalies spanning outward into the Ross Sea region. In JJA, the main area of variability in sea ice concentration tends to reside in the Weddell Sea region further from the Antarctic coast, while less variability tends to hug a little more toward the coast in the Bellingshausen-Amundsen sector.

EOF3 explains just under 10% of the full Antarctic sea ice concentration variability in all four seasons. The sea ice concentration variability in EOF3 (Fig. 4.3) is still prevalent in certain areas around Antarctica, with high values occurring in DJF and SON. In general, the spatial pattern of EOF3 represents two broad regions of opposite signed sea ice variations (i.e., one region of large but weak positive anomalies, and one other region of large but weak negative anomalies). Both regions span about 180° in longitude. The exception is in DJF, where smaller regions of sea ice variability are found. In this season, there is an area of positive sea ice concentration anomalies in the Ross Sea sector throughout DJF that is the second highest in magnitude throughout the whole EOF. The highest positive sea ice concentration anomalies overall occur throughout the King Hakon region with an overall picture of more sea ice concentration variability through the entire year, especially in SON.

EOF4 explains between 5-10% of the Antarctic sea ice concentration variability, with more of a focus on sea ice patterns away from the ice edge (but equatorward from

the ice pack), and a larger portion of variability around East Antarctica (Fig. 4.4). In particular, the pattern of increased variability in the King Hakon sector is well-resolved in EOF4 (Fig. 4.4), with increased sea ice anomalies in the King Hakon sector occurring in the same season, and slightly shifted, from EOF3. Many of the sea ice concentration anomalies in EOF4 are positive and of the same sign, suggesting more uniform variations in the Antarctic sea ice are captured in this mode. One key region of negative sea ice concentration anomalies in EOF4 occurs in DJF along the Weddell Sea sector, which connects towards the Atlantic Ocean off the coast of South America. This EOF also has a region of positive sea ice concentration anomalies towards the East Antarctica sector, located opposite of the Weddell Sea sector.

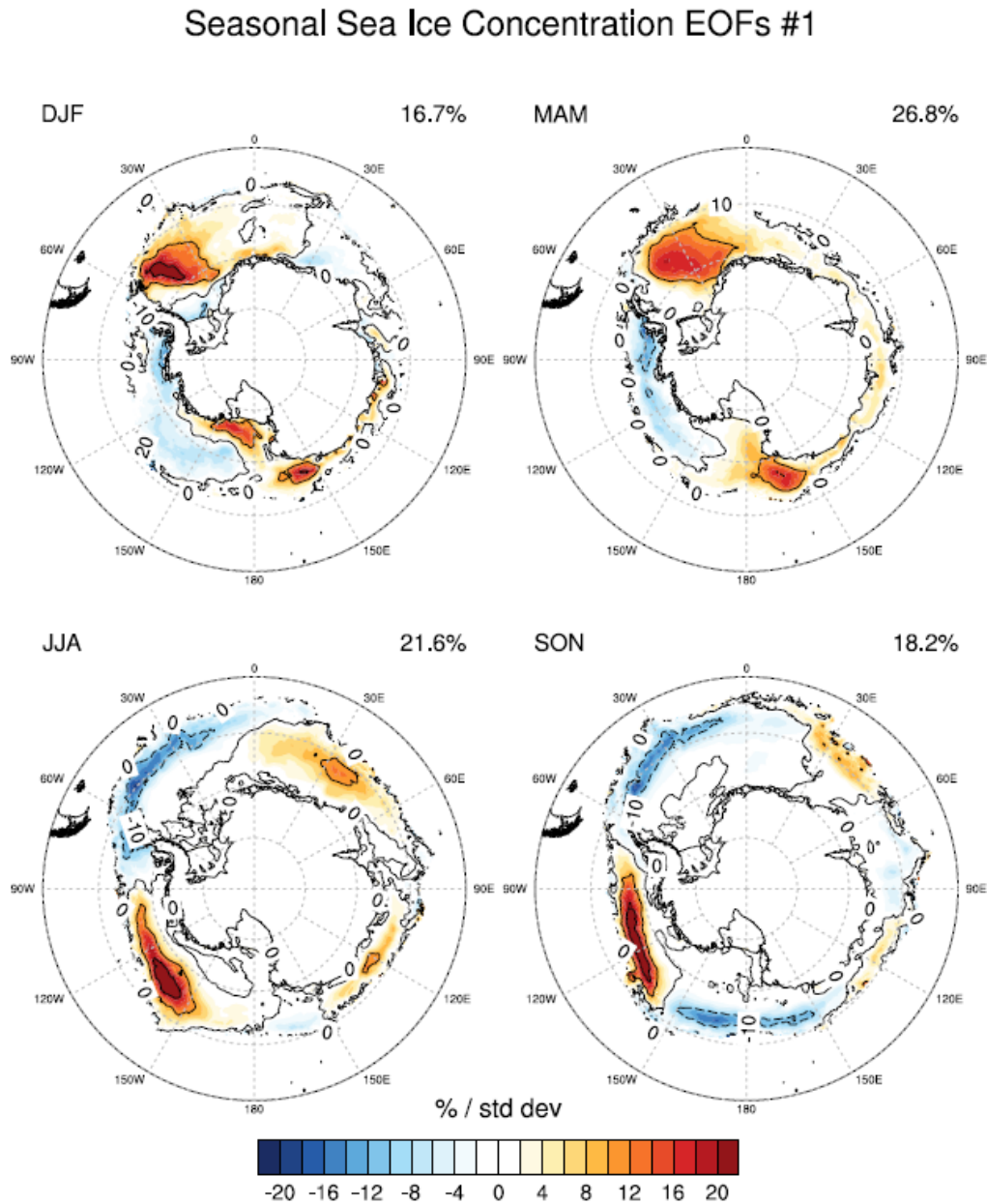
EOF5 continues the pattern of primarily capturing variations poleward from the ice edge and explains roughly 5% of the sea ice concentration variations in all seasons. In the fifth EOF, the overall sea ice concentration anomalies are smaller, where there are little to no areas of high anomalies or low anomalies (Fig. 4.5). The seasons of DJF and SON come the closest to having high/low values, with negative sea ice concentration anomalies closer to the Antarctic Peninsula in DJF and the King Hakon and East Antarctic sectors in SON. The most significant area of positive sea ice concentration anomalies occurring is in JJA; where the Bellingshausen-Amundsen Sea sector receives areas of higher variability; however, these values are comparatively less than in other EOFs.

The overall percent variance explained by the EOFs is also noticeably different throughout the different EOFs. This is especially prevalent in MAM, where EOF 1 shows

a higher value of variance accounted for with the nearest percentage being in EOF 2, where approximately 12% less variance is explained through the latter EOF. However, cumulatively across the five leading EOFs, approximately 50% or more of the total sea concentration variations are explained. The percentage of explained variance is clearly much higher at the ice edge in all seasons, and since this is the region most vulnerable to yearly fluctuations, reconstructing only the five leading patterns will provide a robust estimate of the significant changes in Antarctic sea ice concentrations. The percentage of variance explained across the five leading EOFs is lower in DJF, unsurprisingly, as this corresponds to the period of lowest sea ice extent, with many areas being ice free. Similarly, the reduced explained variance in the pack ice is not of major concern, as this region often has 100% ice cover each year and is not likely to change significantly during the last century. As such, a reconstruction of the first five EOFs appear as a good proxy for the full Antarctic sea ice field, even if only approximately 50% of the full sea ice concentration variance is explained.

4.1.2 Time Series Plots of PCs

Each EOF in Figs. 4.1 – 4.5 can be represented by a principal component (PC) time series that describes the amplitude of the pattern throughout time. High values in a PC time series indicate a year when the sea ice anomaly field takes a strong similarity to the spatial EOF pattern, whereas strongly negative values indicate a year where the sea ice anomaly pattern is nearly opposite the spatial EOF pattern (i.e., same key regions but anomalies of different signs). Near zero values of the PCs indicate the sea ice concentration anomalies have little to no resemblance of the pattern observed in the

Figure 4.1*Spatial Plot of EOF 1*

Note. Each EOF is represented as a regression map of the sea ice concentration and the PC timeseries, in units of % sea ice concentration anomaly per standard deviation of the PC timeseries. The % values given in the upper right of each panel are the percentage of variance explained by the seasonal EOF pattern (with respect to the full sea ice concentration variability).

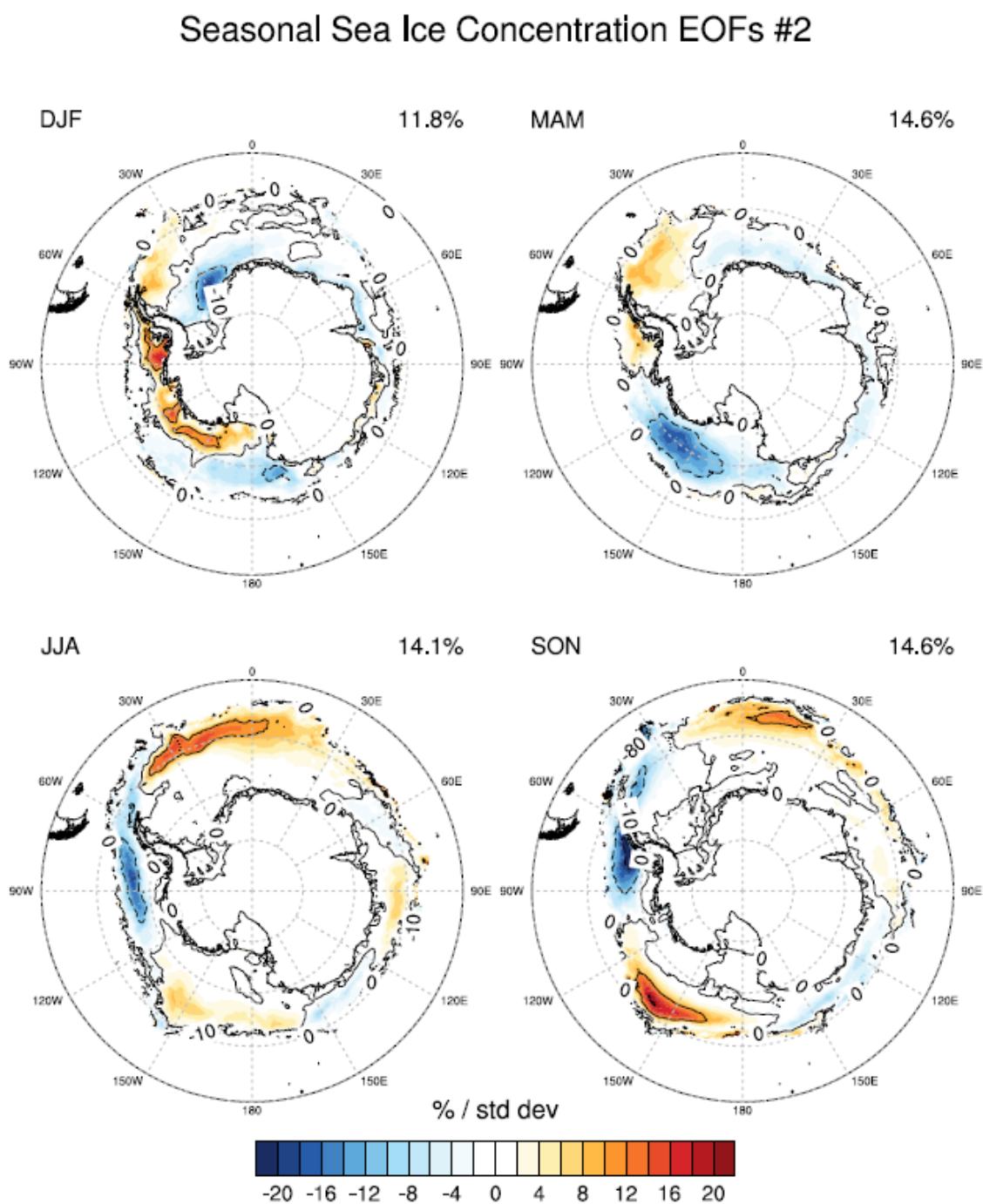
Figure 4.2*Spatial Plot of EOF 2**Note.* As in Fig 4.1, but for EOF2.

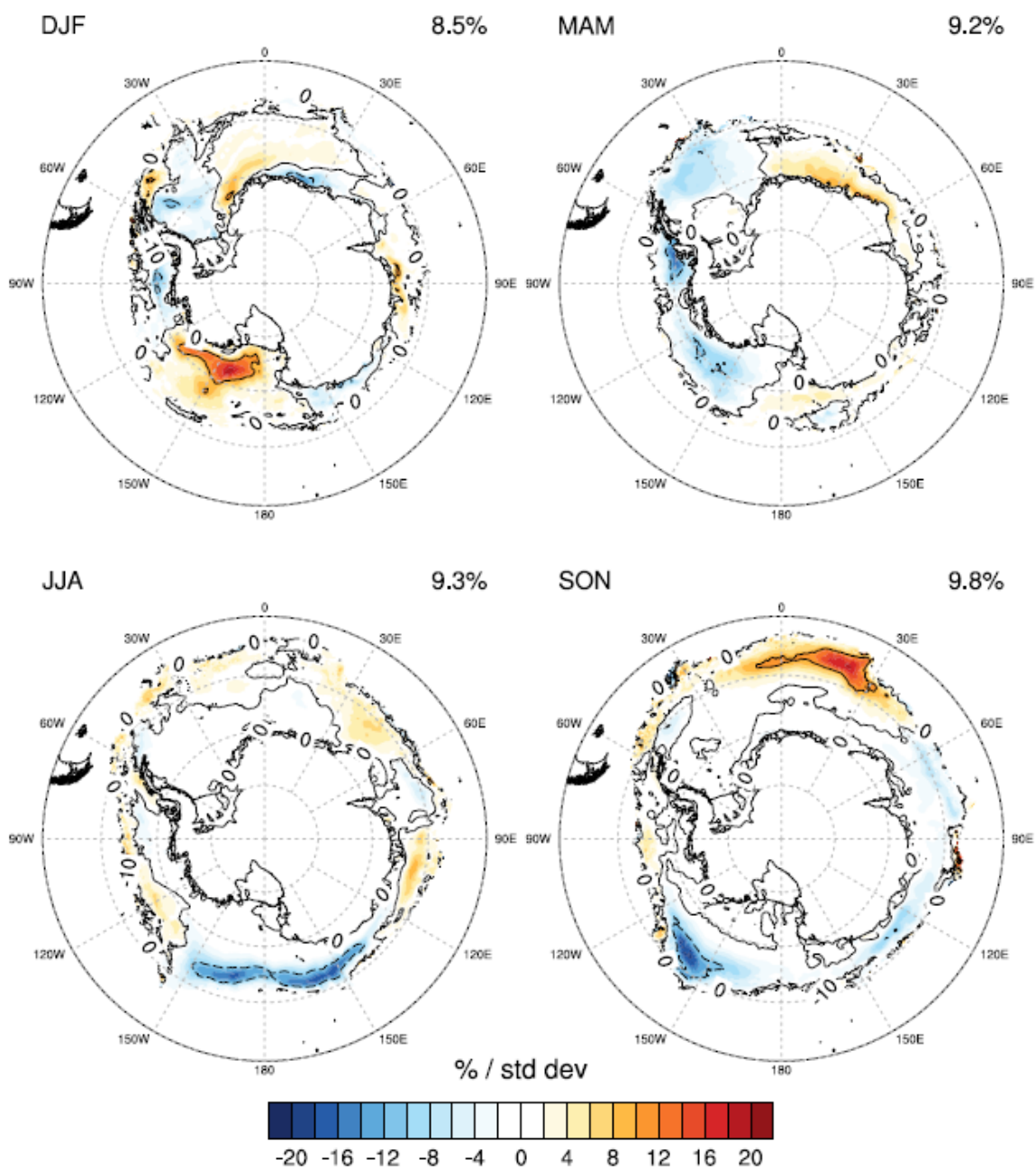
Figure 4.3*Spatial Plot of EOF 3***Seasonal Sea Ice Concentration EOFs #3***Note.* As in Fig 4.1, but for EOF3.

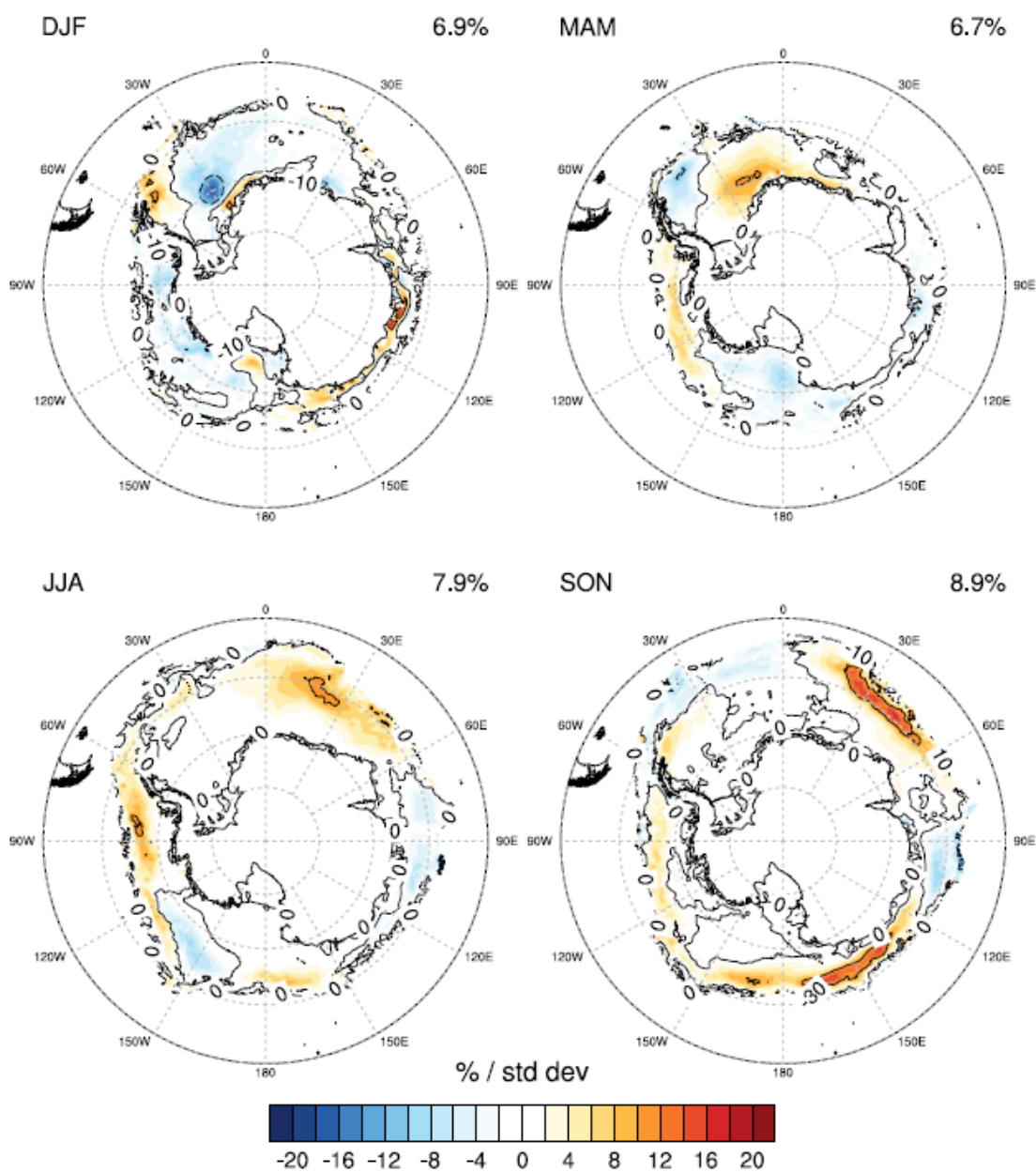
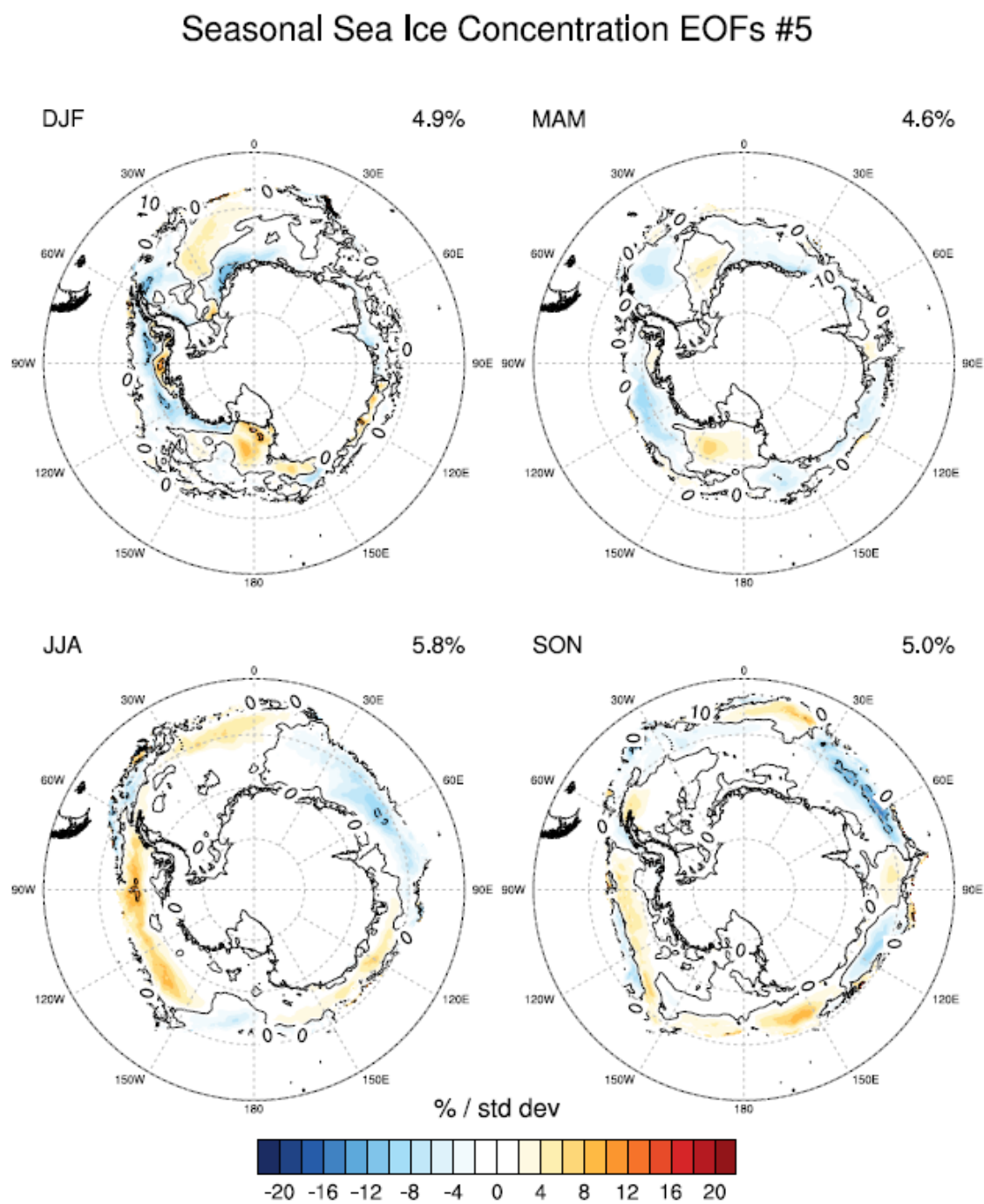
Figure 4.4*Spatial Plot of EOF 4***Seasonal Sea Ice Concentration EOFs #4***Note.* As in Fig 4.1, but for EOF4.

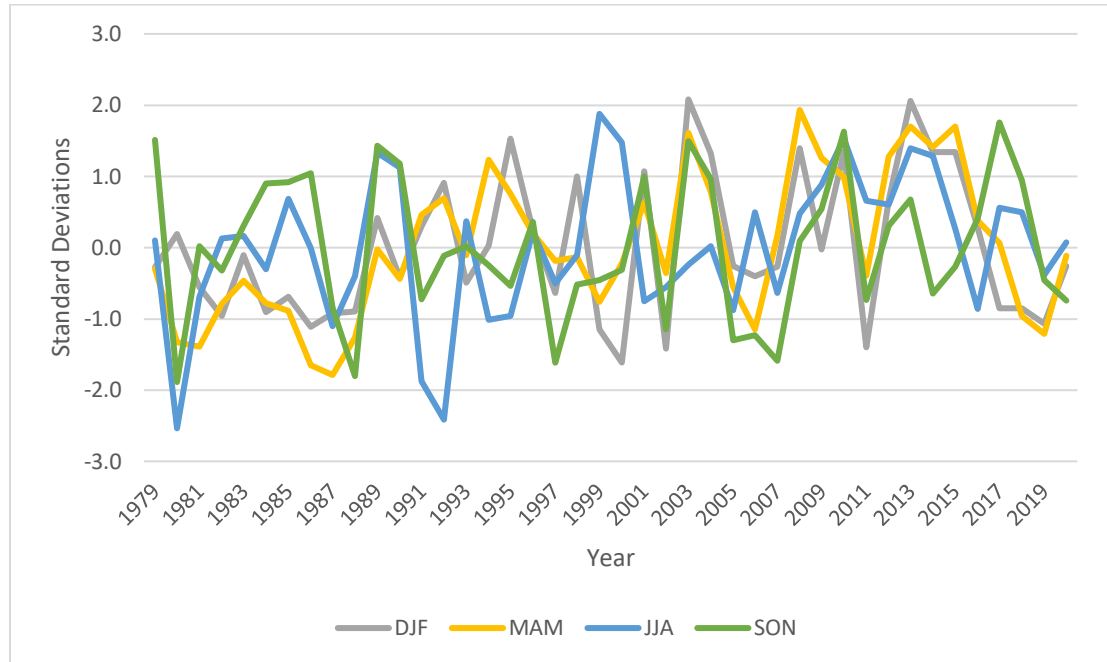
Figure 4.5*Spatial Plot of EOF 5*

Note. As in Fig 4.1, but for EOF5.

spatial EOF. Indeed, the reconstruction approach outlined in Chapter 3 is based on the PC time series (the PCs are reconstructed), so understanding the variations in the PCs is critical to interpreting the sea ice concentration reconstruction. Time series plots of the PCs were made in Excel, where all the PC data by season from 1979-2020 were entered and made into a line graph with each season being demarcated with a distinct color to view differences by season from year to year.

For brevity, an example of the PC time series plot is given for EOF1 (Fig. 4.6); more timeseries plots of the observed PCs will be displayed with the reconstructions in later sections of this chapter. The variable nature of sea ice concentration in time from Fig. 4.6 can be attributed to the different modes of climate variability, as well as some short-term weather variations. The large and quick drops from 1979 through 1981 in JJA and SON before quick rebounds into 1986 before following similar patterns is noticeable up until 1992 where the two seasons deviate away from each other. There is a noticeable difference between MAM and the other three austral seasons towards the beginning of the time series until 2000, where MAM becomes intricately connected with the other three seasons throughout much of the plot up until 2020.

PC2 (not shown) outlines different values throughout the beginning period of 1979 through 1988, which a large deviation in 1986 where SON and JJA experienced extreme deviations in the sea ice anomaly pattern resembling EOF2, while DJF stays at positive values (anomaly pattern strongly resembles EOF2) with a slight decrease and MAM experienced an increase in resemblance to EOF2 for sea ice concentration anomalies through time. The sea ice values collocate close together on the time series after this

Figure 4.6*Seasonal Time Series Plot of PC1 from 1979-2020*

discrepancy until 1998 where MAM experiences a significant drop in resemblance to EOF2, while all other seasons experience increases or constant values from the previous year. In 2015 in PC2, a sudden increase in the amplitude suggests sea ice anomalies resembled EOF2 in MAM, while all other seasons showed marked decreases / dissimilarities to EOF2.

The PC3 (not shown), corresponding to the sea ice concentration anomaly pattern of EOF3 (Fig. 4.3) shows much more irregular behavior where large jumps occur throughout much of the time series; however, many of the seasons showing similar behavior. This is expected since the EOFs (and hence the PCs) are all uncorrelated – increases in the PC values in one mode (PC3, for example) cannot simultaneously occur

with increases in all the other modes, at least not all the time (otherwise the timeseries would have some correlation, and the sea anomaly patterns in the EOFs would not be distinct and separable).

The seasonal time series for PC4 (monitoring the amplitude of EOF 4, not shown) indicates some variability in measurements in the earliest ten years, with less drastic changes towards the late 1980s. This period stays through until about 2000, when MAM begins to behave quite variably with a large deviation between all seasons in 2003. This large deviation is again noticeable in 2016, when the large Antarctic sea ice loss event occurred, where all the seasons except for DJF showed large decreases in sea ice. The sea ice anomaly pattern in EOF4 therefore represents at least some portion of the 2016 decline event.

In the final time series PC5 (not shown), demonstrates large shifts in resemblance to EOF5 throughout the different seasons, especially throughout MAM from 1979-1986 before continuing the large shifts again in 2000-2002. Despite the large shifts by MAM in EOF5 throughout these time periods, the three additional seasons show large shift throughout the 2000-2002 period that show that these large jumps could be due to climate variability from the different mechanisms over Antarctica.

4.1.3 Correlation with Climate Indices

While the temporal variations in the PCs are important for understanding the reconstructions, it is equally important to understand how the sea ice anomaly patterns from the five leading EOFs in Figs. 4.1-4.5 and their time variations in PC1-PC5 are related to patterns of large-scale climate variability (discussed in the literature review).

To understand these relationships, correlations (through time) can be made with each of the PCs and indices of climate as laid out in Chapter 2. To ensure that trends in either the PCs or climate indices do not influence the magnitude of the correlation (and to more appropriately understand the linear interannual relationships), all time series were detrended prior to calculating the correlation coefficient. The result of this analysis can lead towards t-values and p-values that will help show strong, weak, or moderate correlations between a PC timeseries (EOF sea ice anomaly field) and a particular mode of climate variability in each season. The correlation that is determined assists in being able to tell the indices that influence EOF patterns, and thus variability of sea ice concentrations in a season and will help in interpreting the sea ice concentration reconstruction.

Using Excel, detrended correlations (Table 4.1) were completed with each PC compared to climate index values for each season to gather an understanding of significance of climate patterns and how these effects are distributed out throughout each individual sea ice concentration EOF/PC. Seasonal data from 1979-2020 were taken from each PC were correlated with detrended climate indices (SOI, SAM, PDO, IPO, and AMO) from the same period, with the results presented in Table 4.1. Under the null hypothesis of no correlation, green values represent $p < 0.1$, blue values represent $p < 0.05$, and red values represent $p < 0.01$ in Table 4.1. In a few cases, the PC shows no correlation with any climate index (all have $p < 0.1$), indicating the pattern of sea ice concentration anomalies in that EOF does not have a strong linear association to a

particular mode of climate variability for that season. The climate mode with the strongest absolute correlation (or lowest p -value) is given in Table 4.1.

Table 4.1

Seasonal Detrended Correlations of Climate Indices and PCs

PC	DJF	MAM	JJA	SON
1	IPO	None	IPO	SAM
2	SAM	IPO	AMO	SOI
3	SAM	IPO	SAM	PDO
4	AMO	AMO	None	PDO
5	PDO	None	AMO	SAM

Note. Detrended correlation in green represent $p < 0.1$, blue represents $p < 0.05$, and red represents $p < 0.01$, cells with **None** indicate no values of at least $p < 0.1$ for that given PC and season.

In the terms of detrended correlation, SON was most resolved, with three of the five PCs/EOFs having statistically significant values, all of which are Pacific-based climate indicators. The austral spring also experienced moderate and low correlations in the two PCs with the PDO. There is a noticeable absence of statistically significant correlations throughout PC1 and PC5 in MAM; where the IPO maintains statistical significance throughout two of the three remaining EOFs, and the AMO maintains a moderate statistical significance.

The emergence of climate indicators based on EOF begins to become to prevalence with the completion of detrended correlations, as EOF1 is dominated by the IPO and SAM, both of which are Pacific-influenced climate indicator, and are known to influence the dipole pattern present in both EOF1 (Fig. 4.1) and EOF2 (Fig. 4.2). This follows suit for EOF 3, where more statistically significant correlations to the IPO are afforded with a continuation of statistically significant SAM influence, with the PDO showing moderate correlation in the austral spring. The presence of the AMO in EOF 4 throughout half of the year allows for Atlantic influence to be factored in as well, with other EOFs maintaining correlation with the AMO, noticeably in JJA, where low-end statistical significance is achieved.

The most statistically significant climate mode of variability throughout all EOFs was the SAM; where high statistical significance was achieved throughout four different combinations of EOF and season with moderate statistical significance achieved in one additional combination. This supports the notion mentioned in Chapter 2 that the SAM is one of the dominant patterns of natural variability in this region and tends to affect sea ice concentrations significantly.

The results of this detrended correlation analysis to view what EOFs represent was meaningful, as twenty-two out of twenty-five (88 percent) of correlations evaluated revealed some statistical significance in a relationship between the climate modes of variability and EOF pattern. This demonstrates that the PCs will be a meaningful way to creating reconstructions with skillful results as many of the PCs are strongly influenced by the climate indices used as predictors in the reconstruction. However, with the amount

of data that is being inputted into these reconstructions, more analysis will need to be done through the realm of a predictor count and f-test to determine if overfitting, where too much data is put into the reconstruction and can yield incorrect results further into the reconstruction.

4.2 Selection of Retained Predictor Data

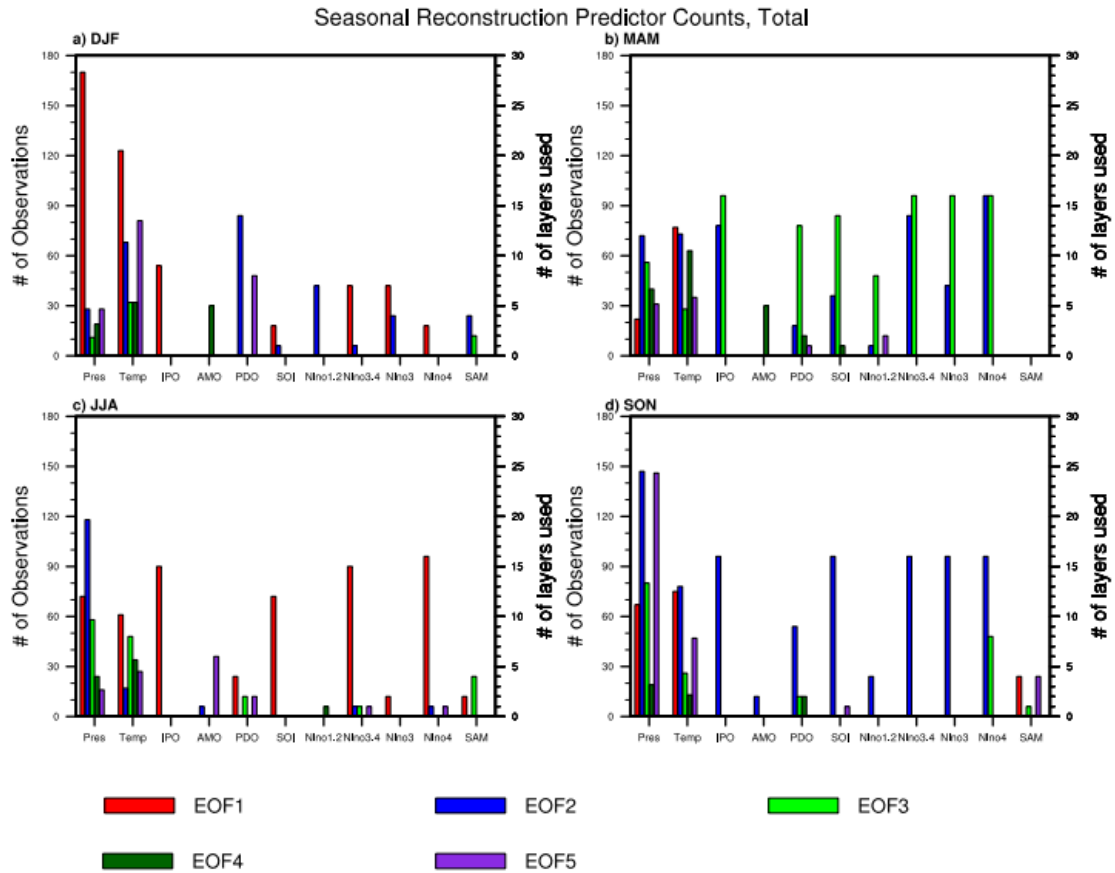
Before examining the reconstruction skill, a visual representation of the number of predictors put into each EOF to analyze the composition of data used in the reconstruction can help understand the reconstruction sensitivity to underlying data. The composition of each ensemble member reconstruction will entail different predictor data, based on the correlation each predictor variable has with the seasonal sea ice concentration PC at four different networks (correlations significantly different from zero at $p < 10\%$, $p < 5\%$, $p < 2.5\%$, and $p < 1\%$) as established in Chapter 3. The seasonal variations, as well as variations across the PCs, can be visualized by the total count of all predictors across all networks to decipher the full composition of data throughout all network sizes of the reconstruction. These predictor count plots also consider lagging of seasons, also explained in Chapter 3, such that the predictor data lead the seasonal sea ice variations by up to one season (at one-month increments). The predictor counts across all data networks is shown in Fig. 4.7.

Throughout all the predictor networks, pressure and temperature observations dominate the reconstructions (Fig. 4.7) with moderate influences of climate indices filling in the reconstructions across different EOFs. The Pacific Ocean relationships with Antarctic sea ice PCs show up for EOF1 in DJF and JJA, and as EOF2 for SON, but not

until EOF3 for MAM, like the expected results in Table 4.1. In these instances, the Pacific Ocean influence is represented by the inclusion of the IPO, PDO, SOI, and all Niño indices. Because the SAM indices are based on reconstructions, no lagging is possible with this index, and the maximum layers it can be included is capped at four (the four different data networks). The SAM does not have any role in the PCs for Antarctic sea ice concentration in MAM (as in Table 4.1) but does appear as an important predictor in other seasons, especially SON (Fig. 4.7). There is a noticeable absence of AMO data in the reconstruction of sea ice concentration, where EOF 4 is the main source of this climate index being analyzed through the reconstruction in DJF and MAM, but it plays a much weaker role in the other seasons.

Consistent with Table 4.1, the influence of the Atlantic Ocean is much weaker, only prevalent in PC4/EOF4 or PC5/EOF5, while the Pacific Ocean influence appears in the leading EOFs (1-3). Pressure and temperature observations are the strongest predictors influence the reconstructions (Fig. 4.7), with a notably weaker role in MAM. Overall, pressure data are included more frequently than temperature observations, which can be tied to the fact that temperature observations may capture more regional variability not tied to Antarctic sea ice variability. This is where a breakdown of predictor counts based on the predictor network size becomes beneficial to the analysis of data distribution in the reconstruction.

Further analysis on the 2.5 percent network separately (Fig. 4.8) shows moderate amount of influence from the Pacific in JJA for EOF1 where IPO, SOI, Niño 3.4, and Niño 4 tend to have significant influence in the reconstruction. At such strong

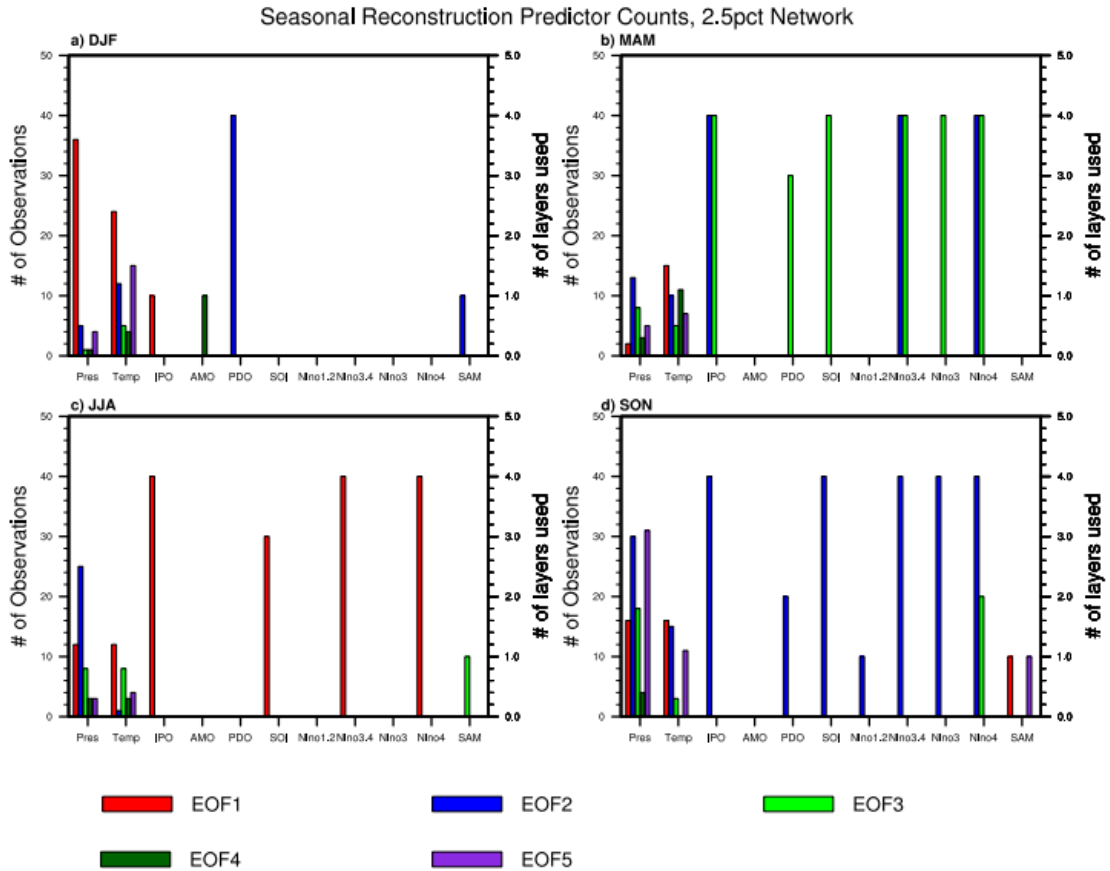
Figure 4.7*Seasonal Predictor Count across all Networks*

Note. Seasonal predictor counts of all network sizes with DJF (top left), MAM (top right), JJA (bottom left), and SON (bottom right). The left axis expresses number of pressure and temperature observations used in the reconstruction while the right axis expresses the number of layers of climate indices used in the reconstruction.

correlations, this smaller data network includes the Pacific Ocean only at EOF2-3 in other seasons outside austral winter. It is also clear that a much smaller subset of temperature and even pressure observations are included in the reconstruction, and the relationship is more uniform across the seasons and even between the number of pressure and temperature observations used as predictor data (Fig. 4.8).

4.3 Performing Reconstructions on Retained Predictors

The reconstructions were computed for each seasonal sea ice concentration PC separately (five PCs, four seasons). For each reconstruction, four data network layers and all available pressure and temperature observations and up to nine additional climate index data layers (based on the data network and their individual correlation) yields a range of five to forty-one (with five being only observations to forty-one being all data layers for each data network) ensemble reconstructions. As a first estimate, the individual ensemble member with highest skill was selected as the highest CE value. However, this ensemble member may have high statistical performance artificially, and it is therefore necessary to ensure its variability is consistent with observed sea ice variability. The completion of an F-test allows for analysis if reconstructions yield different variance compared to the observed, and to select a better suited single ensemble member as the ‘best fit’ reconstruction. Scatter plots were made to examine variance ratios visually. The first series of visualization of variances is given in EOF3 (Fig. 4.9). Variances were compared in multiple separate ways, indicated by the legend and colors, with gray circles indicating no significant variance for any different period (or between the reconstruction during the calibration (1979-2020) and reconstruction (1905-2020) period. Each

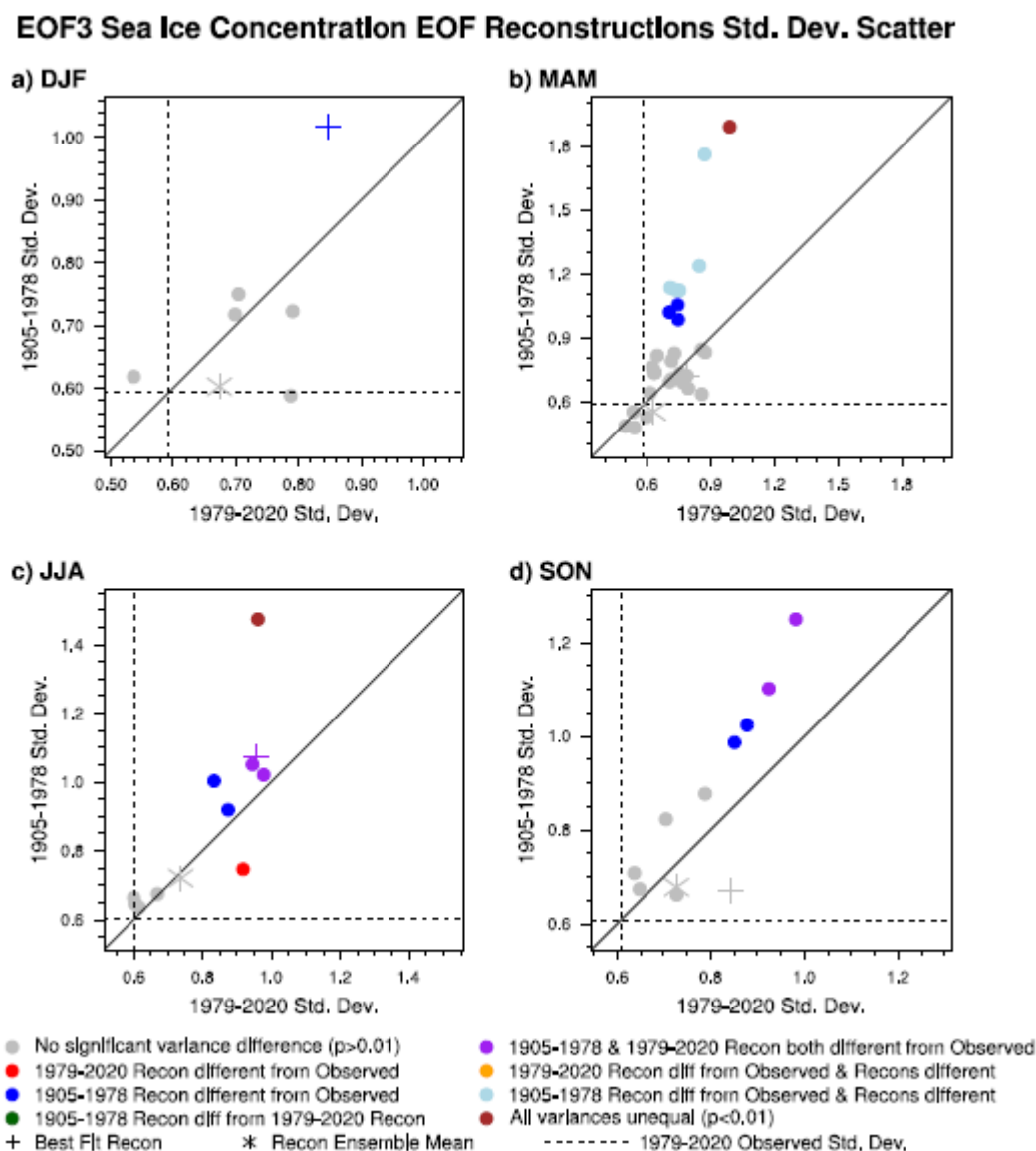
Figure 4.8*Predictor Count across the 2.5 Percent Network*

Note. As in Fig. 4.7, but only across the 2.5 percent network size.

ensemble member is plotted separate, with a plus sign indicating the first guess of the ‘best-fit’ reconstruction (highest CE) and an asterisk indicating the ensemble mean. From Fig. 4.9, overfitting of the best fit reconstruction is potentially prevalent in both DJF and JJA. In DJF (Fig. 4.9a), the reconstruction does well throughout the observed period of 1979-2020; however, when the reconstruction goes back into 1905, the reconstruction variance is much higher than the observed during 1979-2020 (dashed lines), with a $p <$

Figure 4.9

Variance Differences in Reconstructions from 1905-1978 and 1979-2020 for EOF3



Note. The ensemble reconstruction variances as a scatter plot, with the x-axis indicating the variance during the calibration period (1979-2020) and the y-axis indicating the variance during the reconstruction period (1905-1978). The dashed lines indicate the variance of the observed PC, with the solid diagonal line indicating equal reconstruction variance in the two time periods. The color shading corresponds to places where an F-test of equal variances cannot be rejected at $p < 0.01$. The first guess for the best fit reconstruction is indicated by +, and the ensemble mean marked with *.

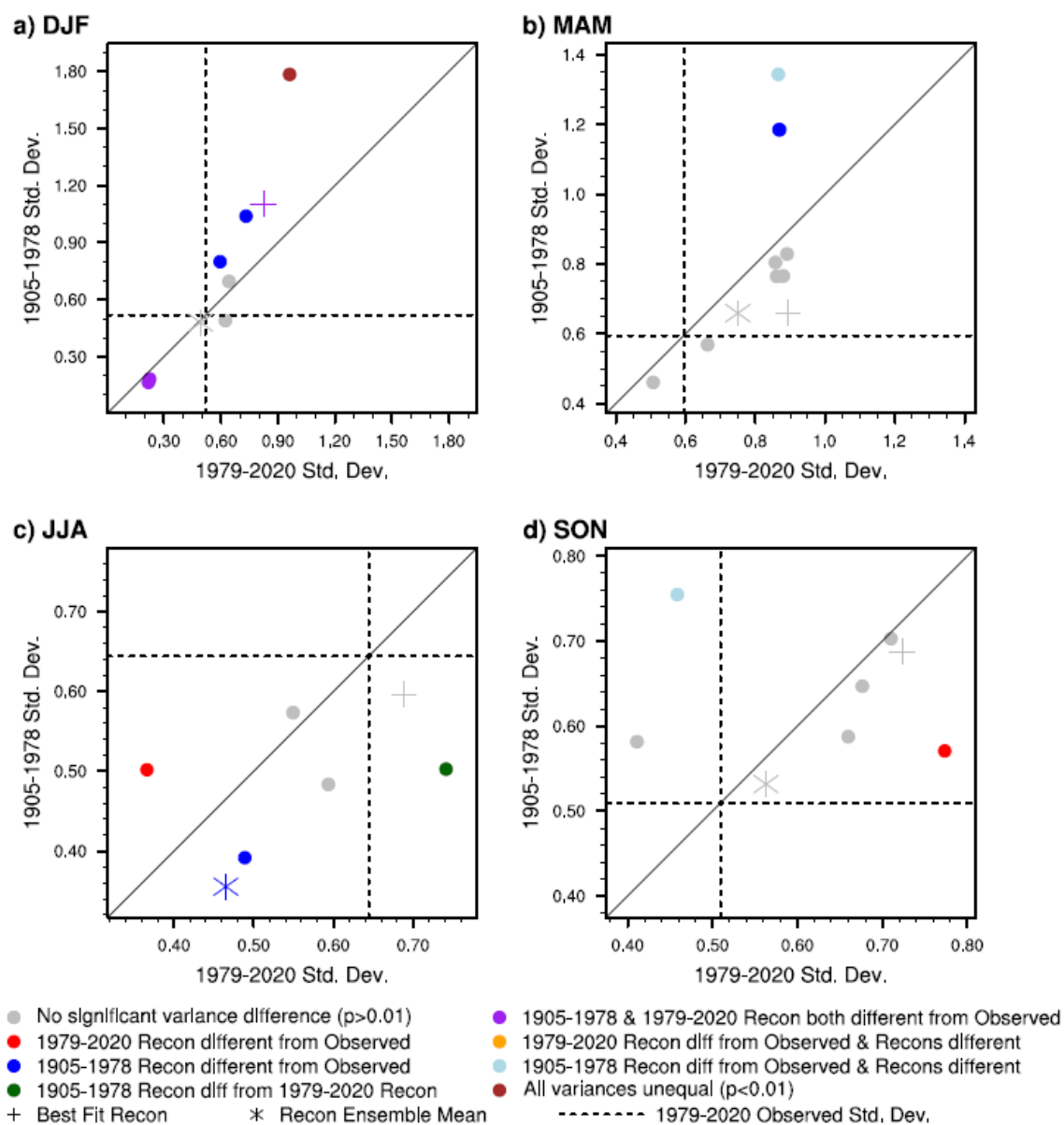
0.01 of equal variances from the F-test. The indication of overfitting suggests that too much information from the predictors makes reconstructions obtain more variance and become noisy in nature, or inversely have too little variance and become dampened where changes in sea ice concentration changes cannot be observed. This indicates that the ‘best fit’ reconstruction must be changed for a better overall reconstruction of the Antarctic sea ice concentration field (i.e., one of the gray markers with comparable skill). Overfitting is also prevalent in JJA (Fig. 4.9c); here, the first guess best-fit reconstruction during both 1905-1978 and 1979-2020 yields different variances than observed values, but the reconstruction is at least consistent over time (i.e., the reconstruction variances between 1905-1978 and 1979-2020 are similar at $p < 0.01$). This could indicate lower reconstruction skill overall, but it is still necessary to investigate the other ensembles (gray markers) that have similar variance, and if one of these reconstructions is performing similarly well (based on its CE) to be selected as the final ‘best-fit’ reconstruction for this season and EOF.

In another example of the F-test plot (Fig. 4.10), the first guess best fit reconstruction passes the F-test metric at $p < 0.01$ in all seasons except DJF (Fig. 4.10a). For example, in JJA, the 1979-2020 and 1905-1978 reconstructions exhibit no statistically different variance between themselves as well as not being different from observed data. The reconstruction named the best fit reconstruction for JJA in EOF 4 would be the first guess (highest CE) reconstruction due to the reconstruction not exhibiting extreme signs of overfitting that could harm the overall skill of the final reconstruction. In DJF, there are two other ensemble reconstructions that exhibit no

Figure 4.10

Variance Differences in Reconstructions from 1905-1978 and 1979-2020 for EOF4

EOF4 Sea Ice Concentration EOF Reconstructions Std. Dev. Scatter



Note. As in Fig. 4.9, except for EOF4.

significant variance difference, which must be further investigated to see if they would be better selections for the best fit ensemble member. In some cases, there were no similarly high CE values in the other reconstruction ensemble members as the first guess (highest CE) member. In these cases, the initial best fit reconstruction was retained as the final best fit reconstruction.

Table 4.2 lists the final best fit reconstructions as a function of the data network and data layer after passing through the F-test procedure. The presence of these twenty best fit reconstructions allows for modes of climate variability to be seen throughout the EOFs, such as a noticeable Niño influence in EOF2 that suggests EOF2 has strong relationships with ENSO events, consistent with Table 4.1 and Fig. 4.7. This logic can be extended to EOF5, which exhibits connections to the PDO, especially at the ten percent predictor network threshold. Observational measurements of pressure and temperature networks also serve as best fit reconstructions for half of the best fit reconstructions, with a marked dominance in MAM, tying in these measurements with modes of climate variability to gather the best possible reconstruction.

Table 4.2*Best Fit Reconstructions by EOF and Season*

EOF	Season	Best Fit Reconstruction (Layer, Network)
1	DJF	IPO (5%)
	MAM	Observations (All)
	JJA	SAM (10%)
	SON	Observations (All)
2	DJF	Observations (All)
	MMA	Niño 4SST (10%)
	JJA	Niño 4SST (10%)
	SON	Observations (10%)
3	DJF	SAM (5%)
	MMA	Observations (10%)
	JJA	Niño 3.4SST (10%)
	SON	Observations (10%)
4	DJF	AMO (5%)
	MMA	Observations (All)
	JJA	Observations (5%)
	SON	PDO (5%)
5	DJF	PDO (10%)
	MMA	Observations (10%)

EOF	Season	Best Fit Reconstruction (Layer, Network)
5	JJA	PDO (10%)
	SON	Observations (5%)

Note. The best fit reconstruction at the conclusion of f-test analysis, named with predictor with network in parentheses.

4.4 Best Fit Seasonal Reconstruction Timeseries

The best reconstructions determined by the highest CE values were put into a timeseries form to be further analyzed throughout this study. These timeseries were divided and produced by EOF into two different forms in Figs. 4.11 – 4.15. The first form (displayed in Figs. 4.11, 4.13, and 4.15) outlines the best fit and ensemble mean reconstructions along with observed sea ice concentration PC values, along with each individual ensemble member time series to observe the range in reconstruction results. The second form of timeseries (displayed in Figs. 4.12 and 4.14) outline the best fit reconstruction, ensemble mean reconstruction, and observed sea ice concentration PC values along with a gray shaded area representing uncertainty in the reconstruction. As discussed in Chapter 3, the reconstruction uncertainty is calculated as the larger of 1.96 times the standard deviation of residuals between the best fit reconstruction and actual data, or two standard deviation ranges of all unique reconstructions. An increased amount of gray area in these figures outline greater uncertainty in the reconstructions, which can also be outlined by low CE values, indicating that reconstructions are not yielding the best results. These figures also include statistical values of cal r, val r, RE, and CE value outlined in Chapter 3, along with ‘correl’ (calculated as correlation between ensemble

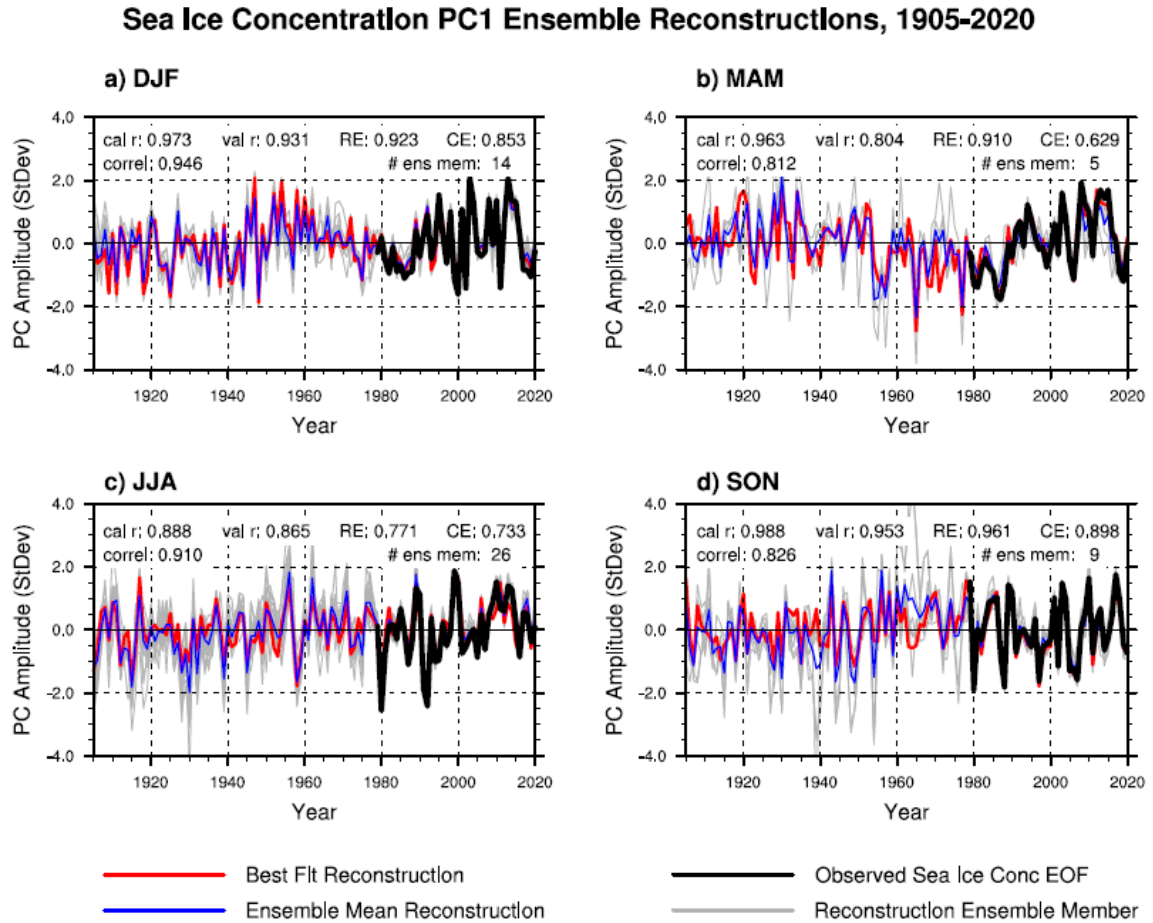
mean and the best fit reconstruction during 1905-2020) values and the number of ensemble members ‘# ens mem’ at the top of these plots.

In a timeseries of EOF1 ensemble reconstruction (Fig. 4.11), the reconstruction is highly skillful in DJF and SON, with remarkably high CE values (>0.85) and calibration correlations above 0.95 accompanying these visual representations. These results outline that PC1 can yield meaningful and skillful results in the reconstruction of sea ice concentrations throughout the austral summer and spring. There is increased uncertainty into MAM, where CE values are the lowest throughout the EOF along with having more variance between the individual ensemble members, especially observed around 1960-1970 where ensemble members have jumps from the best fit and ensemble mean reconstruction lines. MAM also has one of the lowest ‘correl’ values, where the best fit reconstruction and the ensemble mean reconstruction are showing different depictions of PC1 amplitude over time. This can be readily observed in 1920-1925, where the best fit reconstruction observes larger changes in sea ice concentration via PC1 amplitude compared to the ensemble mean reconstruction. This change in sea ice concentration tends to be particularly marked in the best fit reconstruction, with ensemble members showing a large shift in this period as well.

The overall uncertainty is increased for the PC2 seasonal reconstructions (Fig. 4.12), where CE values are overall smaller comparatively to those in PC1. There is, however, a noticeable increase in ‘correl’ values through MAM and SON, with an increase in CE in SON compared to PC1. This decreased uncertainty is shown through the gray shading, where the low amount of shading indicates significantly less

Figure 4.11

Seasonal Best Reconstructions Timeseries for PC1 through 1905-2020



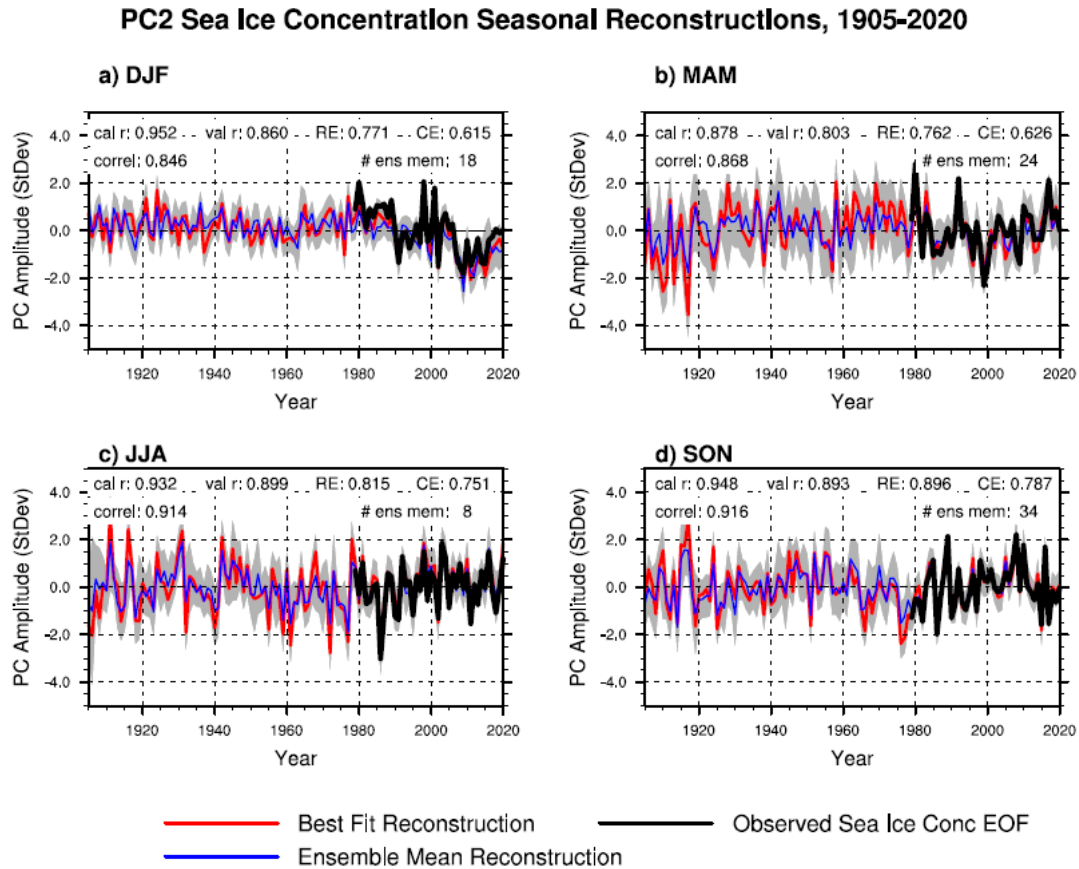
Note. Timeseries representation of all ensemble members, including best fit, ensemble mean, and observed sea ice concentration PC for PC1.

uncertainty in the reconstruction of sea ice concentration in SON for PC2. For this PC, the reconstruction skill is slightly lower in MAM, where a CE value of 0.626 along with the most seasonal shading of uncertainty, shows the increased uncertainty in reconstructions during this season. The most eye-catching difference between PC1 and PC2, however, is observed to be in DJF, where the CE drops significantly, and the ensemble members have a larger spread compared to PC1. It is also noteworthy that a large ensemble (34 out of 41 possible) is used for the reconstruction in SON (Fig. 4.12d), and with the ‘correl’ value of 0.916 the ensemble mean (and therefore most ensemble members) match the variability of the individual best-fit ensemble member.

The drop-off in DJF between the first two PC reconstruction performance is further observed in the PC3 reconstruction (Fig. 4.13a), where CE drops even more significantly with ensembles yielding much different values in the period of 1950-1960. In this period, the best fit and ensemble mean reconstructions show increases in sea ice concentration, while a few individual ensemble members show decreases in sea ice concentration. There is also a significant increase in SON reconstruction uncertainty between PC2 and PC3 through decreased CE values and an increased deviation from the ensemble member reconstructions, along with deviations between the best fit reconstruction and ensemble mean reconstruction expressed through seasonal ‘correl’ values. While all seasons tend to add uncertainty throughout the first three PCs, MAM has consistent uncertainty, where the uncertainty is maintained throughout these PCs rather than increasing reconstruction uncertainty.

Figure 4.12

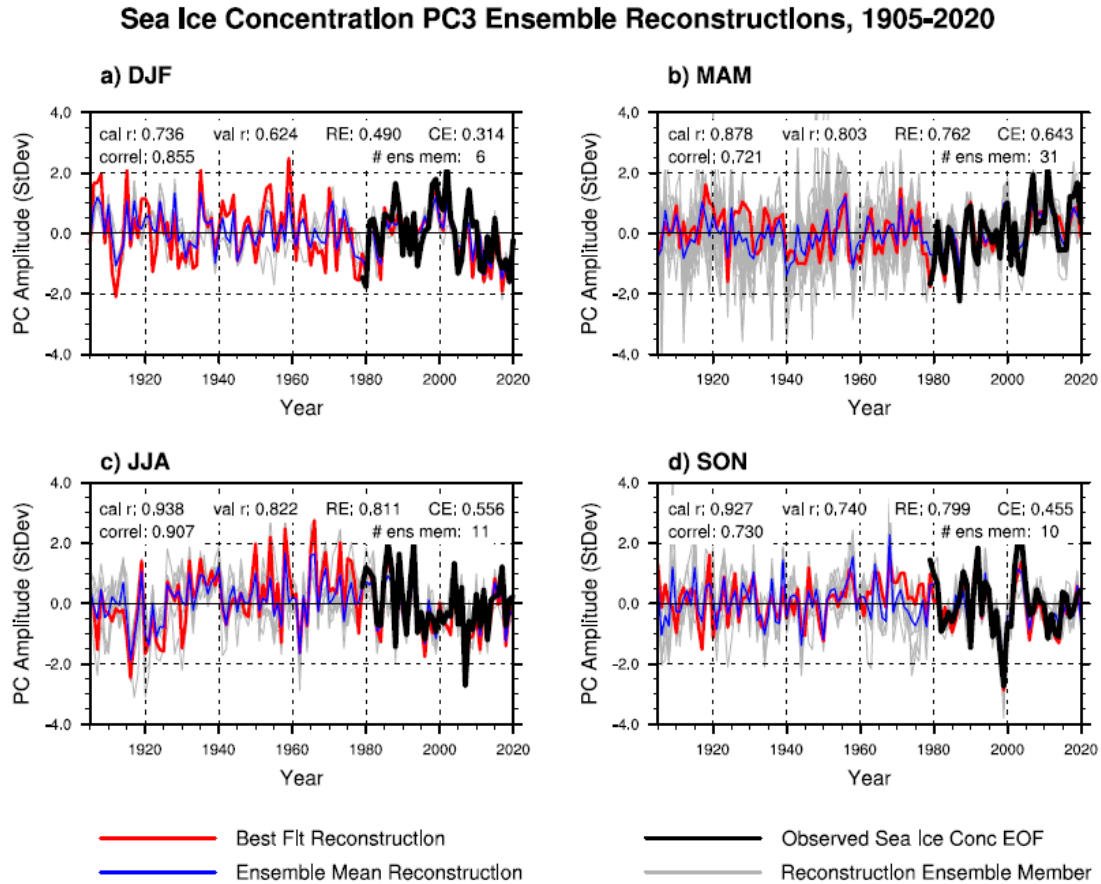
Seasonal Best Reconstructions Timeseries for PC2 through 1905-2020



Note. Timeseries representation with best fit and ensemble mean reconstructions with observed sea ice concentration PC for PC2. The gray shaded area is area of uncertainty.

Figure 4.13

Seasonal Best Reconstructions Timeseries for PC3 through 1905-2020



Note. As in Fig. 4.11, but for PC3.

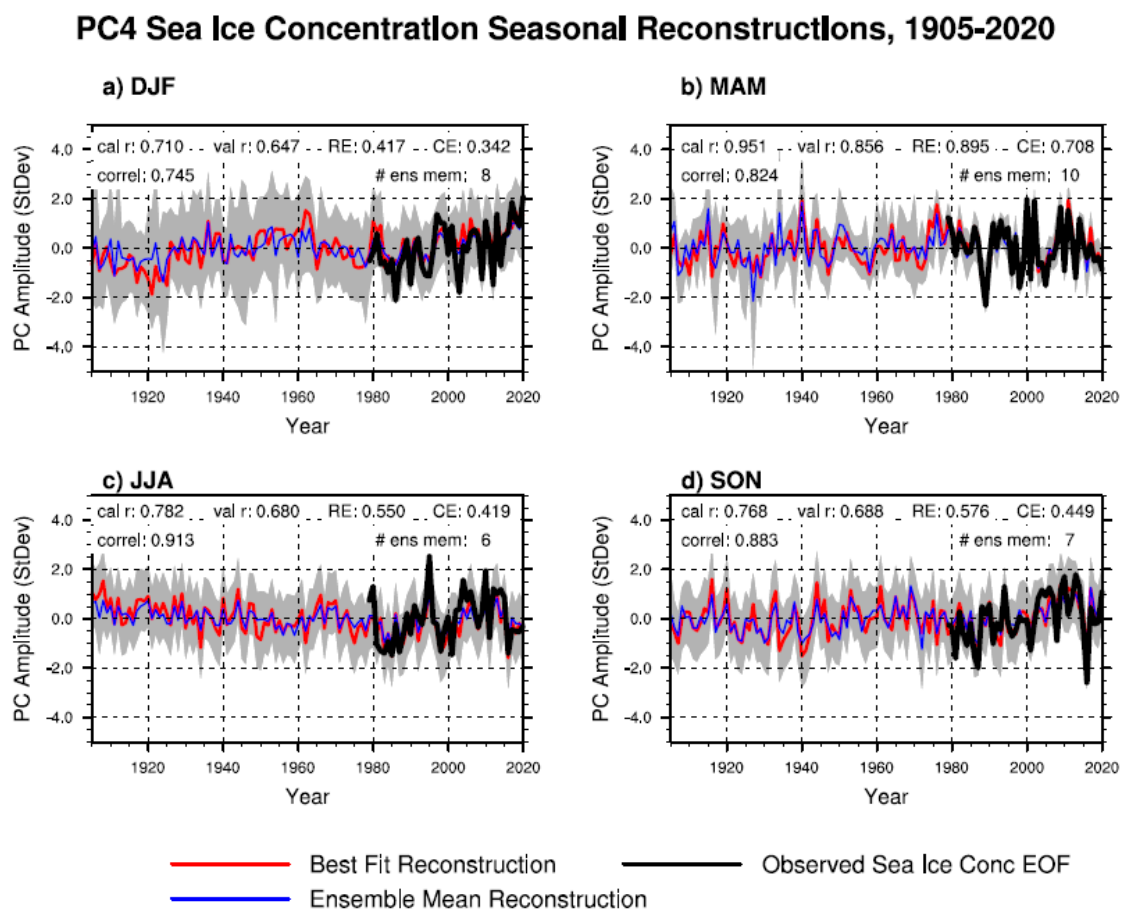
The reconstructions for PC4 are given in Fig. 4.14. Overall, the uncertainty does not remarkably change for the first time between increasing EOF numbers, where most seasonal uncertainty is maintained with slight increases and decreases in uncertainty. Uncertainty in MAM decreases, with less uncertainty shading a notable increase in CE values that begin to outline an increase in reconstruction certainty, along with an increase

in ‘correl’ values from PC3 into PC4, where the best fit and ensemble mean reconstructions show similar trends through time. The austral summer in PC4 decreases uncertainty from PC3, while still having more uncertainty from the first two PCs, with lower CE values; however, the trend of significant increases in uncertainty is not as prevalent in PC4, with only a moderate increase in reconstruction certainty throughout DJF. There is a significant increase in uncertainty throughout JJA, with increased reconstruction uncertainty between the ensemble members compared to previous PCs.

In a timeseries of best fit seasonal reconstructions for PC5 (Fig. 4.15), the uncertainty in DJF, SON, and JJA significantly decreases while uncertainty in MAM slightly increases as observed through the spread of ensemble reconstructions and increases/decreases in CE values for the seasons. The consistent skill in MAM, surprisingly, is also particularly marked by ‘correl’ values, where the highest value of correlation between best fit and ensemble mean reconstructions yield the highest of all PCs. In MAM, the connections with modes of climate variability are much weaker (Table 4.1, Fig. 4.7), so the skill may drop less with EOF number than in other seasons. The highest amount of variance between most ensemble members tends to be around 1920 in DJF and JJA, while SON shows variance in ensemble members throughout the 1940-1970 period. The variance across each ensemble member is higher in PC5 (especially outside of JJA) than all other PCs, where uncertainty tends to increase throughout each PC.

Figure 4.14

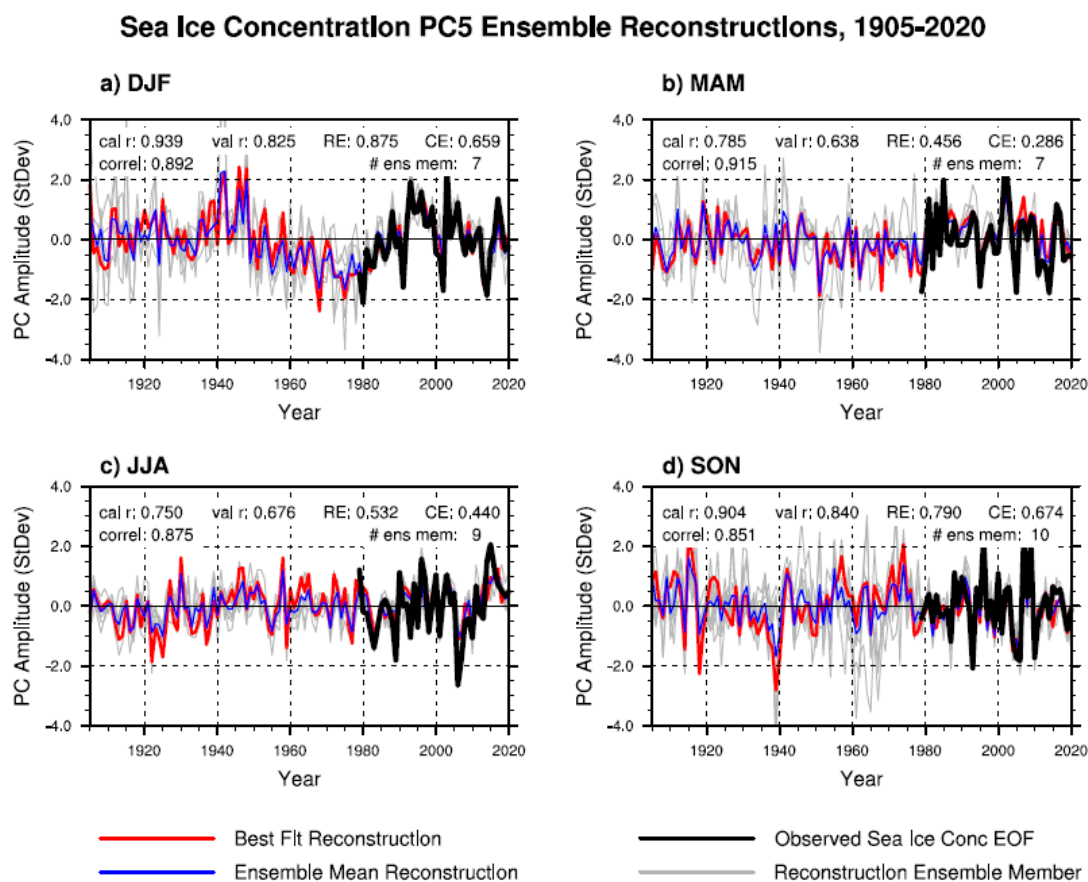
Seasonal Best Reconstructions Timeseries for PC4 through 1905-2020



Note. As in Fig. 4.12, but for PC4.

Figure 4.15

Seasonal Best Reconstructions Timeseries for PC5 through 1905-2020



Note. As in Fig. 4.11, but for PC5.

4.5 Seasonal Boxplots

To gain a visual perspective on reconstruction performance, a boxplot (Fig. 4.16) was created seasonally comparing the four main reconstruction statistical skill metrics (cal r, val r, RE, and CE). The boxplot is created using all ensemble members to see the variance between ensemble members from the five EOFs, four predictor networks, and ten layers (blue boxes) along with the best fit reconstructions (red boxes, five total ensemble members, one per EOF) to evaluate how well the best fit reconstructions statistically represent reconstructions compared to the full range of ensemble members.

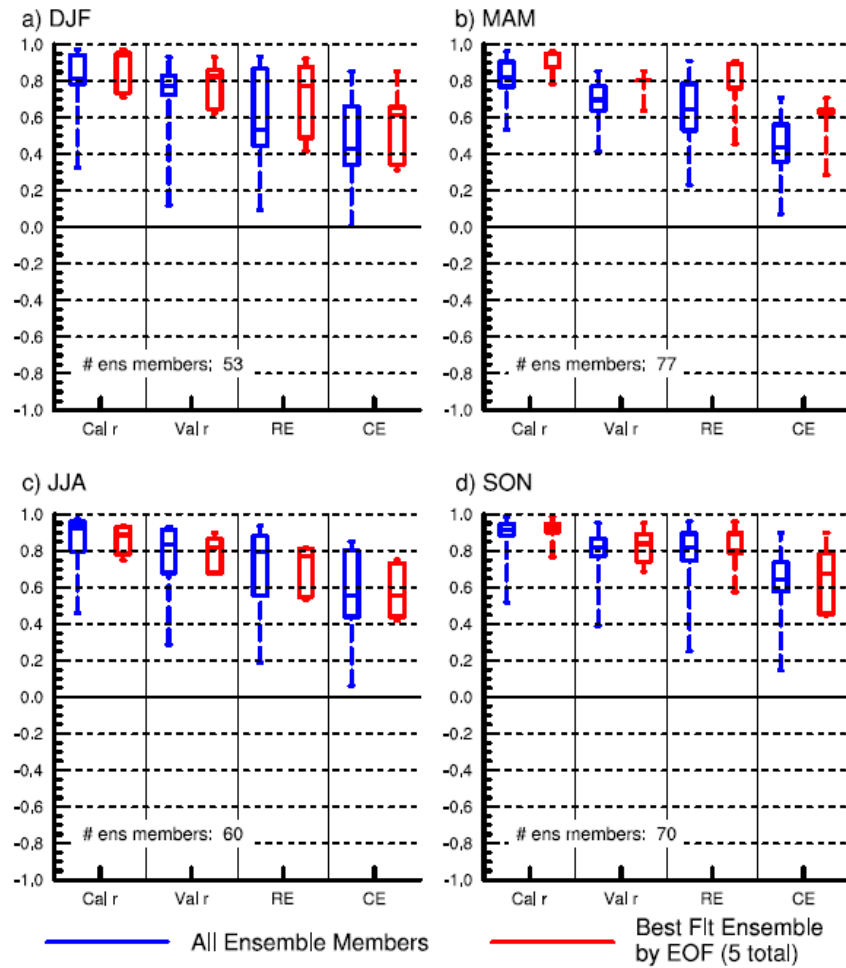
In Fig. 4.16, all skill metrics have positive measures signifying better reconstruction skill than a simple use of the climatological mean, as outlined in Chapter 3. Although some ensemble members receive CE values close to zero, the best fit reconstructions values over all seasons tend to be well above zero. The cal r metric reflects correlation between the observed and reconstruction data; RE values are similarly based on the calibrated reconstruction but are more sensitive to outliers and perform farther below one than the cal r metric. To provide a more robust measure of reconstruction skill, the metrics based on the independently produced validation reconstruction can provide further insight into potential model overfitting, sensitivity to data, and the stationarity of the reconstruction model. Throughout JJA and SON, where higher val r values are present, overfitting is not likely occurring in the best fit reconstructions. There is a higher range of values in val r throughout all ensemble members, with values close to zero in DJF, where reconstruction could be overfitting the

PCs of sea ice concentration. The CE, the hardest skill metric to pass (due to sensitivity to outliers in the validation reconstruction), shows the overall lower values as expected, but

Figure 4.16

Seasonal Boxplots for Each Season's Full Range of Skill for Each Statistical Metric

SIC Reconstruction Seasonal Skill Performance across All Ensembles



Note. The four metrics used to determine reconstruction skill, calibration correlation (cal r), validation correlation (val r), reduction of error (RE), and coefficient of efficiency (CE) for all ensemble members (blue) and best fit ensembles for each EOF (red).

with a median above 0.4 (with 0.5 for best fit reconstructions), which still suggests a robust and meaningful reconstruction of Antarctic sea ice concentration is possible.

4.6 Spatial Performance of Reconstructions

To analyze the skill of reconstructions spatially, plots are made with observational EOF patterns and the reconstructed EOF patterns. As the study only reconstructs the first five EOF patterns of Antarctic sea ice concentration, it does not contain the full portion of Antarctic sea ice variability. However, as discussed earlier in the chapter, the five leading EOFs capture much of the sea ice variability at the ice edge. These observational (leading five EOFs) and reconstructed plots can be then put into a plot with ratio of reconstructed to observed (all EOFs) sea ice concentrations to determine the overall skill of the reconstruction spatially. In both cases, correlations are represented from zero to one, with zero indicating no correlation in measurements to observed data while one indicates a perfect positive correlation between the measurements and the observations.

To understand the maximum skill possible represented in the subset of the leading five EOFs of Antarctic sea ice concentration, Fig.4.17 compares the correlation of these five observed EOFs (merged) with the full sea ice concentration pattern. As suggested earlier, the full Antarctic sea ice concentration observations correlate with the leading five EOFs very well throughout the edges of the continent, especially in the Weddell Sea region throughout DJF and MAM. This region, in MAM, observes the highest amount of correlation between EOFs and observations between all seasonal values with values ranging from 0.9 – 1.0, the highest values of correlation possible. In other seasons, correlations of the combined five leading EOFs with the full ice pattern are >0.7 ,

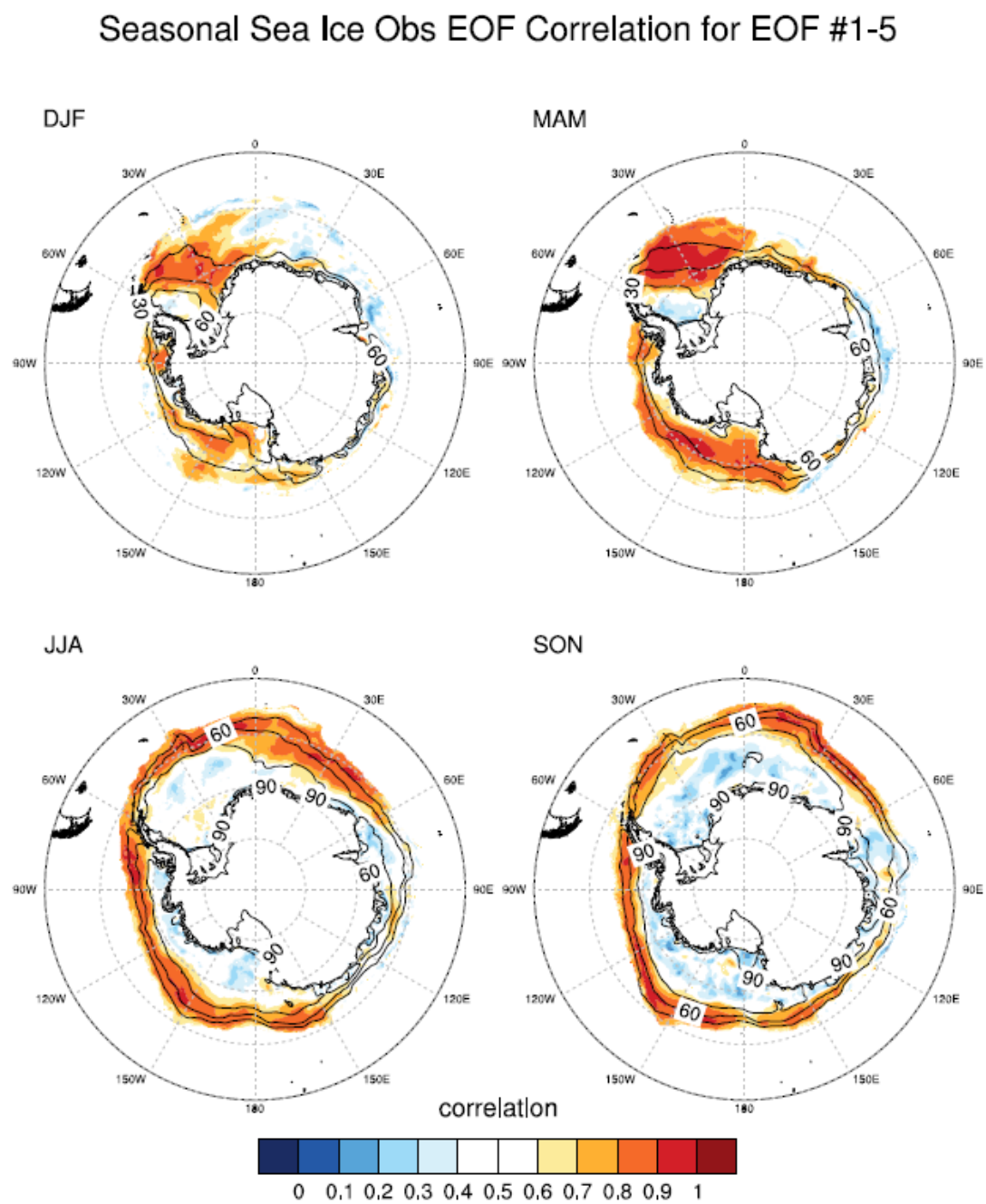
capturing more than ~50% of the year-to-year variability along the ice edge. Thus, the leading five EOFs are an excellent representation of places where the ice is most variable from year to year.

The interior portions of the Weddell Sea around the Antarctic Peninsula observe lower values of correlation, with values of 0.2 - 0.5 throughout DJF and MAM. In JJA and SON, there are noticeable pockets of lower correlation occurring in the interior regions, due to the onset of austral winter and spring allows for the interior areas of the Southern Ocean to become all ice. Importantly, the regions of lower correlation from the five leading EOFs all stem from areas where the climatological (1979-2020) mean ice concentration is >60% (contour lines in Fig. 4.17), representing the pack ice which has much less year to year variability. Thus, the leading five EOF patterns represent the conditions along the ice edge very well, but do not capture the small variations in the pack ice. However, since in JJA and SON the climatological average sea ice concentration exceeds 90% in much of the pack ice (Figs. 4.17), it can be assumed that there is ice present every year within much of the pack ice, and therefore the inability of the leading five EOFs to capture variations in this region is not significant.

Given the high skill of the best fit reconstruction (red boxes in Fig. 4.16), it is expected that the spatial performance of the merged best fit sea ice concentration reconstruction for EOFs 1-5 will look like the correlations depicted in Fig. 4.17. The correlation of the sea ice concentration reconstruction for EOFs 1-5 with the full observed sea ice concentration is shown in Fig. 4.18. The representation of the high correlations at the ice edge, where year to year variations are the strongest and the

Figure 4.17

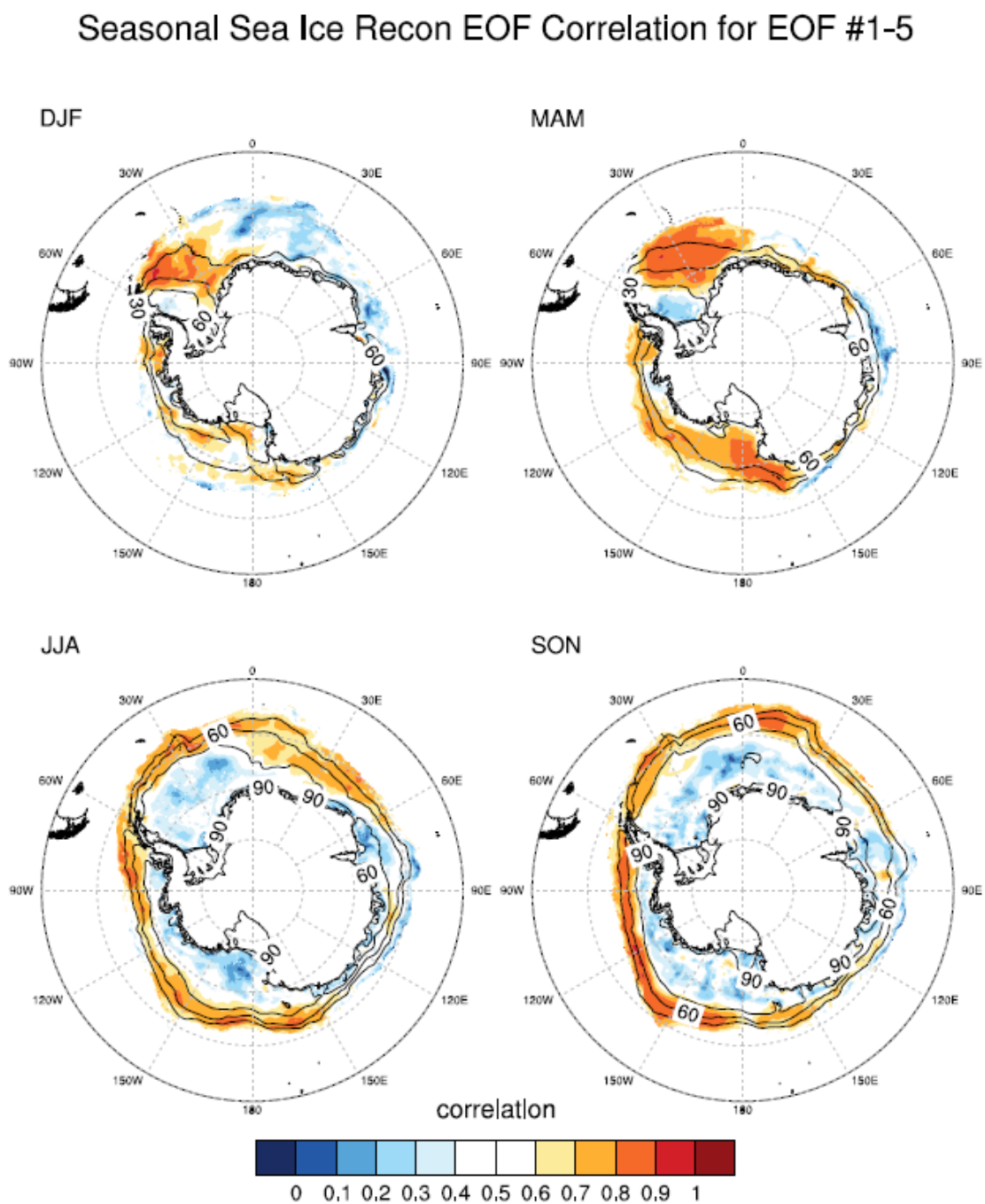
Correlations between Observed Sea Ice Concentration and EOFs 1-5



Note. Correlations between observed and EOFs 1-5.

Figure 4.18

Correlations between Reconstructed Sea Ice Concentration and EOFs 1-5



Note. As in Fig. 4.17, but with reconstructed sea ice concentration instead of observed.

reduced correlations in the pack ice, is well represented (compare Figs. 4.17 and 4.18). The pattern of lower correlations throughout the interior Southern Ocean may at first suggest lower reconstruction performance, however this is an area of similar lower correlation for the observed EOFs (Fig. 4.18). There are remarkable areas of higher correlations extremely close to the Antarctic mainland that can outline polynyas, as discussed in Chapter 2, where sea ice concentrations become more difficult to gather perfect reconstructions due to the variable nature of Antarctic sea ice. In the same figure, the outlined prominence of high correlations occurs again in the Weddell Sea sector during MAM, extending further into the King Hakon region as compared to the observed correlations spatial field (Fig. 4.17). The presence of lower correlations further into the King Hakon region in DJF from the reconstruction (Fig. 4.18) is also shown in observations from the five leading EOFs to the full Antarctic sea ice concentration field (Fig. 4.17).

To provide a clearer representation of the reconstruction skill, the correlations from Fig. 4.18 (shading) are plotted along with contours of the ratio of correlations from Figs. 4.18 (numerator) and 4.17 (denominator). Since Fig. 4.17 represents the highest possible correlation the reconstruction should have with the full observed Antarctic sea ice concentration field, ratio values near one indicate the reconstruction perfectly captures the temporal variability inherent in the first five leading EOFs. Values less than one indicate reconstructions underrepresent the variability present in the leading five EOFs. These results are displayed in Fig. 4.19, which (although a bit noisy) show areas where the reconstruction is producing great skill and where the reconstruction is failing to

produce meaningful results. In MAM, there are overall high values of correlations throughout the Weddell Sea and Ross Sea sectors, with some expansion into the Bellingshausen-Amundsen Sea sector, and correlation ratios (contours in Fig. 4.19) are all above 0.9. This indicates the reconstruction is producing skillful results throughout this season in these sectors; however, there are some areas of the interior Weddell Sea region and into East Antarctica that struggle to produce the same skillful results as the other sectors, but again these are areas of pack ice, and so the ratio still is quite high at 0.7. The reconstruction is therefore capturing 70% of the maximum possible variability from the five leading EOFs, which themselves underrepresent subtle variations within the pack ice.

The pattern of interior correlation struggle in JJA and SON begins to emerge in the correlation between reconstruction and observed sea ice concentrations, where the interior regions of the entire continent do not produce as skillful results indicated by the low correlation, but the ratios often exceed 0.6, indicated even in the pack ice regions the reconstruction is representing similar variability as the observed leading five EOFs. More importantly for a meaningful reconstruction, the presence of increased correlations in conjunction with high correlation ratios is present throughout the outer ring of Antarctica, extended from the Antarctic Peninsula around the continent before reducing closer to the East Antarctic sector. These high correlation ratios can be highlighted into the Ross Sea and King Hakon Sea regions, with some extensions into the Bellingshausen-Ross Sea sector, especially noticeable during SON. The lower performance of sea ice concentration reconstruction around East Antarctica is tied to the fact that the EOFs

relating to East Antarctic sea ice concentration variability do not emerge strongly until EOF 4 or later (Figs. 4.4, 4.5), and the skill is overall lower for these EOFs than for higher EOFs (Figs. 4.11 – 4.15).

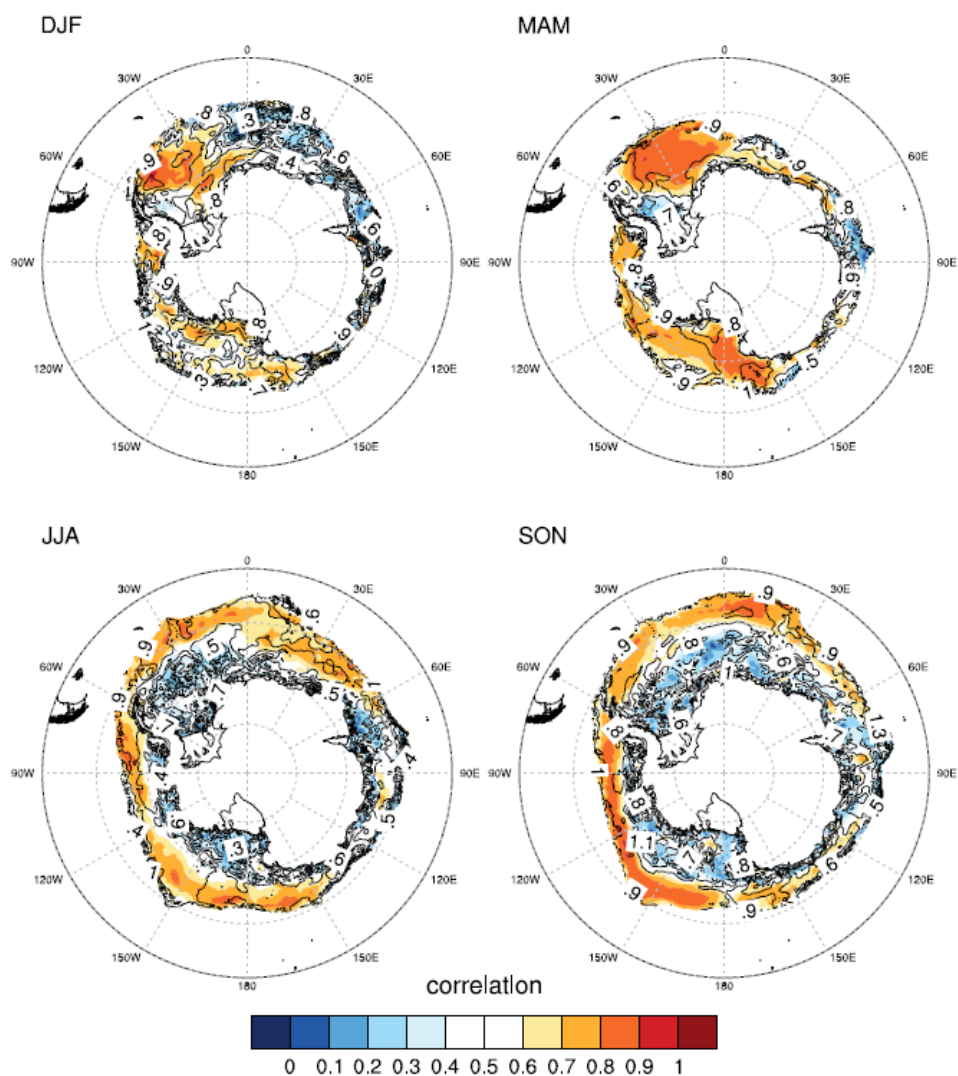
Into DJF, there are more patchy areas of high and lower correlation in Fig. 4.19 compared to the broader regions of correlations in MAM, and to a lesser extent JJA and SON. The spatial variations of these correlations are tied to the fact that there tends to be little Antarctic sea ice in DJF (see contours of climatological ice concentration in Figs. 4.17 and 4.18 for DJF). In many years, much of the Antarctic coastline is ice-free in DJF, making the reconstruction of a discontinuous variable a challenge, even when represented by EOF patterns instead of a full spatial field. There is an overall higher amount of correlation throughout the Weddell Sea region, mirroring trends in MAM, especially near the Antarctic Peninsula, along with stretches into the furthest most regions of the King Hakon Sea sector – since these are areas of higher climatological ice concentration. Further into the King Hakon Sea region, lower correlations in DJF begin to dominate before continuing to stay towards lower values of correlations into East Antarctica. The Bellingshausen-Amundsen Sea and Ross Sea sectors overall have moderate correlation values, with pockets of high correlations occurring throughout the region. The fact that the reconstruction skill drops markedly for lower EOFs in DJF (Figs. 4.11-4.15) reflects some of these lower correlations in Fig. 4.19, along with the large year-to-year variability in where ice occurs around Antarctica in austral summer.

The ultimate result of these spatial plots of Antarctic sea ice reconstruction skill (Figs. 4.18-4.19) is that the spatial performance of reconstructions tends to be

Figure 4.19

Ratio of Correlations between Reconstructed Sea Ice Concentration and Observed

Seasonal Sea Ice Recon EOF Correlation for EOF #1-5, With Ratio to Obs



Note. The ratio of reconstructed to observed sea ice concentration correlations in relation to EOFs, where zero indicates no correlation and one indicates strongly positive correlations.

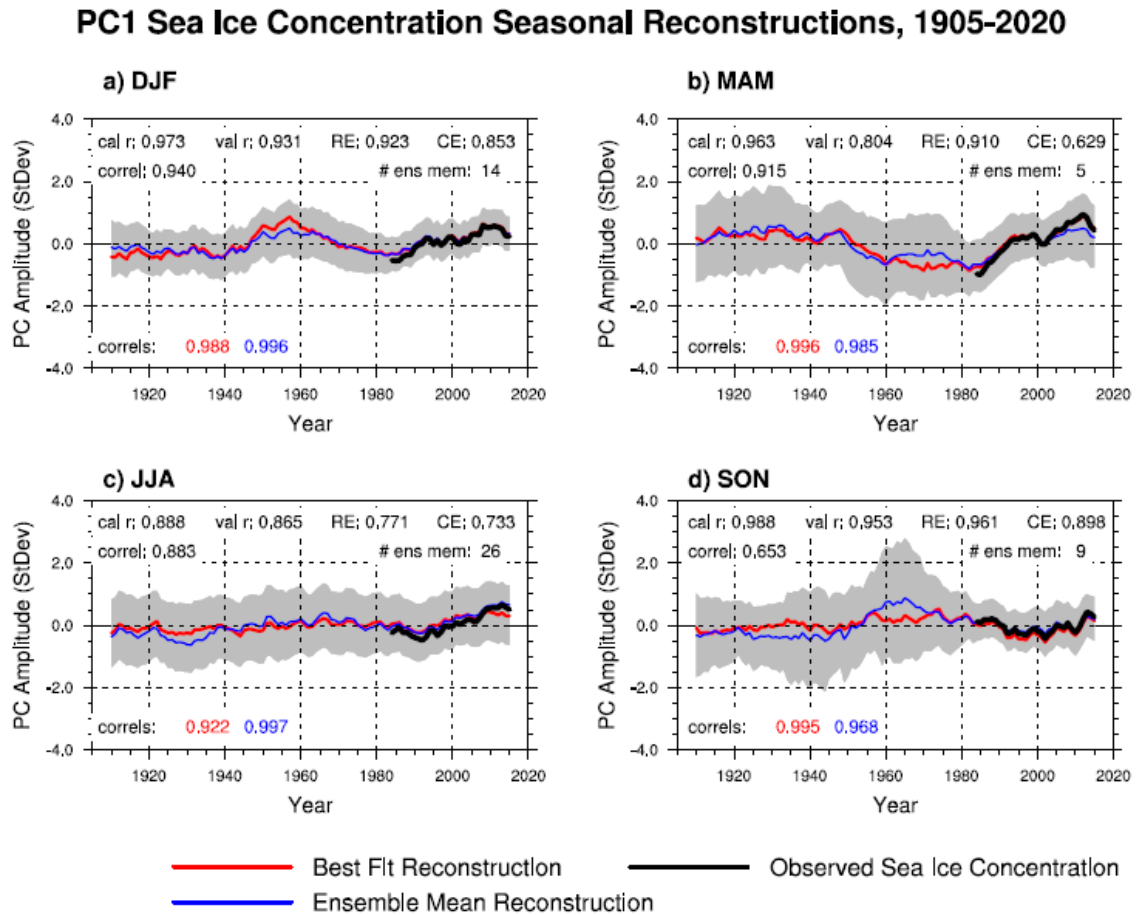
remarkably high throughout the ice edge in the Weddell Sea, Bellingshausen-Amundsen Sea, and Ross Sea sectors throughout much of the year. The skill is lower in the pack ice, but most of these regions can be assumed to be ice covered each year. The King Hakon Sea sector tends to yield lower values, along with East Antarctica, with higher correlations tending to occur throughout JJA and SON around the outer regions of these sectors, extending from the other sectors of Antarctica rather than having isolated pockets of higher correlations throughout the sector.

4.7 Smoothed Reconstructions through Low Frequency Plots

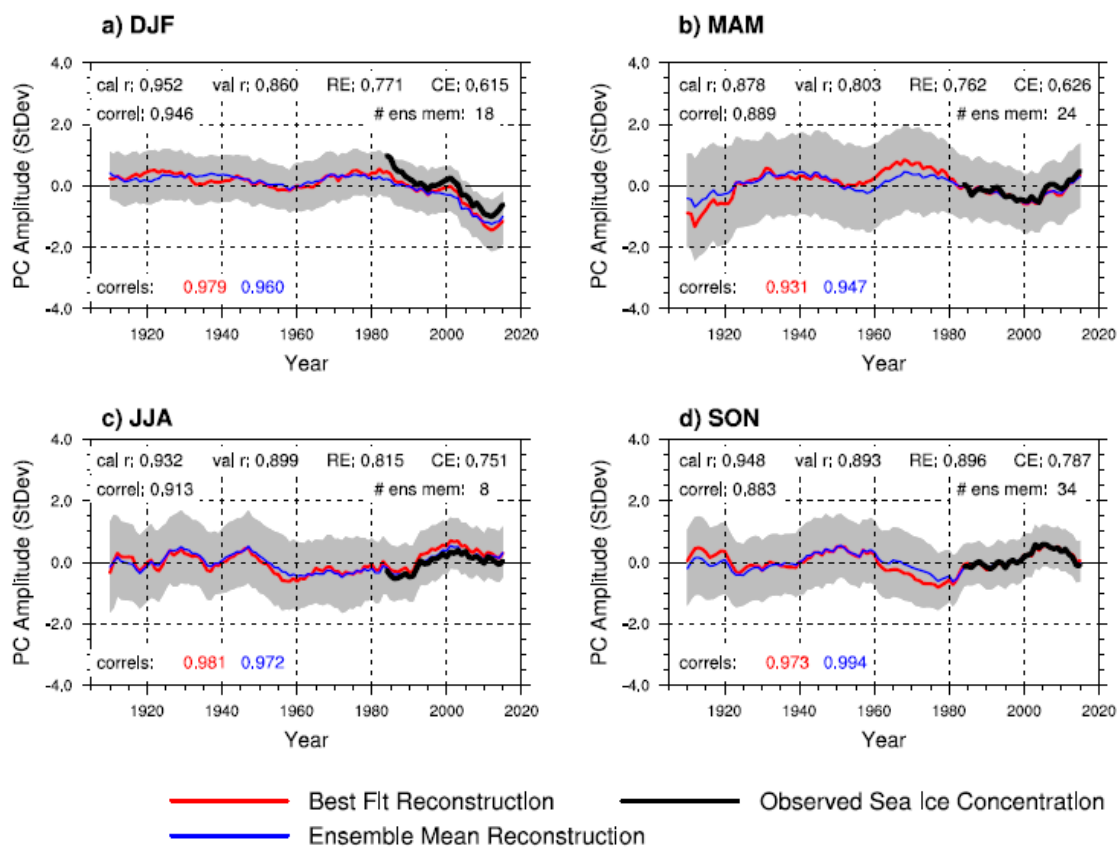
The reconstruction skill assessed thus far has focused on the performance in interannual variability; however, changes in Antarctic sea ice can operate on slower timescales, associated with shifts in the modes of climate variability discussed in chapter 2. As such, it is helpful to investigate the reconstruction skill on lower frequencies than interannual variations, especially since the reconstructions were not explicitly calibrated to resolve low frequency variability. In this section, low frequency plots, used to depict low frequency reconstructions smoothed with an eleven-year running mean along with a 95% confidence interval, are investigated to gain insight on long term changes in sea ice concentrations along with skill in these low frequency variations of sea ice concentration. These plots, one for each PC on seasonal scales, outline the best fit reconstruction, ensemble mean reconstruction, the 95% confidence interval, along with the (smoothed) observed sea ice concentration. These plots also use correlations between the eleven-year smoothed versions of the observed and the best fit/ensemble mean reconstructions, along with the skill metrics as explained in the timeseries plots in section 4.4.

In the PC1 smoothed reconstruction plot (Fig. 4.20), the confidence interval maintains close to the best fit reconstruction throughout DJF and JJA, with correlations between the observed and best fit being 0.988 and 0.922, respectively, indicating high correspondence between the best fit reconstruction and observed sea ice concentration throughout the observed period. The confidence interval spreads out throughout MAM and SON; however, correlations between the best fit reconstruction and observed concentration still are remarkably high (nearly 1.0). The long-term changes in PC1 remain small, with a general range of no more than 1.0 throughout the period of around 1910-2020, especially in JJA where the values of sea ice concentration / PC1 amplitude stay increasingly close to zero, with very few notable variations throughout the entire period. In SON, there is quite a difference in low frequency variation between the ensemble mean and best fit prior to 1979, with the correlation being weak (considering the smoothing) with a value of 0.653 between the two timeseries overall.

In looking at the PC2 plot (Fig 4.21), the correlations between the smoothed best fit reconstruction and the observed sea ice concentration remain high (ranging from 0.931 to 0.981) through all seasons, with especially narrow confidence intervals in DJF, SON, and to a lesser extent JJA. The low frequency changes throughout PC2 tended to follow patterns of PC1, where long term changes were not as extreme; however, there was noticeable decreases in PC amplitude in DJF starting around 1990, where steady declines into approximately 2017 occurred before rebounds into 2020. This period of a dip in PC amplitude was only prevalent in DJF patterns, indicating climate modes influencing the EOFs differently.

Figure 4.20*Low Frequency Timeseries of PC1*

Note. A smoothed reconstruction timeseries of PC1, where the red line indicates the best fit reconstruction, the blue line indicates the ensemble mean reconstruction, the black line indicates observed sea ice concentration, and the gray shaded area represents the 95% confidence interval between all ensemble members. The skill metrics at the top of each panel are for the interannual reconstruction, while the ‘correl’ value is for the low frequency cross correlations between the best fit and ensemble mean reconstructions during 1905-2020. The ‘correls’ values at the bottom are the correlations of the smoothed observed and best fit reconstruction (red) and ensemble mean (blue) during 1979-2020.

Figure 4.21*Low Frequency Timeseries of PC2***PC2 Sea Ice Concentration Seasonal Reconstructions, 1905-2020***Note.* As in Fig. 4.20, but for PC2.

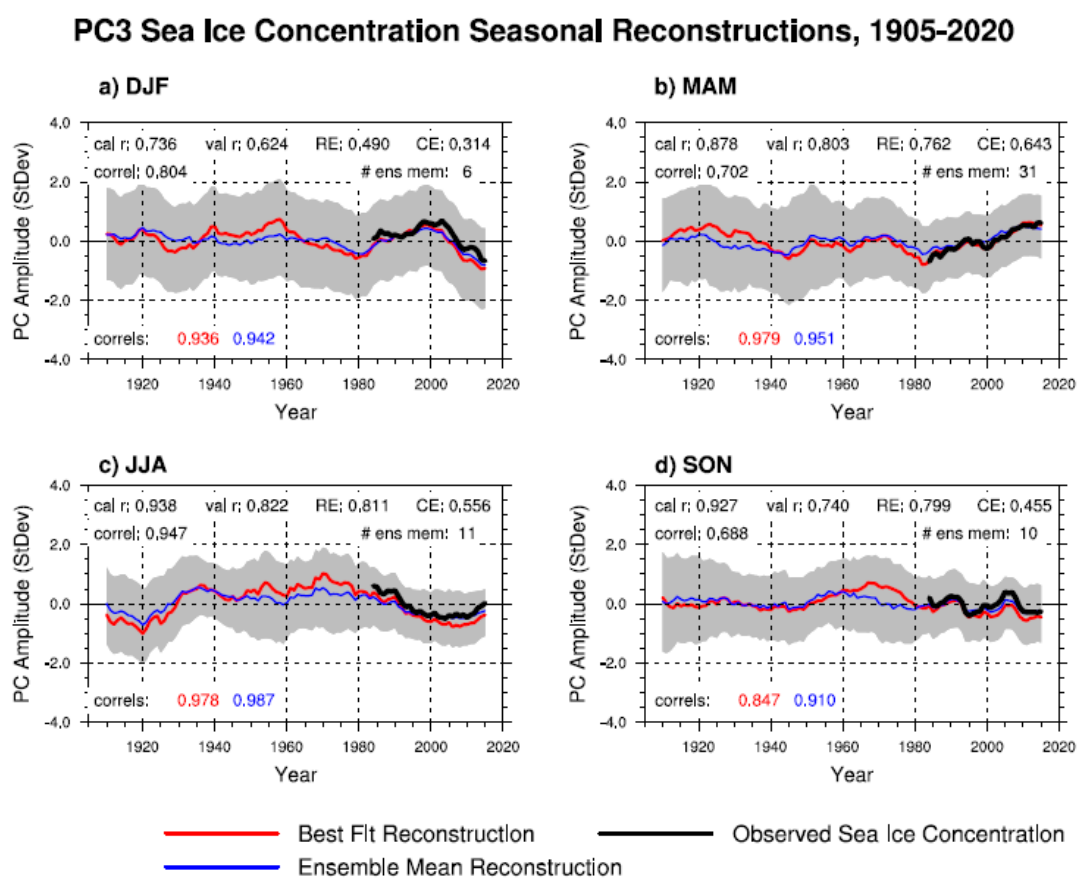
Through the analysis of a smoothed timeseries of PC3 (Fig. 4.22), the increased uncertainty in the reconstruction begins to become more prevalent with the confidence interval beginning to widen throughout all seasons, especially in DJF and MAM throughout much of the time from 1905-2020. The correlations between the best fit reconstruction and the observed sea ice concentration, however, tends to be remarkably high still throughout the plots. This could be misleading, as the correlation between the two lines in is lower in all but JJA, suggesting quite different low frequency variability in the individual ensemble members. In JJA, there is considerable skill in depicting the low frequency variations, with more pronounced low frequency shifts in PC amplitude for PC3 in austral winter. The long-term changes in PC3 overall tend to be slightly more extreme compared to the first two PCs, especially with the decreases in DJF, along with the large jumps in sea ice concentration in 2010 for SON. However, ranges tend to be like the previous two PCs.

In Fig. 4.23, the smoothed reconstruction timeseries of PC4 outlines higher uncertainty throughout all seasons, apart from MAM when confidence intervals tend to remain narrow. The correlations between the best fit reconstruction and the observed sea ice concentration data are remarkably high (0.919), but again in MAM the correlations between the ensemble mean and best fit reconstruction are a bit lower (0.797), and above 0.9 in all other seasons. The most noticeable low frequency change is in 2014 for SON, where there is an increase in PC amplitude that is also seen in the JJA time series as well. PC4 shows overall more significant long-term changes, with higher ranges of sea ice

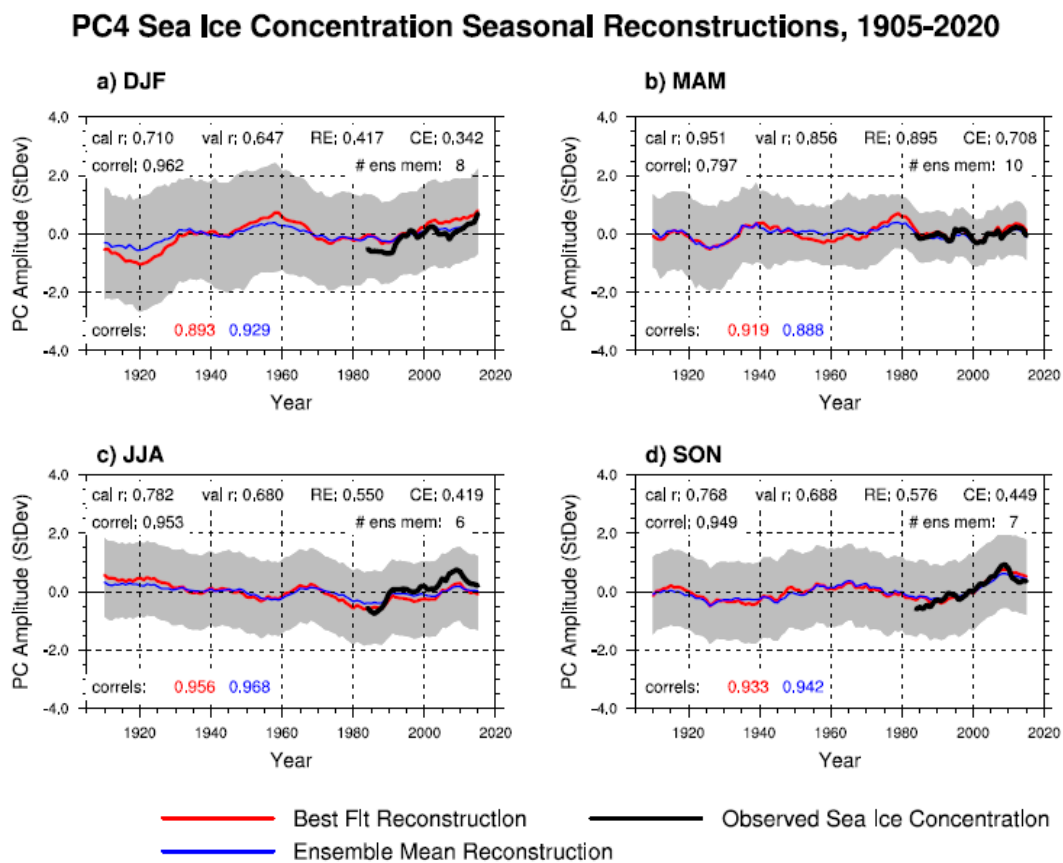
concentration values / PC amplitude throughout the period plotted in the overall time series.

Figure 4.22

Low Frequency Timeseries of PC3

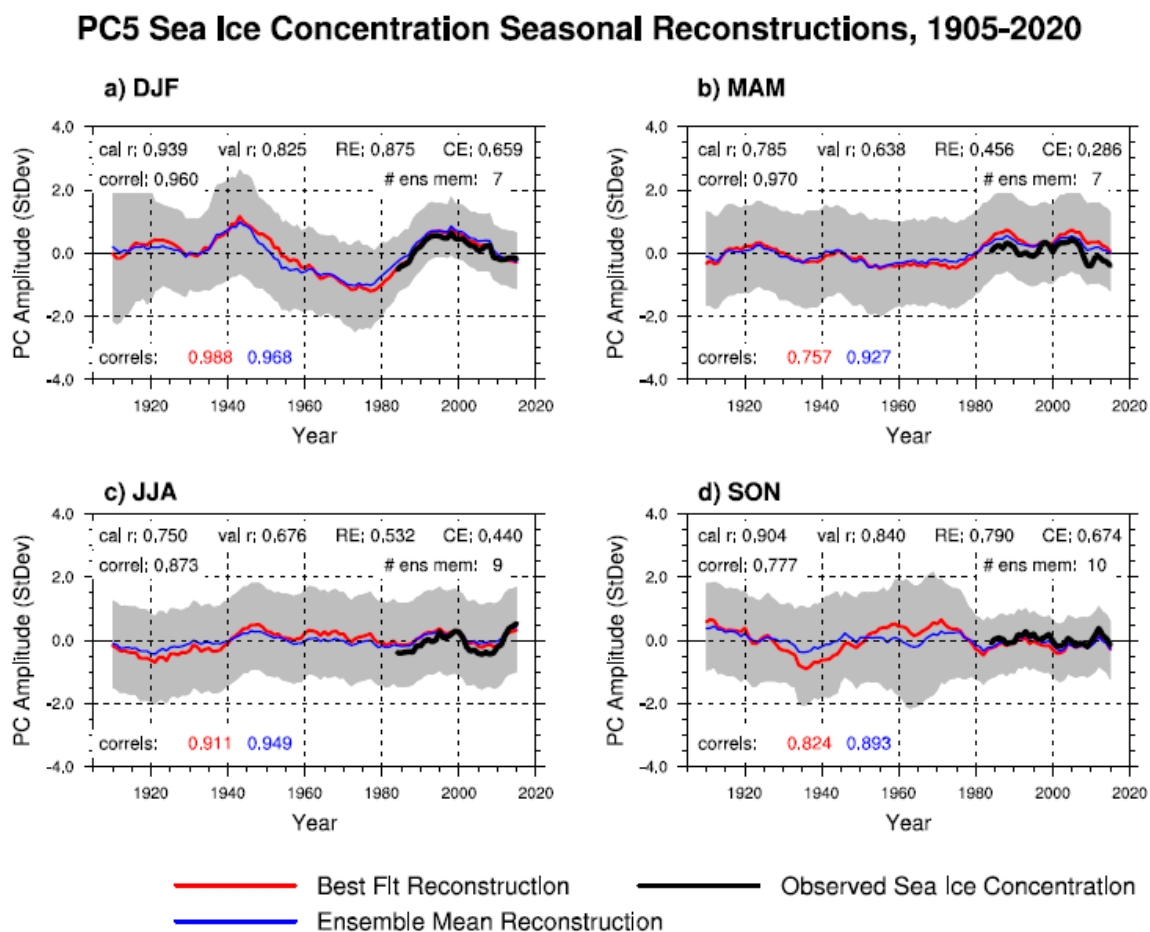


Note. As in Fig. 4.20, but for PC3.

Figure 4.23*Low Frequency Timeseries of PC4**Note.* As in Fig. 4.20, but for PC4.

The final low frequency PC time series (Fig. 4.24) outlines much higher long-term changes, especially noticed in DJF where values shift from a maximum value of positive one to negative 1.5 throughout a period of less than forty years. This change occurs amongst higher certainty than compared to the rest of the seasonal plots for the same PC and is consistent between the best fit and ensemble mean reconstructions. These long-term changes are mirrored in SON in the best fit reconstruction, although at much smaller magnitude with less certainty in the reconstruction. The correlations between the best fit and observed have noticeably dropped in MAM and SON, while being retained in DJF and JJA, with high values of correlations being observed with some high values of reconstruction uncertainty.

The presence of these smoothed reconstruction timeseries plots allowed for the skill in low frequency situations, where there was an increased skill in these situations while being able to observe long term changes and the skill in reconstructions in reproducing these long-term changes. These long-term changes being able to be accurately depicted in the reconstruction allows for changes to be observed throughout the period of 1905-2020 with greater confidence, given that the reconstructions reproduce both interannual variations and low frequency shifts in PC amplitude. These changes are analyzed in greater detail in the following sections.

Figure 4.24*Low Frequency Timeseries of PC5**Note.* As in Fig. 4.20, but for PC5.

4.8 Sea Ice Trends

Now that the reconstructions have been determined dependable, it is possible to conduct preliminary analysis on them to understand changes in Antarctic sea ice concentrations since 1905, longer than previously ever estimated. There is a particular interest in linear trends of sea ice concentration, given a companion study by Fogt et al. (2022) determined a notable regime shift in Antarctic sea ice extent trends in the mid-20th century. Here linear trends in the reconstructed Antarctic seasonal sea ice concentration PCs are examined along with their 95% confidence intervals in Table 4.3. These trends were calculated for the entire period of 1905-2020 to observe overall trends over all five EOFs, before being broken into pre-satellite era trends (1905-1978) and satellite era (1979-2020). This allows for fifteen trend values for each season (three for each EOF, with five EOFs), or a total of sixty sea ice trends. These values are reflected in Table 4.3 with *, **, and *** representing trends that are statistically different from zero at $p < 0.1$, $p < 0.05$, and $p < 0.01$, respectively. The trends denoted in red text are significant at $p < 0.01$.

The austral summer (DJF) had a total of ten statistically significant trends (Table 4.3), with six of these ten trends being significant at $p < 0.01$. Strongly negative trends were observed in PC2 over the entire period along with the post-satellite era. The presence of strongly positive trends over the entire period, pre-satellite era, and post-satellite era appears in PC4, with the highest trends occurring in the post-satellite era. The final strongly significant trend in DJF was observed in the pre-satellite era for PC5, where strongly negative trends were observed. There were also significant trends at $p <$

0.05 throughout the entire period for PCs 1 and 3, with these trends emerging in the pre-satellite era for PC1, and post-satellite era for PC3. Positive trends in the seasonal sea ice PCs reflect changing sea ice concentration conditions that align more prominently with the sea ice anomalies for each EOF plotted in Figs. 4.1 - 4.5.

The next season analyzed in Table 4.3 is the austral winter (JJA), where seven out of the fifteen trends showed significant trends, with only three of these being significant at $p < 0.01$. In JJA, strongly negative trends were observed in the overall trends in PC4 as well as in the pre-satellite era. The other strongly significant sea ice trend appeared in PC3, where strongly positive trends were observed in the pre-satellite era. Three additional trends were significant to $p < 0.05$, with positive overall trends in PC5 reaching this significance level stemming from the pre-satellite trends (significant to the same level), along with negative trends in the post-satellite era of PC3. Along with these significant trends, the overall trend in PC1 was weakly significant ($p < 0.1$), where significant positive trends were observed.

In the austral autumn (MAM), there were a total of five trends deemed significant out of the fifteen total, with four of these trends being significant at $p < 0.01$, with PC1 showing a strong negative trend in the pre-satellite era before changing into strongly positively trended in the post-satellite era. The pre-satellite era in PC2, along with the post-satellite era in PC3, were marked with strongly significant positive trends throughout their timeframes. The only form of statistical significance outside of these strongly positive and negative trends that were significant at $p < 0.01$ occurred in the overall trends from 1905-2020 for PC5.

The final season, the austral spring (SON), had the least amount of statistically significant trends, with four of fifteen trends being deemed statistically significant. The only form of a strongly positive or negative trend ($p \ll 0.01$) occurred in the satellite period of PC4, with strongly positive trends occurring. This positive trend also extended into the total trends; however, this trend was only moderately significant in comparison, being significant to $p < 0.05$. There were two additional statistically significant trends, significant to $p < 0.10$ in the pre-satellite era for PC1 and PC3.

The combination of all seasons shows statistically significant trends in twenty-six of the sixty sea ice trend patterns, or 43.33%, with fourteen of the twenty-six being statistically significant at $p \ll 0.01$, indicating these trends have less than a 1% chance of there being no trend present. The austral winters and summers (DJF and JJA, respectively) tend to keep trends positive or negative throughout PCs, while MAM and SON see changes in the post-satellite, pre-satellite, and overall trend eras. This information combined exhibits that sea ice concentration reconstructions can produce valuable information regarding trends within the four seasons by EOF – and suggests that there are numerous changes in the sea ice concentration anomalies throughout the 20th century, which will be explored further in the following sections.

Table 4.3*Sea Ice Trends $\pm$ 95% Confidence Intervals*

DJF			
PC	1905-2020 Trend	1905-1978 Trend	1979-2020 Trend
1	0.063 $\pm$ 0.053**	0.127 $\pm$ 0.103**	0.195 $\pm$ 0.246
2	-0.126 $\pm$ 0.049***	-0.034 $\pm$ 0.081	-0.602 $\pm$ 0.237***
3	-0.061 $\pm$ 0.053**	-0.044 $\pm$ 0.113	-0.248 $\pm$ 0.208**
4	0.142 $\pm$ 0.048***	0.188 $\pm$ 0.096***	0.390 $\pm$ 0.195***
5	-0.032 $\pm$ 0.054	-0.181 $\pm$ 0.103***	0.087 $\pm$ 0.235
JJA			
1	0.050 $\pm$ 0.054*	0.064 $\pm$ 0.100	0.173 $\pm$ 0.271
2	0.034 $\pm$ 0.054	-0.039 $\pm$ 0.113	0.135 $\pm$ 0.216
3	-0.007 $\pm$ 0.055	0.196 $\pm$ 0.100***	-0.251 $\pm$ 0.218**
4	-0.104 $\pm$ 0.051***	-0.181 $\pm$ 0.089***	0.171 $\pm$ 0.262
5	0.068 $\pm$ 0.053**	0.127 $\pm$ 0.110**	0.111 $\pm$ 0.218
MAM			
1	-0.016 $\pm$ 0.055	-0.188 $\pm$ 0.100***	0.380 $\pm$ 0.217***
2	0.037 $\pm$ 0.054	0.198 $\pm$ 0.108***	0.031 $\pm$ 0.205
3	-0.006 $\pm$ 0.055	-0.073 $\pm$ 0.102	0.483 $\pm$ 0.225***
4	0.032 $\pm$ 0.054	0.066 $\pm$ 0.093	-0.078 $\pm$ 0.300
5	0.073 $\pm$ 0.053***	-0.076 $\pm$ 0.099	-0.101 $\pm$ 0.226
SON			
1	-0.004 $\pm$ 0.055	0.083 $\pm$ 0.093*	0.024 $\pm$ 0.298
2	-0.009 $\pm$ 0.055	-0.091 $\pm$ 0.108	0.119 $\pm$ 0.240
3	-0.041 $\pm$ 0.054	0.086 $\pm$ 0.094*	-0.126 $\pm$ 0.279
4	0.065 $\pm$ 0.053**	0.066 $\pm$ 0.104	0.390 $\pm$ 0.230***
5	-0.028 $\pm$ 0.054	0.012 $\pm$ 0.110	0.022 $\pm$ 0.236

Note. Cells with * represent $p < 0.1$, ** represent $p < 0.05$, *** represent $p < 0.01$, red text indicates $p < 0.01$. The units of the trends are standard deviations per decade.

4.9 Time-Varying Trends

While Table 4.3 is a great way to first examine the added value of the longer reconstructions in sea ice concentration changes since the early 20th century, it could easily mask shorter-term trends that occurred over different time periods. To understand how the trends vary throughout time, the creation of time-varying trend plots can assist in

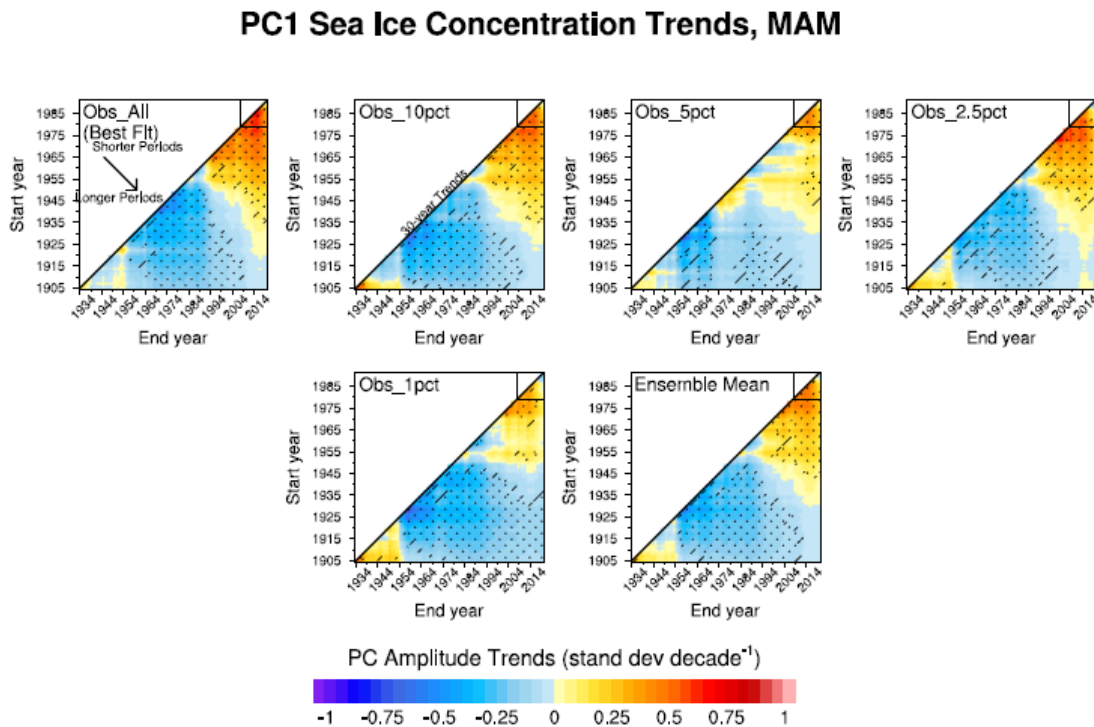
being able to assess the trends in across each PC reconstruction ensemble members, along with the statistical significance of these trends through time. By looking at time-varying trends for PC1 for MAM (Fig. 4.25), a notable shift in the trend sign occurs throughout the 20th century, partially reflected in Table 4.3: trends were strongly negative across all ensemble members throughout the early 20th century, and then become strongly positive just prior to and including the modern satellite era (represented as the small boxes in the upper right corners of each plot). This pattern of trend reversal is present in other PC1 trends where declines in concentration are observed around 1960 and prevalent across all ensemble members, ultimately suggesting that the regime shift in total Antarctic sea ice extent discussed by Fogt et al. (2022) is prominent in the pattern from EOF1. As EOF1 is strongly related to Pacific Ocean variability and/or the SAM (excluding MAM), this further implies that the regime shift in Antarctic sea ice trends has a marked tropical component.

In contrast, an additional time-varying trend plot for JJA in PC3 (Fig. 4.26), there is considerable variability between the ensemble reconstructions, with no clear and consistent regime shift, and less agreement between the best fit reconstruction and the ensemble mean. The best fit reconstruction shows positive PC trends only for a short period between the mid-1940s – early 1990s; otherwise, the best fit reconstruction shows weakly negative trends that become stronger (and more statistically significant) for periods ending after 2000. Other ensemble members may suggest a regime shift from positive trends in the early 20th century to negative trends in the later part, but the positive trends are weak in many members, and only marginally significant in the ensemble mean

($p < 0.10$). Overall, PC3 highlights some dependence on the ensemble member in terms of the changes in sea ice concentration in this season, making it much harder to interpret this PC through time.

Figure 4.25

PC1 Time Varying Trends for Sea Ice Concentration Trends for MAM



Note. The starting year for linear trends is given on the y-axis, and the ending year on the x-axis. The solid diagonal black line indicates thirty-year trends, with shorter periods closer to the top left of the plot and longer periods to the bottom right of the plot. Cross-hatching represents trends statistically different from zero at $p < 0.10$, while stippling indicates trends statistically different from zero at $p < 0.05$.

A final example of even more deviation between the ensemble members is provided by the time-varying trend plot for DJF PC4 (Fig. 4.27). In this reconstruction

the ‘Obs_All’ member is very distinct from the remaining members. The ensemble reconstructions that include the AMO show decadal variability that is consistent with phase shifts in the AMO. Since the AMO 5% network ensemble member was selected as the best fit reconstruction, it is more likely that this PC changes in response to shifts in the Atlantic Ocean, which are not fully captured by the observational network. Further, the lower skill of higher numbered (PC3-5) sea ice concentration PC reconstructions in DJF further complicate the interpretation of the time-variations in the trends in this PC. It is interesting also that many of the trends in Fig. 4.27 become maximized close to the diagonal line (representing exactly 30-year period), suggesting short-term variability that could easily be influenced by outliers.

4.10 Spatial Reconstruction Trends

Additional work beyond this thesis is needed to interpret the time-variations in the trends across different PCs and in different seasons. However, the PC reconstructions can be merged to give a full time-varying spatial field of Antarctic sea ice concentration (as examined before in the spatial reconstruction skill, section 4.6). For this project, a spatial depiction of sea ice concentration trends will be plotted for three different time periods; one from 1979-2016 to analyze trends before the 2016 Antarctic sea ice loss event explained in Chapter 2, along with the total trends from 1979-2020, before extending the reconstruction to the pre-satellite era of 1905-1978. The analysis of these three time periods will be able to make deductions on the skill of the reconstructions pertaining to sea ice concentration and investigate regional patterns of Antarctic sea ice concentration on much longer timescales than previously done.

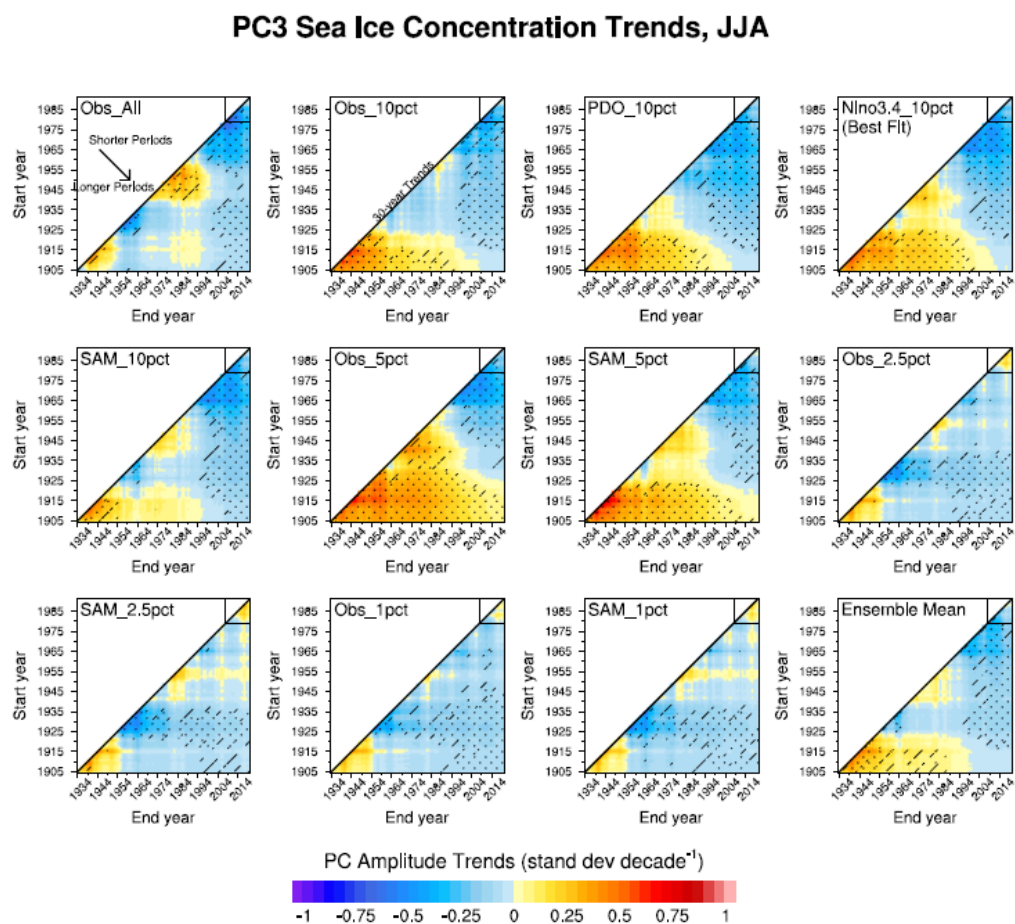
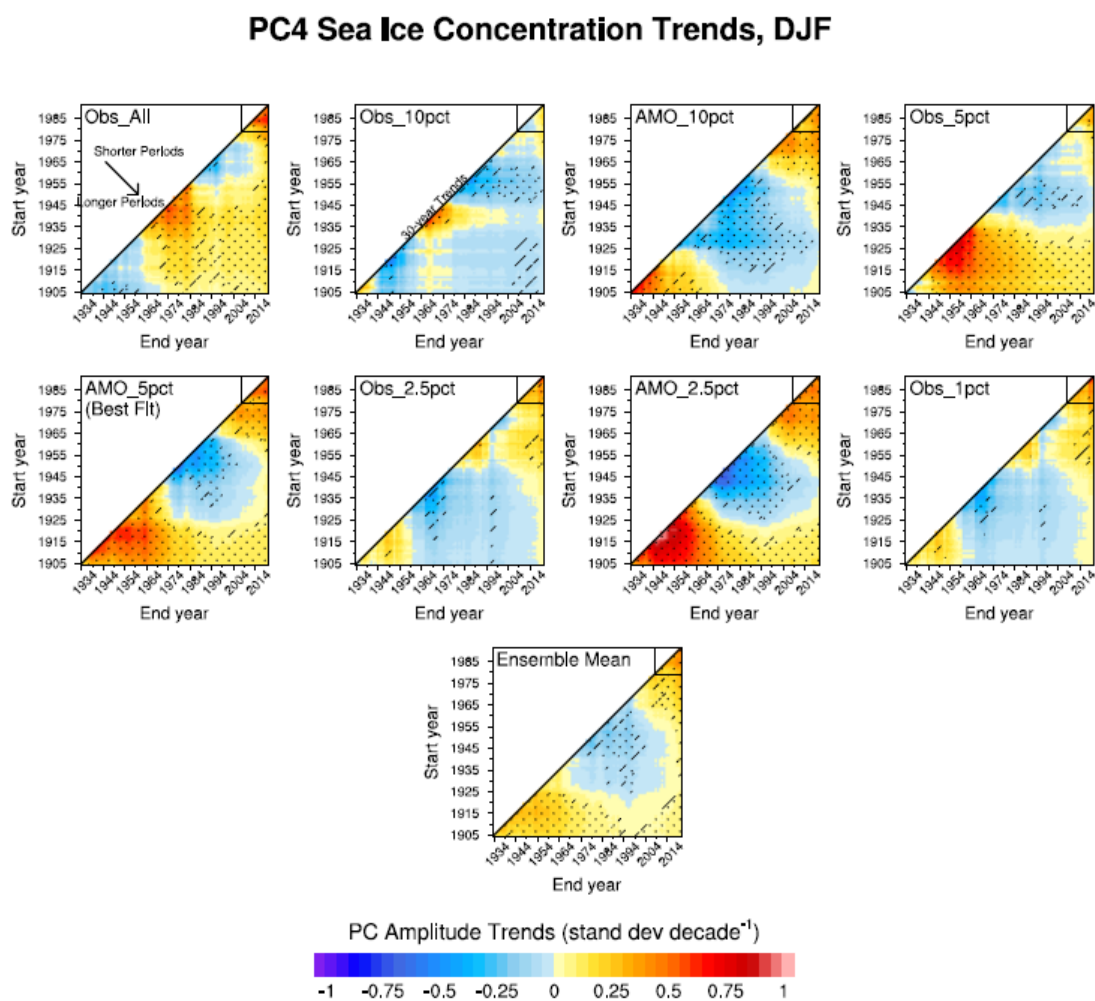
Figure 4.26*PC3 Time Varying Trends for Sea Ice Concentration Trends for JJA**Note.* As in Fig. 4.25, except for JJA in PC3.

Figure 4.27*PC4 Time Varying Trends for Sea Ice Concentration Trends for DJF**Note.* As in Fig. 4.25, except for DJF in PC4.

In the time post-satellite era, yet prior to the 2016 event, (Fig. 4.28) Antarctic sea ice concentration trends were strongly positive throughout the Weddell Sea in DJF. This pattern continues, albeit slightly weaker, in MAM, with a similar positive trend observed in the Ross Sea. There were also slightly positive trends into East Antarctica and the King Hakon Sea regions. In contrast, there are significant negative trends in the Bellingshausen-Amundsen Sea sectors that slightly extend further into the eastern Ross Sea sector for the same timeframe. In DJF and MAM, many of these trends are significantly different from zero at $p < 0.05$.

Throughout the same timeframe, in JJA and SON increases in Antarctic sea ice concentration / positive trends extend outward from the Ross Sea sector and East Antarctica sector. These trends are especially significant ($p < 0.05$) in JJA, with some small areas of significance in SON, albeit for trends at a much lower magnitude. The statistical significance of the trends in the pack ice region poleward from the ice edge is not very meaningful – the ice concentration is very high here continuously (see contours in Fig. 4.17), and the reconstruction skill is lower as well (Fig. 4.18). There are also slightly positive trends further off into the King Hakon Sea sector with significant trends in JJA, with slightly less area and significance in SON.

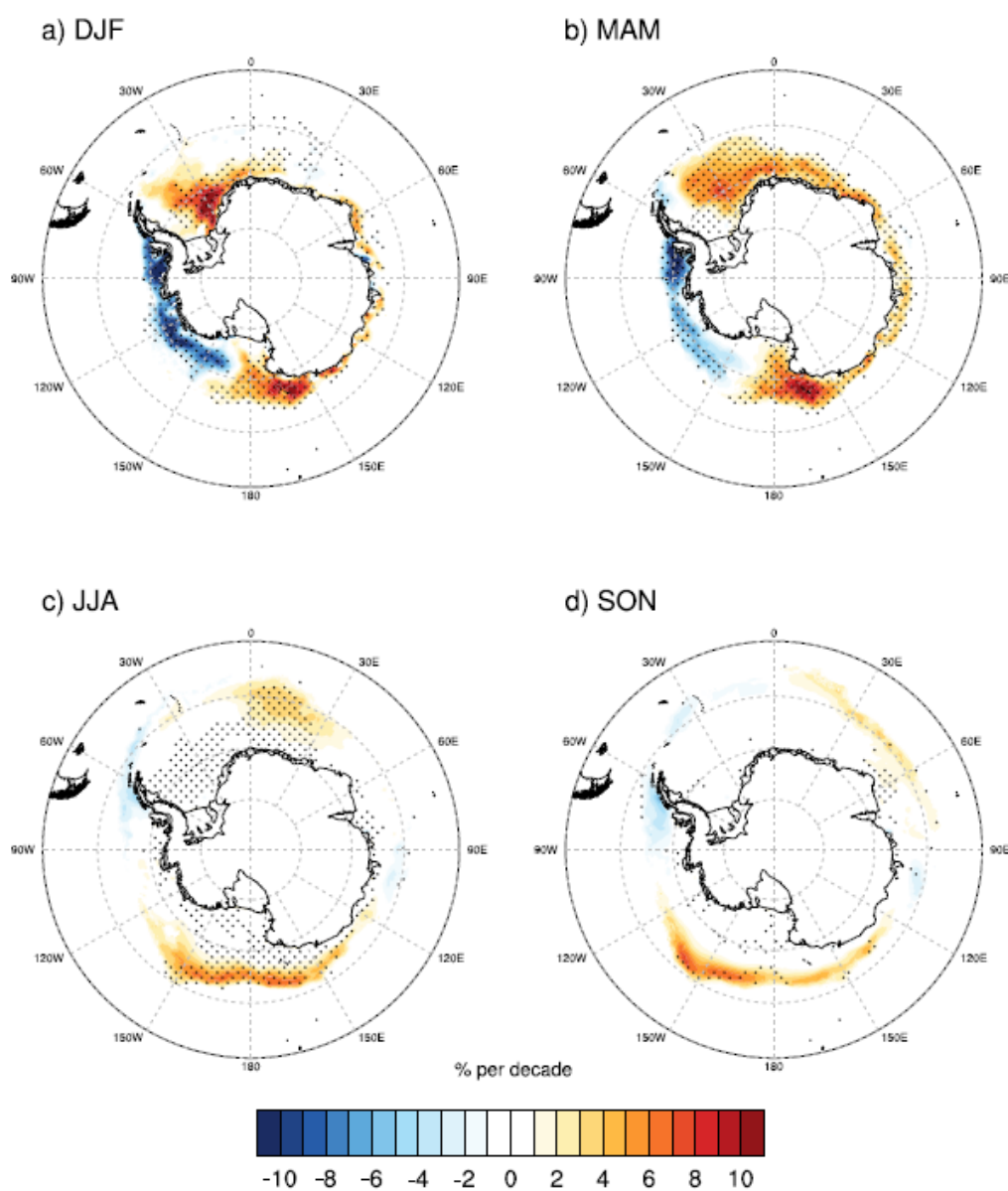
When considering the 2016 event and extending the trends further into 2020 (Fig. 4.29), the positive trends in the Weddell Sea become significantly less prominent and lose a little of their statistical significance, with this loss in positive trends becoming much more emergent into MAM, where most of the statistical significance in this sector is erased and the positive trends drop significantly in this sector. There is also a

significant drop in MAM trends throughout the Ross Sea sector; while statistical significance is maintained, the trend magnitudes still dropped noticeably with the addition of a few more years. The other seasons, however, maintained most of their normal characteristics from the 1979-2016 spatial plots, maintaining statistical significance with slight change throughout the trends themselves. These trends closely match the observed trends (especially for trends that are captured in the first five EOFs), indicating that the reconstructions can depict realistic trends spatially. This provides greater confidence when examining these trends back through the early 20th century prior to satellite observations.

The presence of these statistically significant trends that were able to reconstruct sea ice concentration trends throughout the observed post-satellite era can then be applied to pre-satellite era, such as in Fig. 4.30 for the period 1905-1978. In contrast to the observed pattern in Fig. 4.29, Antarctic sea ice concentrations prior to the satellite era show a marked decline in MAM and JJA in most regions, except for increases in the Bellingshausen Sea (Fig. 4.30). This pattern confirms the regime shift in Antarctic sea ice, discussed by Fogt et al. (2022) for sea ice extent, and hinted at in the time-varying trends for EOF1 (Fig. 4.25). Specifically, the sea ice concentration reconstruction shows some modest increases throughout the Weddell Sea sector, particularly close to the Antarctic Peninsula, with some areas close to the Antarctic continent in the Weddell Sea and King Hakon Sea sectors in DJF. In the austral autumn (MAM), the reconstruction shows increased positive trends near the Antarctic Peninsula in the Bellingshausen-Amundsen Sea sector with decreasing trends throughout the Ross Sea and King Hakon

Figure 4.28*Spatial Seasonal Sea Ice Concentration Trends for 1979-2016*

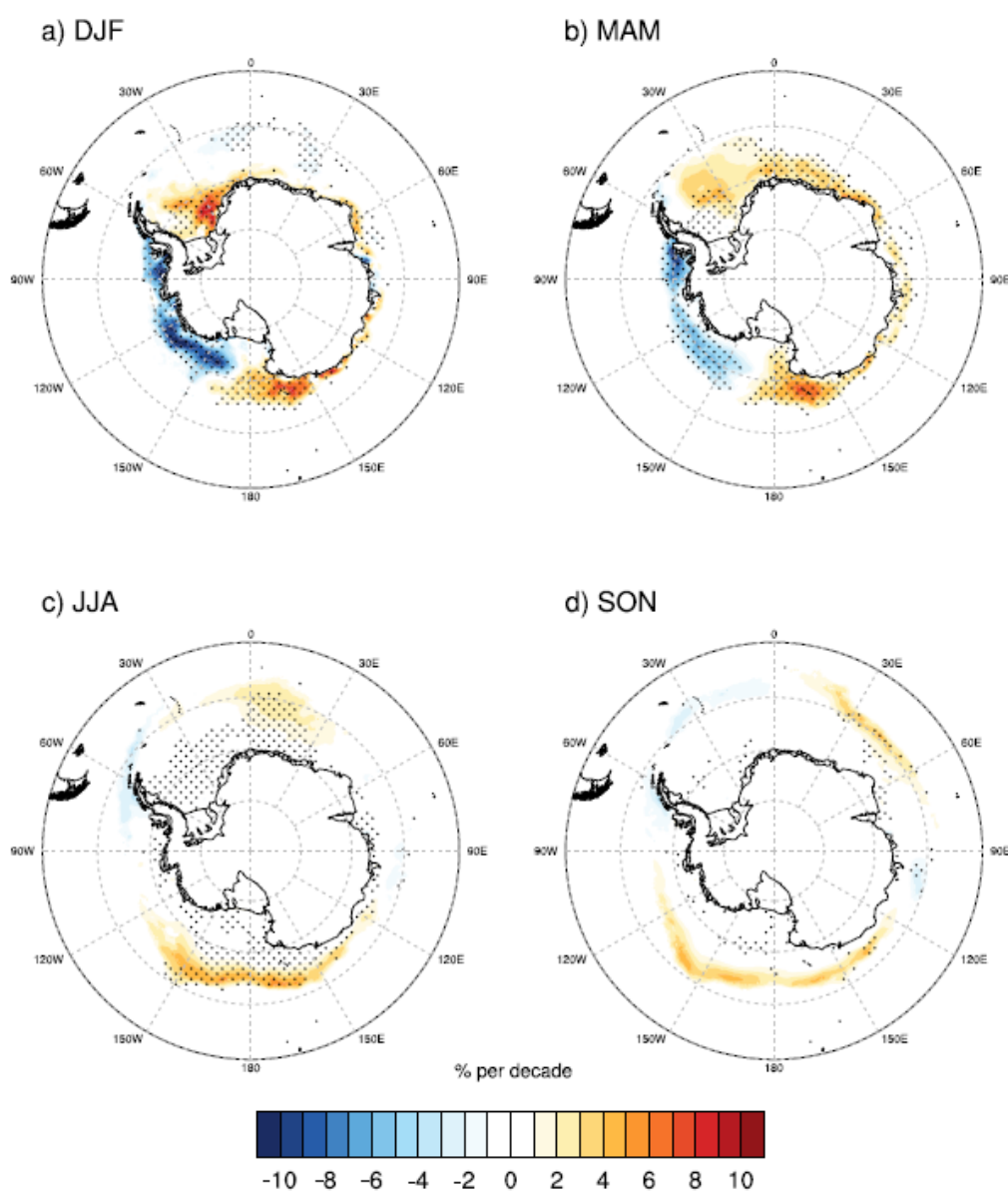
Sea Ice Concentration Trend, 1979-2016



Note. Stippling indicates trends that are statistically different from zero at $p < 0.05$.

Figure 4.29*Spatial Seasonal Sea Ice Concentration Trends for 1979-2020*

Sea Ice Concentration Trend, 1979-2020

*Note.* As in Fig. 4.28, but for the timeframe of 1979-2020.

regions. This is mirrored slightly in SON; however, there is also an increase of trends into the King Hakon Sea sector throughout SON.

The biggest takeaway from the sea ice concentration trend reconstruction (Fig. 4.30) is the amount of statistical significance in these pre-satellite era trends throughout the reconstruction. In SON, the least statistically significant reconstruction, there is still a fair amount of area around the continent marked with statistical significance in the reconstruction. This significance is increased throughout the other three seasons, especially in MAM and JJA (where the regime shift discussed in Fogt et al. (2022) was the strongest), where much of the area is considered statistically significant. This exhibits the skill spatial reconstruction of sea ice concentrations can yield by producing meaningful results in a spatial reconstruction of concentrations in the pre-satellite era. Indeed, the sea ice reconstruction, conducted entirely independent from the sea ice extent reconstruction, highlight the Antarctic sea ice regime shift with impressive skill.

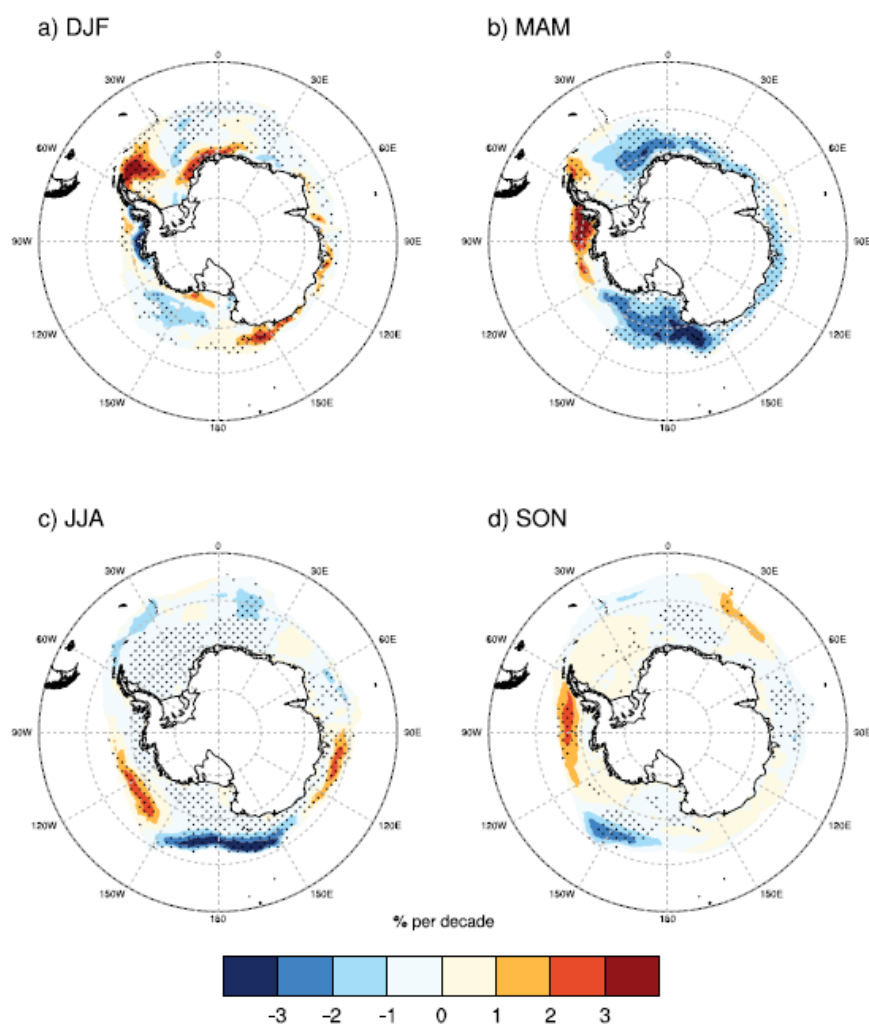
4.11 Spatial Reconstruction Correlations to Climate Indices

In using spatial plots of the reconstruction, sea ice concentration anomalies can be correlated with climate indices to understand the important relationships between Antarctic sea ice and large-scale climate variability throughout the 20th century. This correlation, ranged from negative one (indicating inverse relationships) to positive one (indicating direct relationships) throughout the spatial field allows for analysis of where the reconstruction is aligning with modes of climate variability. Through seasonal correlations of the sea ice concentration reconstruction with the SAM index from 1979-2016 (Fig. 4.31), DJF indicates areas of high correlation to SAM values in the Ross, King

Figure 4.30

Reconstruction of Spatial Seasonal Sea Ice Concentration Trends for 1905-1978

Sea Ice Concentration Trend, 1905-1978



Note. As in Fig. 4.28, but for the timeframe of 1905-1978 and a scale of $\pm 3\%$.

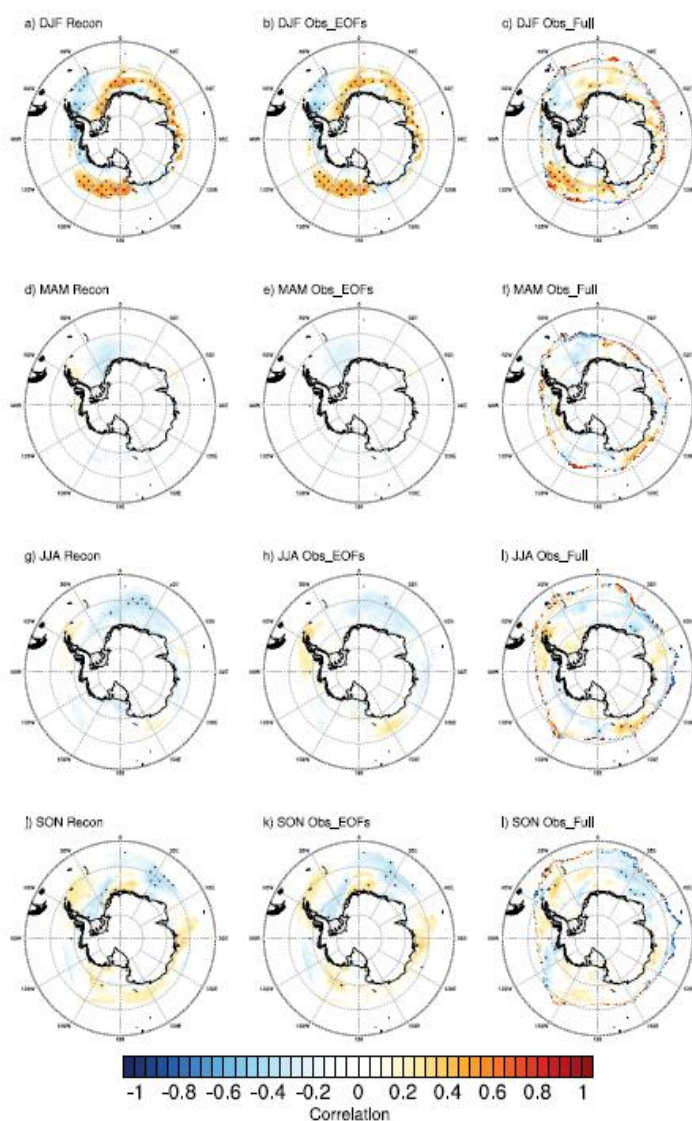
Hakon, and Weddell Sea sectors with statistically significant ($p < 0.05$) correlations occurring in these sectors; however, significant negative correlations occur into some areas of the Weddell Sea sectors towards the Antarctic Peninsula, where values from the reconstruction are showing opposite variability compared to SAM index values in this same region. These positive and negative correlations do not appear in the correlation as significantly throughout the MAM and JJA reconstructions or even observations, with little significance occurring over the King Hakon region with slight negative correlation between SAM values and sea ice concentration reconstruction values. There are slight areas of both positive and negative correlations occur throughout the austral spring (SON), with areas of East Antarctica observing slight positive correlation to SAM values, while the King Hakon region observes slight negative correlations with more statistical significance.

To better resolve the 2016 Antarctic event outline in Chapter 2, spatial correlations between the reconstruction and SAM values were also created to see if larger-scale ice losses and gains will be reconstructed with a significant amount of skill while not eliminating the predictor data that influences the reconstruction of sea ice concentrations. In correlations between the SAM with sea ice concentration reconstructions from 1979-2020 (Fig. 4.32), the overall reconstruction of sea ice concentration yields more significant correlations throughout DJF, with areas experiencing generally higher values of correlation with more areas of statistical significance. The observed EOFs yield similar results of correlations throughout the entire spatial field that mirrors observational sea ice concentration, albeit at larger scales

Figure 4.31

Correlation between SAM Values with Sea Ice Concentration Patterns from 1979-2016

SAM Correlation with Ice Concentration, 1979-2016



Note. A: Correlations with the reconstruction of sea ice concentration seasonally; B: Correlations with the observed EOF of sea ice concentration seasonally; C: Correlations with observations of sea ice concentrations seasonally. Stippling indicates correlations significantly different from zero at $p < 0.05$.

where sea ice concentration in the reconstruction is more smoothed out. In MAM, the slightly negative correlations have disappeared from the Weddell Sea sector, with almost zero correlations observed throughout much of the spatial area around Antarctica with no statistical significance remaining consistent between the 1979 – 2016 (Fig. 4.31) to the 1979 – 2020 (Fig. 4.32) plot. The overall plots between JJA and SON remain consistent throughout the additional timeframe; however, there are some noticeable areas throughout SON in the King Hakon Sea sector that yield higher (or less negative) values of correlation with less area of statistical significance in these plots (Fig. 4.32) compared to those ending in 2016 (Fig. 4.31). This overall increase in correlations shows meaningful results can come despite substantial changes in sea ice concentration values over short periods of time.

The reconstruction creates the unique opportunity to examine how these relationships with the SAM have changed throughout the 20th century. Notably, the significant correlations with SAM and sea ice concentrations extend through the pre-satellite era (Fig. 4.33), especially in the austral summer, with statistically significant ($p < 0.05$) correlations throughout the Ross Sea and King Hakon Sea regions with the SAM. There are significant correlations throughout the Weddell Sea sector with a significant negative correlation between reconstructed sea ice concentration values and SAM values between 1905-1978 with some significant negative correlations extending throughout the Bellingshausen-Amundsen Sea sector. The overall spatial correlation plot between SAM and sea ice concentration in DJF outlines a high area of statistical significance, outlining these sea ice concentration reconstructions can be skillful to

investigate SAM relationships with Antarctic sea ice prior to satellite observations. There is also the presence of some significant correlations throughout the other seasons, most notably the further extent of the King Hakon sector in SON and the Ross Sea sector in JJA along with an area in the Bellingshausen-Ross sea sector in JJA, where negative correlations show statistical significance.

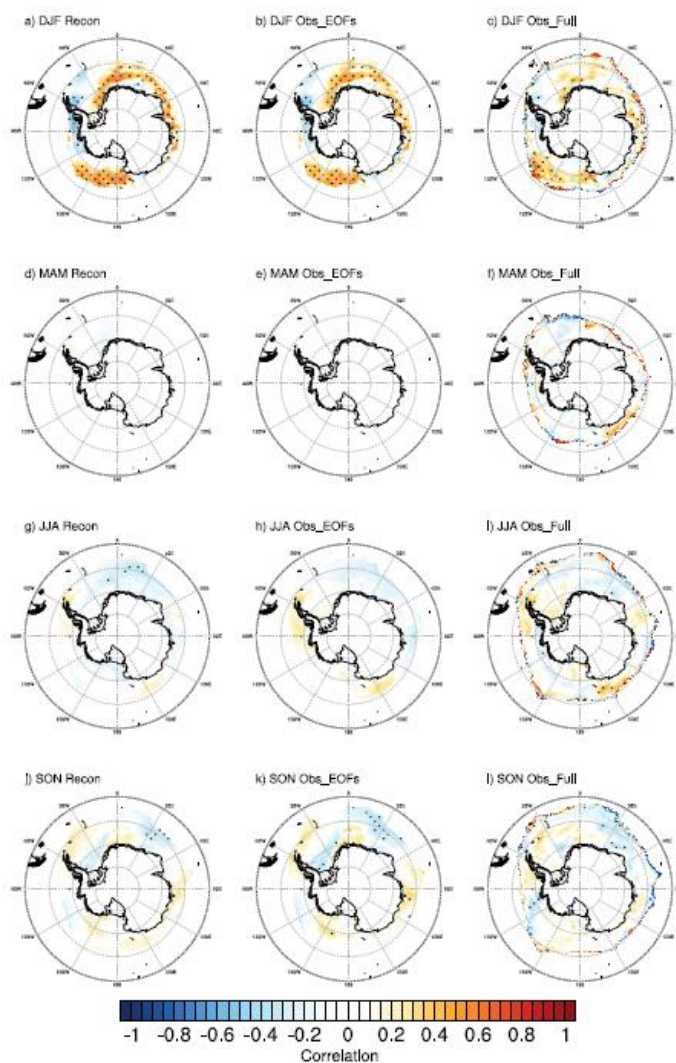
These correlations of climate indices into reconstructions can be extended into the IPO, which was outlined in Table 4.1 as having significant relationship with sea ice concentration EOFs. In a spatial seasonal correlation between the IPO and sea ice concentration reconstructions and observations (Fig. 4.34), there are positive correlations throughout the Weddell sea region in MAM, JJA, and SON in the reconstructions, many of which exhibit statistical significance. In these seasons, there are also significant negative correlations between the IPO and sea ice concentration reconstruction in the Ross Sea and King Hakon Sea sectors; however, these negative correlations are more marked in the leading 5 EOFs and reconstruction than in the full observed data, especially throughout MAM and JJA in the Ross Sea region. These negative correlations being resolved in both the reconstruction and full observational data of sea ice concentrations in the Ross Sea further demonstrates the skill in reconstruction throughout SON. This skill is not as resolved throughout DJF, where correlations in reconstruction, observational EOFs, and observations are consistent; however, there is little statistical significance throughout the reconstruction spatial plot from 1979 – 2016.

The IPO also exhibits statistical significance prior to the satellite-era (Fig. 4.35), where high areas of statistical significance are displayed in MAM, JJA, and SON. In

Figure 4.32

Correlation between SAM Values with Sea Ice Concentration Patterns from 1979-2020

SAM Correlation with Ice Concentration, 1979-2020

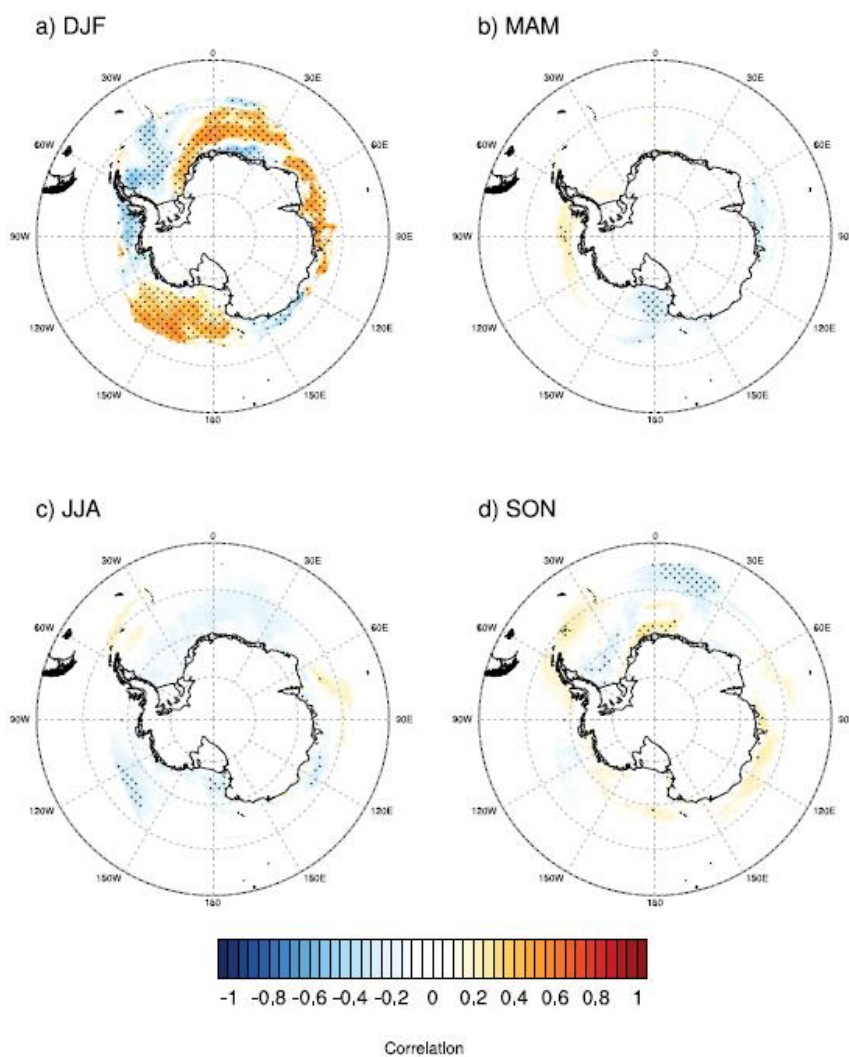


Note. As in Fig. 4.31, except for the period of 1979-2020.

Figure 4.33

Correlation between SAM Values with Sea Ice Concentration Patterns from 1905-1978

SAM Correlation with Ice Concentration, 1905-1978

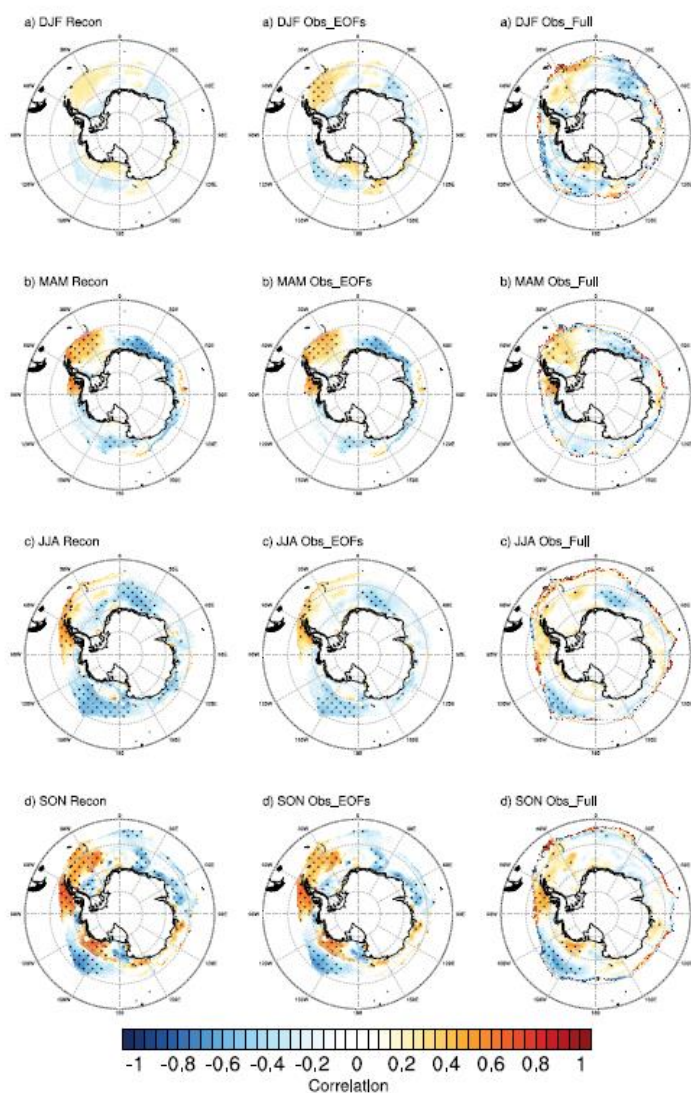


Note. As in Fig. 4.31, except for the period of 1905-1978.

Figure 4.34

Correlation between IPO Values with Sea Ice Concentration Patterns from 1979-2016

IPO Correlation with Ice Concentration, 1979-2016



Note. As in Fig. 4.31, except for correlations with the IPO rather than SAM.

MAM, areas of statistical positive correlations between sea ice concentration reconstructions and IPO values through 1905 – 1978 are present within the Weddell Sea and Bellingshausen-Amundsen Sea sectors, especially near the Antarctic Peninsula. The highest correlations (which also demonstrates statistical significance) are collocated towards the peninsula itself throughout all three seasons. This pattern is especially prevalent in SON, where the highest correlations are observed near the peninsula; however, there are pockets of high correlations into the Ross Sea sector, where pockets of high and low correlations tend to deviate more dramatically whilst maintaining statistical significance. These changes in correlations while maintaining significance outlines that the IPO provides skillful and meaningful reconstructions through MAM, even in cases where the sea ice concentrations tend to behave more erratically. The statistical significance is also outlined throughout the King Hakon region with areas extending into the furthest areas of the Ross Sea sector exhibiting negative significant correlations throughout all seasons except for the austral summer. There is a notable absence of statistical significance in DJF, similar to the 1979-2016 period (Fig. 4.34), where the IPO will not provide extravagantly meaningful results; however, the prevalence in statistical significance throughout MAM, JJA, and SON outline the ability of providing skillful results in sea ice concentration reconstruction. This also demonstrates a persistent influence of the IPO throughout the 20th century, outside of DJF – which is more influence by the SAM (Fig. 4.31).

A final spatial correlation between the Niño 3.4SST network and sea ice concentration values through reconstruction and observation (Fig. 4.36) can be observed

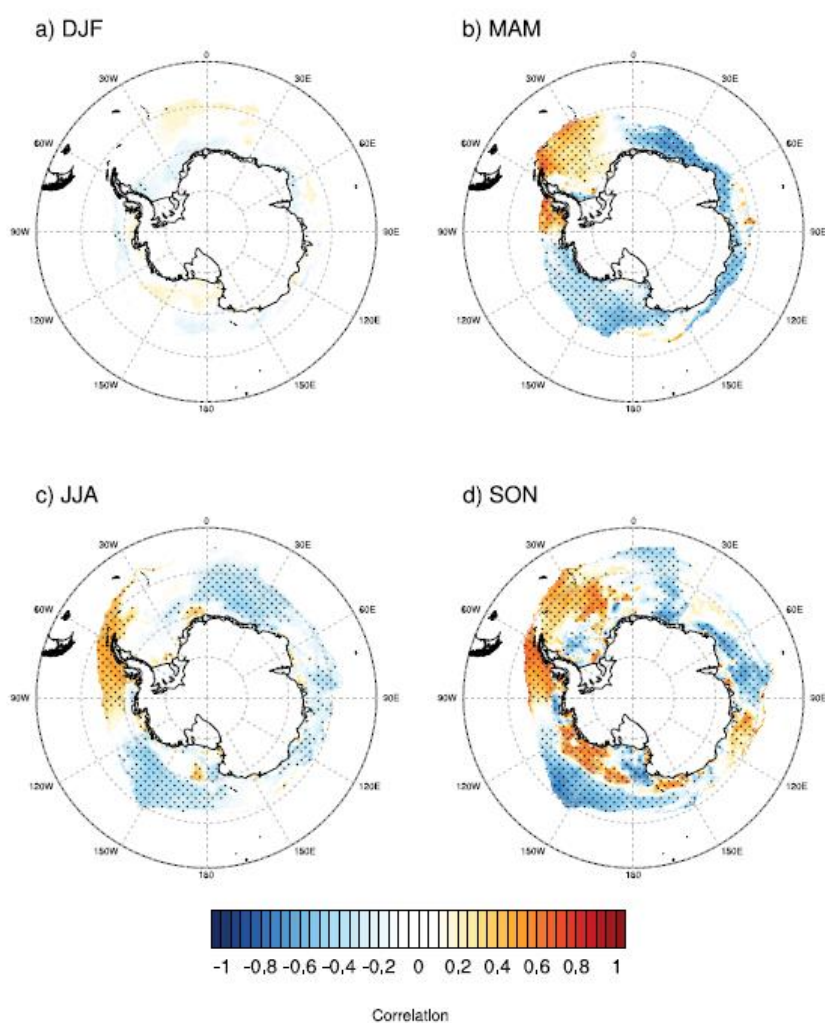
to further assess the skill of reconstructions and the role of tropical Pacific variability on Antarctic sea ice variations throughout time. In Fig. 4.36, these relationships near the Antarctic Peninsula are very well resolved, with high correlation values and statistical significance throughout many points in the region among all seasons in the plots. This statistical significance is least resolved in DJF; however, correlations between the reconstruction and observations have similar positive values with little statistical significance in observations, which can explain the lack of statistical significance in the reconstruction. The correlations between Niño3.4 and reconstructions can also be found to be negative in the King Hakon region throughout most seasons with the exterior portions of the Ross Sea sector observing this pattern, finding the most statistical significance in JJA and SON, when sea ice tend to be less resolved due to the ice freezing closer to the continent and providing extremely high values of concentration.

The relationships with the Niño 3.4 SSTs can also be observed by shifting the time to the pre-satellite era (Fig. 4.37), where significant correlations are observed throughout all seasons. In analyzing Niño3.4SST correlations, patterns similar to the IPO emerge throughout all seasons, where much of the area in SON and MAM outline areas of statistical significance while less areas exhibit statistically significant trends in DJF. Positive correlations become present throughout the Weddell Sea and Bellingshausen-Ross Sea sectors throughout SON with interior portions of the Ross Sea and East Antarctica sectors exhibiting similar correlations, while negative correlations becoming prevalent in the King Hakon region and the outer edges of the Ross Sea region. The Niño3.4SST and IPO also outlines an overall negatively correlated King Hakon region

Figure 4.35

Correlation between IPO Values with Sea Ice Concentration Patterns from 1905-1978

IPO Correlation with Ice Concentration, 1905-1978

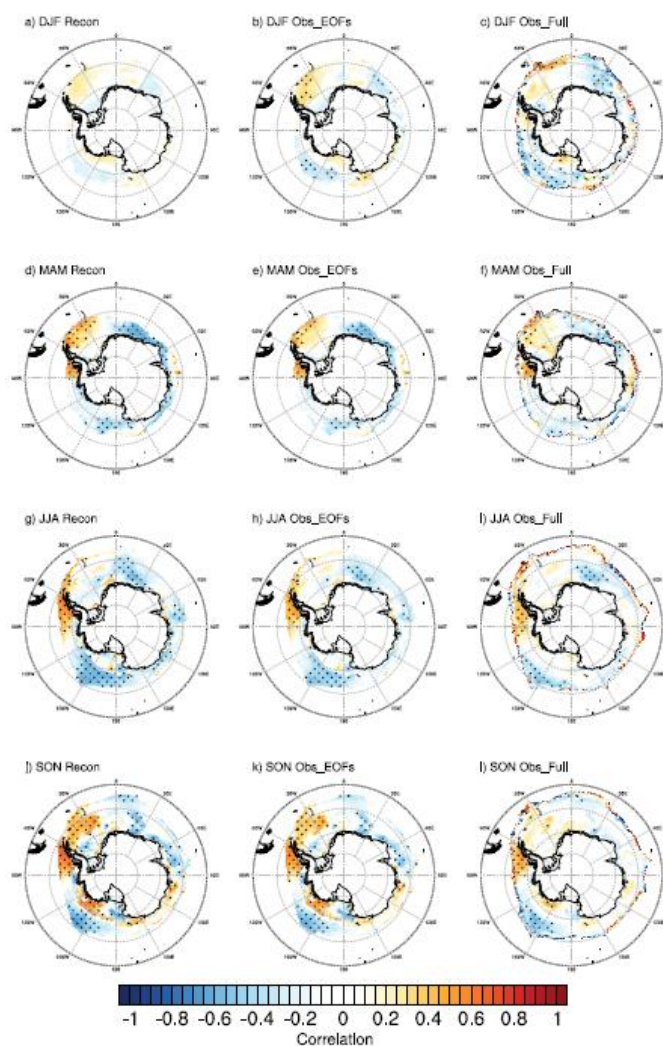


Note. As in Fig. 4.34, except for a period of 1905-1978.

Figure 4.36

Correlation between Niño3.4SST Values with Sea Ice Concentration Patterns from 1979-2020

Nino3.4 Correlation with Ice Concentration, 1979-2020

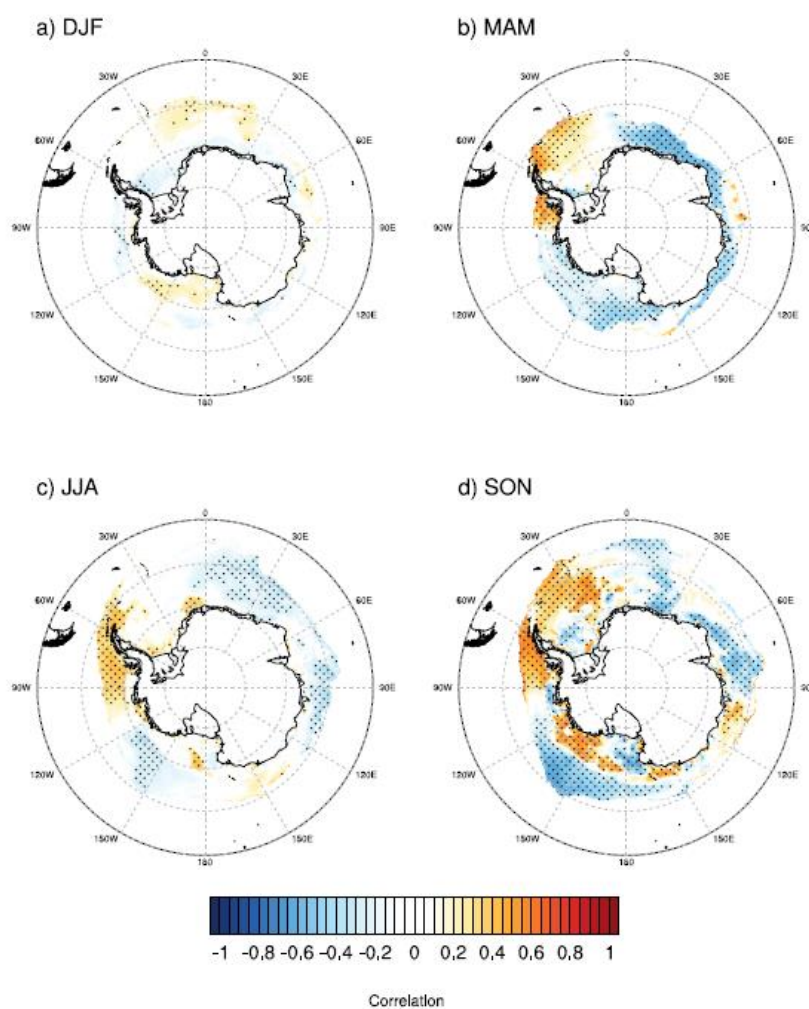


Note. As in Fig. 4.31, except for correlations with the Niño3.4 SSTs rather than SAM.

Figure 4.37

Correlation between Niño3.4 SST Values with Sea Ice Concentration Patterns from 1905-1978

Nino3.4 Correlation with Ice Concentration, 1905-1978



Note. As in Fig. 4.36, except for the period of 1905-1979.

throughout MAM along with low correlations throughout the Ross, Weddell, and King Hakon Sea sectors throughout DJF outlining Pacific influences in ENSO and the IPO as explained in Chapter 2. The statistical significance of these correlations between Niño3.4SST and reconstructed sea ice concentrations outline another case of meaningful and skillful reconstruction results.

Using these spatial correlation plots, the skill of reconstructions can be observed further through understanding the relationships of Antarctic sea ice with large scale climate variability throughout the 20th century. In these plots, overall, the reconstruction spatial plots show consistency with the observed sea ice concentration plots, where observed correlations with climate modes tend to be resolved well throughout the seasonal reconstructions, including in the pre-satellite era. The ability to reconstruct sea ice concentration and find similar patterns in observed sea ice represents the ability to reconstruct spatial fields with meaningful results, and suggest these relationships are stationary throughout the 20th century.

4.12 Chapter Summary

Overall, the results from reconstructing sea ice concentrations can conclude that reconstructions throughout the 20th century are possible on seasonal scales with skillful results.

Due to the complexities of the spatial sea ice concentration field, it was decomposed into the five leading EOF patterns for each season, and the PCs of these were reconstructed. The use of EOFs accounts for a certain percentage of the total variance in the spatiotemporal sea ice concentration anomalies, with distinct patterns that

are strongly tied to climate variations and easier to reconstruct separately than the full field at once. In these, EOF1 explains the greatest percentage of the variance (between 16.7% - 25% per season) while EOF5 explains ~5% of the variance in sea ice concentration anomalies throughout all seasons but manages to capture variations around the ice edge. These PCs can be related to patterns of large-scale climatic variability, outlined in Chapter 2, using detrended correlations to depict these connections in each PC. Through the detrended correlation analysis, influences from SAM and IPO become prevalent throughout many PC/season combinations, with a smaller influence from AMO (significant to $p < 0.1$) prominent throughout PC3 and throughout JJA. Throughout the austral spring, these PCs were well related to Pacific-based climate modes. These detrended analyses ultimately returned that 88 percent of correlations revealed statistical significance between a climate mode of variability and pattern in the PC, suggesting these PCs can produce meaningful reconstructions due to the influence of these climate indicators on the spatial EOF patterns.

The reconstruction model uses a subset of the available predictor data based on the correlation statistical significance. Investigating which predictor data are strongly correlated to each PC (and therefore included in the reconstruction) helps to understand aspects of reconstruction skill. Using predictor counts, the individual station pressure and temperature observations dominate the reconstruction predictor network, with modes of climate variability influencing the different EOFs. There is a relationship between Pacific modes of variability throughout the reconstruction that agrees with results from the detrended correlation analysis, where these modes of variability serve as important

predictors for the first three EOFs. The absence of AMO data in the reconstruction is noticeable, with weaker influences prevalent in EOF4 and EOF5.

The reconstruction performance can be further analyzed using boxplots of various skill metrics comparing using all ensemble members throughout the reconstruction. Throughout all seasons, positive measures in the skill metrics indicate higher reconstruction skills compared to a reconstruction using the climatological mean, with best-fit ensemble members passing the most difficult skill metric (the coefficient of efficiency, CE) with values above zero. This skill is echoed in the calibration correlation (cal r) metric, where correlation between observed data and reconstruction data tends to be high throughout all seasons, with many values close to 1, indicating a strong positive correlation between the observed and reconstructed data. These reconstruction skills were performed spatially, where correlations are observed to be providing great skill around the ice edge and into the interior Weddell Sea region, while failing to provide great skill closer to the Antarctic continent. These areas of weakness could be explained, especially in the austral winter and spring, where pack ice is present throughout the entire year.

The changes in sea ice concentration changes can also be assessed using sea ice trends of PCs along with their 95% confidence intervals. These trends outline statistical significance throughout ~43% of sea ice trends being deemed statistically significant, with over half of these trends occurring at $p < 0.01$. These trends can be observed at smaller scales through time-varying trend plots, where variability between ensemble members outline short-term changes. For instance, PC4, strongly related to variability from the Atlantic Ocean, has trends that change throughout time, suggesting different

influences on Antarctic sea ice from the Atlantic Ocean in DJF. In outlining correlations between seasonal modes of variability and sea ice concentration trends, such as SAM in DJF for 1905-1978, statistical significance of $p < 0.05$ is observed throughout the spatial field. This is especially prevalent in the King Hakon and Ross Sea sectors, where it can be observed that sea ice concentrations can be skillful in resolving SAM influences in the pre-satellite era.

The results of this chapter prove that reconstruction of Antarctic sea ice concentration is skillful and has potential to be understood in a broader context through further research questions. These reconstructions provide results that suggest reconstructions of sea ice concentrations can be provide skillful results and can be understood at a greater context, with the ability to reconstruct sea ice concentration into the pre-satellite era. These reconstructions can also provide more information on climatic variability around the Antarctic continent that can be employed in further simulations to gather a long-term picture of Antarctic sea ice variability.

Chapter 5: Summary and Conclusions

This thesis used statistical relationships between predictor data networks to compose, evaluate, and analyze skillful reconstructions of Antarctic sea ice concentrations throughout the twentieth century. This thesis, then, set out to assess these reconstructions and see the exact skill and if meaningful results could be extracted for the understanding of long-term Antarctic sea ice variability. The Antarctic sea ice cryospheric interactions have great importance, where historical records can assess what interactions are taking place and how modes of climate variability can influence the overall sea ice behavior around Antarctica. The use of principal component analysis techniques with sea ice concentration can provide information on how different modes of climate variability influence these concentrations in a historical context.

The use of observation-based climate reconstructions has been used to gather pressure reconstructions (Fogt et al. 2016) prior to direct measurements along with, recently, sea ice extent (Fogt et al. 2022) patterns. In these studies, principal component regression techniques provided skillful results, with the latter study indicating the possibility of sea ice concentration reconstructions becoming inherently possible. This thesis sought out to investigate the ability for skillful Antarctic sea ice concentration reconstructions with the three following research questions:

1. How does the skill of seasonal Antarctic sea ice concentration reconstructions vary spatially throughout the 20th century?
2. How do large-scale patterns of climate variability influence the sea ice concentration skill over the Southern Ocean?

3. From using the sea ice concentration reconstructions, how are the Antarctic sea ice changes since 1979 understood in a longer historical context spanning the entire 20th century, and to what magnitude have large-scale patterns of climate variability influenced Antarctic sea ice concentrations throughout the 20th century?

The skill of seasonal Antarctic sea ice concentration reconstructions (question 1) was investigated using principal component analysis coupled with observational measures of Antarctic sea ice concentration to generate reconstructions. These reconstructions were analyzed using seasonal boxplots along with the skill metrics of calibration correlation (cal r), validation correlation (val r), coefficient of efficiency (CE), and reduction of error (RE).

Prior to understanding the spatial skill of the reconstruction, understanding the performance of overall ensemble reconstructions of each sea ice concentration PC is vital. The performance between all ensemble members returned measures of above 0.4 throughout JJA and SON, with the lowest value of CE falling at ~0.35 in DJF and ~0.3 in MAM, with the CE values yielding the lowest values; however, all the medians of all skill metrics were greater than 0.5, indicating the skillful results of the reconstruction. In passing the CE skill metric, the reconstruction of sea ice concentration yields extremely meaningful results and can be expanded spatially, far outperforming estimates of past sea ice concentration based on the climatological mean.

The individually reconstructed PCs can be merged and compared to the maximum possible performance using only the leading five EOF patterns of observed sea ice concentration to understand the reconstruction skill spatially. In doing so, it was found

that the seasonal skill of sea ice concentration reconstructions varies spatially throughout the 20th century, where the ice edge is relatively well resolved, with correlation values ranging between 0.7-1.0 throughout JJA and SON, throughout all sectors except for East Antarctica. There is also high skill in the reconstruction throughout the Weddell Sea sector throughout all seasons, where these highest values of correlation between observed and reconstructed sea ice concentrations are observed in MAM, with correlation values approaching 0.9; however, the interior portions of this sector fail to produce as meaningful results. This pattern is amplified into JJA and SON, where the reconstruction produced less skill in the interior Weddell Sea sector with these patterns becoming prevalent amongst all sectors throughout these seasons. This could indicate that the reconstruction fails to provide meaningful results throughout these seasons closer to the Antarctic coast; however, the prospect of pack ice can influence these reconstructions closer to the continent where the climate modes of variability will not play as large of a role in sea ice concentration trends, and where the sea ice cover is nearly uniform every year.

By use of seasonal plots correlated with different climate modes of variability, positive correlations between sea ice concentration reconstructions, sea ice concentration from the leading five observed EOFs, and the full sea ice concentration observations were found to be closely tied together. In looking at the SAM index (which held dominant influence in EOF patterns) prior to the 2016 Antarctic sea loss event, the reconstructions of concentrations along with the observed over 1979-2016 outlined many areas of statistically significant correlations of ~0.5 throughout the Ross Sea sector, along with

regions of the King Hakon sector in DJF. In extending the end time to 2020, even more areas of statistical significance with the reconstruction yielding similar correlations throughout the spatial field compared to the observed sea ice concentration.

Overall, in direct answer to the first research question, the Antarctic sea ice concentration reconstruction provides skillful reconstruction throughout the ice edge in the Weddell Sea, Ross Sea, and Bellingshausen-Amundsen Sea sectors while lower in areas where pack ice is present. These skillful reconstructions can be observed year-round for the ice edge, along with higher correlation values in the outer regions East Antarctica and King Hakon sector in JJA and SON.

The influence of patterns of climate variability (outlined in chapter 2) on reconstruction skill over the Southern Ocean (question 2) becomes more prevalent in looking at detrended correlation analyses between the indices of major modes of climate variability with the seasonal Antarctic sea ice concentration PCs that were reconstructed. In determining this relationship, it was found that Pacific-based climate modes of variability, such as the tropical Pacific SSTs and IPO dominate the reconstruction of the PCs, with moderate influence from the SOI, PDO and the Atlantic-based AMO. The SAM played a much more important role in PC1 overall. The completion of the detrended correlation analysis showed 88% of correlations reached statistical significance at $p < 0.10$ with at least one mode of climate variability.

The role of patterns of climate variability in influencing the reconstruction skill was also determined by analyzing the distribution of predictor data that were included in each ensemble reconstruction. In the analysis of predictor counts that resolve lagging of

seasons, station temperature and pressure measurements dominated overall. However, the Pacific modes of climate variability show up throughout the first three EOFs, that explain a majority of the variance, with the Atlantic relationships encompassing the final two EOFs, which explain less of the variance in sea ice concentration, especially outside the Pacific sector. If this is broken down into further networks, moderate influences from Pacific modes appear throughout JJA for EOF1 with smaller subsets of pressure and temperature observations being prevalent in this season. Thus, in answer to research question 2, the Pacific modes of climate variability strongly influence the reconstruction skill, while Atlantic modes are more common in higher EOF patterns that explain smaller portions of the total sea ice concentration variability. Temperature and pressure observations provide the most influence on the reconstruction skill over modes of climate variability, especially at smaller data network sizes ($p < 0.025$, $p < 0.01$).

With the validity of the seasonal Antarctic sea ice concentration reconstructions determined, they can now be used to investigate changes in sea variability throughout the 20th century in response to research question 3. In this thesis, this was primarily assessed through linear sea ice trends and 95% confidence intervals, where the confidence intervals are used in observing the range of values where 95% confidence the slope is occurring from the entire period of 1905-2020, with subsections of the pre-satellite era (1905-1978) and the satellite era (1979-2020). These trends were calculated over different time periods to observed how trends changed in each EOF pattern / PC timeseries. From the linear trends statistical significance ($p < 0.05$) was achieved throughout DJF (where ~67% of the trends were strongly significant) with positive trends

over the entire period and the post-satellite period. These trends also appeared in EOF4 and EOF5. This was extended into the austral winter (JJA), where ~47% of trends were deemed statistically significant. These trends were observed throughout time-varying trend plots for PCs, where noticeable shifts throughout the early 20th century as discussed in Fogt et al. (2022) were reflected from previous sea ice trends, with negative trends occurring throughout all ensemble members. The time-varying trends in the austral summer for PC4 outline shifts in the Atlantic Ocean, while also outlining that trends throughout the PC are maximized close to the 30-year period, suggesting that short-term variability can be influenced by outliers in the reconstruction.

To investigate how modes of climate variability are related to sea ice concentrations, correlations with the SAM, IPO, and Niño3.4 SSTs were conducted with the spatial Antarctic sea concentrations throughout the 20th century. In looking at SAM, which dominates most of the EOF patterns compared to other modes of variability, in the pre-satellite era there are pronounced areas of statistical significance observed with correlation values ranging from -0.4 to 0.6, where this mode of climate variability does carry some influence into reconstruction skill throughout the Southern Ocean. This is mirrored when looking at similar plots for the Niño3.4SST and IPO correlation plots for the pre-satellite era.

The IPO also demonstrated significant influence in EOF patterns and provides statistical significance throughout SON, JJA, and MAM, with areas of positive correlations throughout the Weddell Sea and Bellingshausen-Amundsen Sea sectors, located close to the Antarctic Peninsula, especially throughout SON with high pockets of

correlation into the outer reaches of the Ross Sea sector. However, these correlations tend to fluctuate more significantly in the Ross Sea between high and low values while maintaining their significance statistically, indicating areas of extreme IPO variability in this region resolved through the reconstruction. The high area of statistical significance in MAM demonstrates that the relationship of the IPO on Antarctic sea ice persists throughout the 20th century and is stationary in time.

The Antarctic sea ice concentration correlations Niño 3.4SST provide similar results to the IPO and SAM in the pre-satellite era, with significant correlations persisting before 1979, observed among all seasons, with MAM and SON outlining large areas of statistical significance with less significance into DJF. These positive correlations are especially prevalent into the interior portions of the continent while also outlining negatively correlated areas in the King Hakon sector into DJF. These significant correlations outline the ability of ENSO to continually influence Antarctic sea ice throughout the 20th century. Therefore, in response to research question 3, Antarctic sea ice has fluctuated throughout the 20th century, with periods and regions of statistically significant positive and negative trends that vary in space and time. However, the relationships between patterns of large-scale climate variability across the Southern Hemisphere and Antarctic sea ice concentrations remain consistent throughout the 20th century. Further work is needed to understand these new aspects of Antarctic sea ice variability and change afforded by the new reconstructions presented in this thesis.

The result of this thesis presents the skill in sea ice concentration reconstructions and has greatly advanced knowledge pertaining to the reconstruction of these

concentrations into the 20th century. Through the many statistical analyses, these reconstructions have been deemed statistically significant to a greater extent than the climatological mean, where values can now be assessed to understand the variability of Antarctic sea ice variability. This research also opens the door to further research into climate variability in the Southern Ocean, along with providing framework for further reconstruction studies.

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